

LINEAR ALGEBRA I & II

Lecture Notes

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Contents

1	Introduction	0
1.1	Fibonacci	0
1.1.1	Motivation	0
1.1.2	Making the Problem Bigger	1
1.1.3	Prior Knowledge	3
1.1.4	Symmetry	4
2	Foundations	7
2.1	Logic	7
2.1.1	Statements	7
2.1.2	Negation	7
2.1.3	Truth Table	8
2.1.4	And and Or Operations	8
2.1.5	Logical Implications	9
2.1.6	Logical Equivalence	10
2.1.7	Predicate Logic	11
2.2	Set theory	13
2.2.1	Sets	13
2.2.2	Set-Builder Notation	14
2.2.3	Empty Set	14
2.2.4	Sets as Elements	14
2.2.5	Subsets	15
2.2.6	Operations on Sets	15
2.2.7	Cartesian Product	16
2.2.8	Power Sets	17
2.2.9	Cardinality of a Set	17
2.2.10	Russell's Paradox	17
3	Foundations of linear algebra	19
04	Maps: Sets and Functions	19
04.1	Cardinality	19
04.2	Functions: basic notions	19
04.3	Injective, Surjective, Bijective Maps	20
04.4	Inverse Map	21
04.5	Composition of Maps	21
04.6	Image and Inverse Image	22
04.7	Graph of a Function	23
04.8	Bases	23
05	Fields	23
06	Linear equations: Systems of Linear Equations	24
06.1	Row reduction	24
06.2	Parametric form of the solution	24
06.3	Elementary row operations	26
06.4	Elementary Row Operations and Row Equivalence	27
06.5	Row-Reduced Echelon Form	30
06.6	Worked example	30
06.7	Another worked example	31
07	Vector spaces: Definition of Vector Spaces	31
07.1	Linear Subspaces	33
07.2	More examples of vector spaces and subspaces	35
07.3	Examples of Vector Spaces	36
07.4	Spaces of Functions	36
07.5	Polynomials	37
07.6	Spaces of Matrices	38
4	Vector spaces and bases	40

12.1	Dimension of subspaces	49
13	Row and col spaces	52
13.1	Row and Column Spaces	53
14	Sums of vector spaces	55
14.1	Sums of Vector Spaces	56
5	Linear maps and matrices	60
15.a	Linear maps: Homomorphisms of Vector Spaces	60
15.b	Linear maps: Kernel and Image of a linear map	64
16	Linear maps and bases: Writing linear maps using bases/coordinates	70
17.a	Matrices: Matrix multiplication	76
17.b	Matrices: Invertible matrix	79
17.b.1	Change of basis	82
17.c	Matrices: Equivalent and similar matrices	84
17.d	Matrices: Back to linear equations	89
17.e	Matrices: Elementary row operations	91
18.a	New vector spaces out of old ones: $\text{Hom}(V, W)$	96
18.a.1	Dual Space	101
18.a.2	The dual map	103
18.a	New vector spaces out of old ones: Semester 2 Setup	107
18.a.1	Reflexivity	107
18.a.2	naturality	108
18.b	New vector spaces out of old ones: Direct sums	108
18.c	New vector spaces out of old ones: Quotient spaces	110
18.c.1	Quotient spaces & complements	115
18.c.2	Relations to the dual space	116
6	Determinants and eigen theory	119
20.a	Determinants: Introduction	119
20.b	Determinants: Uniqueness of the determinant function	127
20.c	Determinants: Lecture 34 Placeholder	131
20.c.1	Cramer's rule	136
20.c.2	Determinants and Endomorphisms	137
21.a	Eigenvectors: Eigen Vectors	138
21.a.1	Lecture 36 Placeholder	142
21.a.2	The characteristic polynomial	143
21.b	Eigenvectors: Direct sums of eigenspaces	147
21.c	Eigenvectors: Geometric and algebraic multiplicities	151
21.c.1	Diagonalisability	152
21.c.2	Trigonalization	155
21.d	Eigenvectors: Minimal polynomial & Cayley–Hamilton theorem	155
21.d.1	Minimal polynomial & Cayley–Hamilton theorem	160
7	Inner product and spectral theory	167
22.a	Euclidean-Hermitian spaces: Euclidean and Hermitian (unitary spaces)	167
22.a.1	Orthogonal & unitary matrices	174

Chapter 1

Introduction

Lecture 1: A Nice Introduction

17 Sep 2025

1.1 Fibonacci

1.1.1 Motivation

In this introduction we solve a problem with the help of linear algebra. The problem is classical, but it already contains many of the central ideas of the course.

In school one quickly learns how to handle arithmetic and geometric sequences. If $a, d \in \mathbb{R}$, then the arithmetic sequence is given by

$$a_0 = a, \quad a_n = a_{n-1} + d \quad \text{for } n \geq 1,$$

and one easily finds the explicit formula

$$a_n = a + nd \quad \text{for all } n \geq 0.$$

Similarly, if $a, q \in \mathbb{R}$, then the geometric sequence is defined by

$$a_0 = a, \quad a_n = a_{n-1}q \quad \text{for } n \geq 1,$$

and here the explicit formula is

$$a_n = aq^n \quad \text{for all } n \geq 0.$$

Once one knows an explicit formula, one can compute very far terms quickly. More interestingly, one can ask structural questions about such sequences.

Example. Let $a_n = a + nd$ be an arithmetic sequence. One can ask:

1. Does the sequence contain infinitely many prime numbers?
2. How large can the first prime in the sequence be?

3. Can one find long arithmetic progressions consisting only of prime numbers?

Theorem (Dirichlet, 1837). Let $a, d \in \mathbb{N}$ with $a \geq 1$, $d > 1$, and $\gcd(a, d) = 1$. Then the arithmetic sequence

$$a_n = a + nd$$

contains infinitely many prime numbers.

Remark. For example, there are infinitely many primes of the form $1 + 4k$ and infinitely many primes of the form $3 + 4k$, where $k \in \mathbb{N}$.

Interesting as these questions are, they do not really lead us into linear algebra. Instead we turn to the Fibonacci sequence:

$$a_0 = 0, \quad a_1 = 1, \quad a_n = a_{n-1} + a_{n-2} \quad \text{for } n \geq 2.$$

The natural question is the same as before: can we find an explicit formula for the n -th term? That is not obvious at all. The strategy of this lecture is to make the problem larger before we solve it.

1.1.2 Making the Problem Bigger

Definition (Knowing a sequence well). We say that we KNOW a sequence well if we have an explicit formula for its n -th element.

Arithmetic and geometric sequences are known well in this sense. Our goal is therefore to know the Fibonacci sequence well.

Definition (Generalized Fibonacci sequences). Let $a, b \in \mathbb{R}$. We define the sequence $F_{a,b}$ recursively by

$$a_0 = a, \quad a_1 = b, \quad a_n = a_{n-1} + a_{n-2} \quad \text{for } n \geq 2.$$

If we choose $a = 0$ and $b = 1$, then we recover the standard Fibonacci sequence. So the original problem is to know $F_{0,1}$ well.

The new and much larger goal is:

$$\text{Know } F_{a,b} \text{ well for all } a, b \in \mathbb{R}.$$

This new problem looks harder, but it comes with useful structure.

Definition (The space of Fibonacci sequences). A sequence A is called a FIBONACCI-SEQUENCE if there exist $a, b \in \mathbb{R}$ such that $A = F_{a,b}$. We denote the space of all Fibonacci-sequences by

$$\text{Fib} := \{F_{a,b} \mid a, b \in \mathbb{R}\}.$$

Exercise 1.1.5

Show that a sequence $(a_0, a_1, a_2, \dots)$ is a Fibonacci-sequence if and only if

$$a_n = a_{n-1} + a_{n-2}$$

for all $n \geq 2$.

Now let

$$F_1 = (a_0, a_1, a_2, \dots), \quad F_2 = (b_0, b_1, b_2, \dots)$$

be two sequences in Fib. We add them component-wise:

$$F_1 + F_2 := (a_0 + b_0, a_1 + b_1, a_2 + b_2, \dots).$$

Claim. If $F_1, F_2 \in \text{Fib}$, then $F_1 + F_2 \in \text{Fib}$.

Proof of claim. Write

$$F_1 + F_2 = (c_0, c_1, c_2, \dots),$$

where $c_n = a_n + b_n$. Since F_1 and F_2 are Fibonacci-sequences we have

$$a_n = a_{n-1} + a_{n-2}, \quad b_n = b_{n-1} + b_{n-2}$$

for all $n \geq 2$. Hence

$$c_n = a_n + b_n = (a_{n-1} + a_{n-2}) + (b_{n-1} + b_{n-2}) = c_{n-1} + c_{n-2}.$$

By Exercise 1.1.5 this shows that $F_1 + F_2$ is again a Fibonacci-sequence.

Exercise 1.1.6

Check that the argument above actually proves

$$F_{a,b} + F_{c,d} = F_{a+c,b+d}.$$

Apart from addition we also have multiplication by a scalar. If $\alpha \in \mathbb{R}$ and $A = (a_0, a_1, a_2, \dots)$, define

$$\alpha A := (\alpha a_0, \alpha a_1, \alpha a_2, \dots).$$

Exercise 1.1.7

Show that if $\alpha \in \mathbb{R}$ and $A \in \text{Fib}$, then $\alpha A \in \text{Fib}$. Moreover, show that

$$\alpha F_{a,b} = F_{\alpha a, \alpha b}.$$

If one knows two sequences

$$F_1 = (a_0, a_1, \dots), \quad F_2 = (b_0, b_1, \dots)$$

well, then one also knows $F_1 + F_2$ well, because the n -th term is simply the sum of the two known formulas. The same is true for scalar multiples.

Exercise 1.1.8

If one knows $F_1, \dots, F_k$ well, then one also knows

$$\alpha_1 F_1 + \dots + \alpha_k F_k$$

well for all $\alpha_1, \dots, \alpha_k \in \mathbb{R}$. Such expressions are called LINEAR COMBINATIONS.

So the structure of Fib allows us to transfer knowledge from some sequences to others. But do we already know any sequence in Fib well?

Exercise 1.1.9

Find one Fibonacci-sequence that we already know well.

We do know $F_{0,0}$ well: every term is 0. Unfortunately this is not enough, because addition and scalar multiplication never produce anything new from the zero sequence.

Exercise 1.1.10

Assume $(a, b) \neq (0, 0)$. Show that the set

$$\{F_{\alpha a, \alpha b} \mid \alpha \in \mathbb{R}\}$$

contains infinitely many elements, but still does not equal Fib.

This shows that one sequence is not enough. On the other hand, if we knew $F_{1,0}$ and $F_{0,1}$ well, then every sequence in Fib would be known well, since

$$F_{c,0} = aF_{1,0} + bF_{0,1}.$$

Remark The facts that two sequences suffice, one sequence does not suffice, and that any third sequence can be redundant all reflect that Fib has dimension 2. In that sense Fib behaves like a plane.

1.1.3 Prior Knowledge

How does one find a solution to a problem like this? Usually by combining prior knowledge with whatever symmetry the problem has.

Exercise 1.1.12

Show that the only arithmetic Fibonacci-sequence is $F_{0,0}$.

So arithmetic sequences do not help. We therefore try geometric sequences. Let

$$G_q := (1, q, q^2, q^3, \dots),$$

where $q \neq 0$. This is a Fibonacci-sequence if and only if

$$q^n = q^{n-1} + q^{n-2}$$

for all $n \geq 2$. Since $q \neq 0$, we may divide by q^{n-2} and obtain the quadratic equation

$$q^2 = q + 1.$$

Hence

$$q = \frac{1 \pm \sqrt{5}}{2}.$$

Define

$$\varphi := \frac{1 + \sqrt{5}}{2}, \quad \psi := \frac{1 - \sqrt{5}}{2}.$$

Then the two geometric sequences

$$G_\varphi = (1, \varphi, \varphi^2, \varphi^3, \dots), \quad G_\psi = (1, \psi, \psi^2, \psi^3, \dots)$$

belong to Fib, and we know both of them well.

Exercise 1.1.13

Verify that

$$\frac{1}{\varphi - \psi} G_\varphi + \frac{1}{\psi - \varphi} G_\psi = r_0,$$

Remark. To carry this out, one has to solve a small system of linear equations. Solving such systems is one of the first major topics of linear algebra.

From the previous exercise we obtain the explicit formula for $F_{0,1} = (F_0, F_1, F_2, \dots)$:

$$F_n = \frac{\varphi^n - \psi^n}{\varphi - \psi} = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right).$$

We have therefore achieved our original goal.

Exercise 1.1.15

Assume that the limit

$$\lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}}$$

exists. Compute it.

1.1.4 Symmetry

One might wonder whether the discovery of the two geometric sequences was simply lucky. A better explanation is that they come from the symmetry of the space Fib.

If X is a geometric space, then a symmetry of X is usually a map $T : X \rightarrow X$ that preserves the relevant structure. The same idea makes sense for Fib.

Definition (Symmetry of Fib). A map $T : \text{Fib} \rightarrow \text{Fib}$ is called a SYMMETRY of Fib if for all $A, B \in \text{Fib}$ and all $\alpha \in \mathbb{R}$ we have

$$T(A + B) = T(A) + T(B), \quad T(\alpha A) = \alpha T(A).$$

Here are three very simple examples:

1. The identity map $\text{id} : \text{Fib} \rightarrow \text{Fib}$, $A \mapsto A$.
2. For $\alpha \in \mathbb{R}$, the map $M_\alpha : \text{Fib} \rightarrow \text{Fib}$, $A \mapsto \alpha A$.
3. For $B \in \text{Fib}$, the translation map $A_B : \text{Fib} \rightarrow \text{Fib}$, $A \mapsto A + B$.

Exercise 1.1.17

Show that id and M_α satisfy the symmetry rules above for every $\alpha \in \mathbb{R}$, while A_B fails to do so whenever $B \neq F_{0,0}$.

These examples are not yet very interesting, because they exist in essentially every vector space. The real structure of Fib comes from the fact that its elements are sequences.

Exercise 1.1.18

Try to find a more interesting symmetry $T : \text{Fib} \rightarrow \text{Fib}$ that uses the sequence structure.

1

There is a natural translation map

$$S : \text{Fib} \rightarrow \text{Fib}, \quad (a_0, a_1, a_2, \dots) \mapsto (a_1, a_2, a_3, \dots).$$

Exercise 1.1.19

Verify that $S(A) \in \text{Fib}$ whenever $A \in \text{Fib}$, and show that S is indeed a symmetry.

A first instinct is to study fixed points of S .

Exercise 1.1.20

Show that $F_{0,0}$ is the only fixed point of S .

So fixed points are too restrictive here. We therefore weaken the condition slightly.

Definition (Eigensequence). Let $T : \text{Fib} \rightarrow \text{Fib}$ be a symmetry. A sequence $A \neq F_{0,0}$ is called an EIGENSEQUENCE of T if there exists $\alpha \in \mathbb{R}$ such that

$$T(A) = \alpha A.$$

The scalar α is called the EIGENVALUE.

We now search for eigensequences of the shift map S . Suppose

$$A = (a_0, a_1, a_2, \dots) \neq F_{0,0}$$

satisfies $S(A) = \alpha A$. Then

$$(a_1, a_2, a_3, \dots) = (\alpha a_0, \alpha a_1, \alpha a_2, \dots),$$

so

$$a_n = \alpha a_{n-1} \quad \text{for all } n \geq 1.$$

Hence A must be geometric:

$$A = (a_0, a_0\alpha, a_0\alpha^2, a_0\alpha^3, \dots).$$

Because $A \in \text{Fib}$, its terms must also satisfy $a_2 = a_1 + a_0$. Substituting the geometric form gives

$$\alpha^2 a_0 = \alpha a_0 + a_0.$$

Since $A \neq F_{0,0}$, we must have $a_0 \neq 0$, and therefore

$$\alpha^2 = \alpha + 1.$$

So the only possible eigenvalues are φ and ψ . This explains why the two geometric sequences

$$G_\varphi \quad \text{and} \quad G_\psi$$

appear naturally.

1 To summarize, the n -th Fibonacci number is given by

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right).$$

Exercise 1.1.22

Find an explicit formula for the n -th element of $F_{a,b}$ in terms of a , b , and n .

Exercise 1.1.23

Show that

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \ddots}}}.$$

Remark. The idea for this introduction came up while explaining why linear algebra is fun. It is a good example of how a concrete question can lead directly to vector spaces, linear combinations, symmetry, and eigenvalues.

Chapter 2

Foundations

Lecture 2: Logic & Math.

19 Sep 2025

2.1 Logic

2.1.1 Statements

Definition (Mathematical and logical statements). A MATHEMATICAL STATEMENT (or LOGICAL STATEMENT) is a sentence that can be either true or false.

In particular, it cannot be both true and false at the same time.

Example. Consider the following statements:

A: “For every real number x , we have $x^2 \geq 0$.”

B: “Every prime number $p \geq 3$ must be odd.”

C: “Every odd number $p \geq 3$ must be prime.”

Statement A is true, statement B is true, and statement C is false.

2.1.2 Negation

Definition (Negation). Let A be a statement. The NEGATION of A is the statement asserting that A does *not* hold.

We write

$$\neg A$$

for the negation of A . Sometimes one also writes $\overline{A}$.

Remark. Read $\neg A$ as “not A ”.

2.1.3 Truth Table

The truth values of A and $\neg A$ are related as follows:

A	$\neg A$
T	F
F	T

2.1.4 And and Or Operations

Definition (Conjunction). Let A and B be statements. The statement

$$A \wedge B$$

is called the CONJUNCTION of A and B . It means that A holds and B holds.

In other words, both A and B are true.

Remark. In L^AT_EX, one writes $A \wedge B$ as `A\wedge B`.

Informally, one sometimes also writes $A \& B$.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Example. Let $x \in \mathbb{R}$. Then

$$(x^2 = 1) \wedge (x \geq 0)$$

means the same as the statement $x = 1$.

Definition (Disjunction). Let A and B be statements. The statement

$$A \vee B$$

is called the DISJUNCTION of A and B . It means that A holds or B holds, that is, at least one of the two statements is true.

Remark. In everyday language, “either A or B ” often suggests that exactly one of the two statements holds.

In mathematics, however, $A \vee B$ is *inclusive*: it means “either A or B , or both”. In

particular, it may happen that only A holds, only B holds, or that both A and B hold.

Remark. In $\text{\LaTeX}$, one writes $A \vee B$ as $A\text{\textbackslash}vee B$.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Example. Let $x \in \mathbb{R}$. The statement

$$(x > 1) \vee (x < 2)$$

is always true.

Example. Again for $x \in \mathbb{R}$, the statement

$$(x < 1) \vee (x > 2)$$

is equivalent to

$$\neg(1 \leq x \leq 2).$$

2

2.1.5 Logical Implications

Definition (implication). Let A and B be statements. We can form the new statement

$$A \Rightarrow B.$$

It states that whenever A holds, then B holds too.

Equivalently: if A is true, then B is true.

By definition, the truth values are given by:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

Remark. The statement

$$A \Rightarrow B$$

is the same as

$$\neg A \vee B.$$

Example. The statement

$$(0 = 1) \Rightarrow \text{"Earth is flat"}$$

is a true statement.

2.1.6 Logical Equivalence

Definition (Logical equivalence). Let A and B be statements. The statement

$$A \Leftrightarrow B$$

is called the LOGICAL EQUIVALENCE of A and B .

By definition,

$$A \Leftrightarrow B$$

is the statement

$$(A \Rightarrow B) \wedge (B \Rightarrow A).$$

Remark. To say that $A \Leftrightarrow B$ is true is the same as saying that A holds *if and only if* B holds.

One also says that A and B are equivalent. This is often abbreviated by “iff”.

Example. For a fixed real number x , the statement

$$(x^2 > 0) \Leftrightarrow (x \neq 0)$$

is true. If $x \neq 0$, then both component statements are true; if $x = 0$, then both are false.

Example. The statement

$$(0 = 1) \Leftrightarrow \text{"Earth is flat"}$$

is true as well, because both component statements are false.

Remark. Causality is irrelevant!

Claim. Let A and B be statements. Then

$$(A \Rightarrow B) \Leftrightarrow (\neg B \Rightarrow \neg A).$$

Exercise 1.2.1

Show that

$$\neg(\neg A) = A.$$

Show also that

$$\neg(A \wedge B) = \neg A \vee \neg B, \quad \neg(A \vee B) = \neg A \wedge \neg B.$$

Spell out the last two identities in words.

2.1.7 Predicate Logic

Definition (Predicate). A PREDICATE is a statement involving some variables from a set.

Typical examples are

$$P(x), \quad P(x, y), \quad \dots$$

Example. For example, let

$$I(n) = "n = n^2".$$

This can be true or false depending on the value of the variable.

Example. Here is a false statement:

$$\forall n \in \mathbb{Z} : n = n^2.$$

Remark. The symbol

$$\forall$$

means “for all”, or simply “all”.

Example. Here is a true statement:

$$\exists n \in \mathbb{Z} : n = n^2.$$

Read this as: there exists $n \in \mathbb{Z}$ such that $n = n^2$.

Remark. The symbol

$$\exists$$

means “there exists”.

Example. Another true statement is

$$\forall n \in \mathbb{Z} : ((n = n^2) \Rightarrow ((n = 0) \vee (n = 1))).$$

Remark. The symbols $\forall$ and $\exists$ are called QUANTIFIERS.

Example. We can combine quantifiers. For example:

$$\forall y \in \mathbb{R} : (y \geq 0 \Rightarrow (\exists x \in \mathbb{R} : x^2 = y)) \quad (\text{TRUE})$$

Remark. In practice, one often writes this as follows:

$$\forall 0 \leq y \in \mathbb{R}, \exists x \in \mathbb{R} \text{ s.t. } x^2 = y. \quad (*)$$

Remark. Be careful about the order:

$$\forall x \in X, \exists y \in Y : A(x, y)$$

is *not* the same as

$$\exists y \in Y, \forall x \in X : A(x, y).$$

Try this on (*).

Example. Let X be the set of students at ETH and let Y be the set of courses.

For $x \in X$ and $y \in Y$, let $A(x, y)$ be the statement “student x takes course y ”.

Then

$$\forall x \in X, \exists y \in Y : A(x, y)$$

is true.

But

$$\exists y \in Y, \forall x \in X : A(x, y)$$

is not true.

Remark. The second statement would mean that there exists a course such that all students of ETH take this course.

Remark. Shorthand notation:

$$\exists x \in X, \exists y \in X \rightsquigarrow \exists x, y \in X,$$

and similarly

$$\forall x \in X, \forall y \in X \rightsquigarrow \forall x, y \in X.$$

Remark. Negation of quantifiers:

$$\neg(\forall x \in X : A(x)) \Leftrightarrow \exists x \in X : \neg A(x),$$

and

$$\neg(\exists x \in X : A(x)) \Leftrightarrow \forall x \in X : \neg A(x).$$

Definition (Unique existence). The statement

$$\exists! x \in X : A(x)$$

means that there exists an $x \in X$ for which $A(x)$ holds, and that this x is unique.

Remark. Informally, this means: there exists $x \in X$ with $A(x)$, and among all $y \in X$ for which $A(y)$ holds, this x is the unique one.

Remark. Another way to write this is

$$(\exists x \in X : A(x)) \wedge (\forall x, y \in X : (A(x) \wedge A(y)) \Rightarrow x = y).$$

Example. The statement

$$\exists! x \in \mathbb{R} : x^2 = 4$$

is not true.

But

$$\exists! x \in \mathbb{R} : x \geq 0 \wedge x^2 = 4$$

is true.

Lecture 3: Set Theory

24 Sep 2025

2.2 Set theory

2.2.1 Sets

Definition (Set). A SET is a collection of different objects. These objects are called ELEMENTS of the set.

Example.

$$M = \{1, 3, \text{moon}\} = \{3, \text{moon}, 1\} \quad (\text{order is NOT important}).$$

$$M = \{1, 1, 1\} = \{1\} \quad (\text{repetitions do NOT matter}).$$

Remark. If α is an element of M , we write $\alpha \in M$.

If α is *not* an element of M , we write $\alpha \notin M$.

2.2.2 Set-Builder Notation

Remark. We can describe a set by specifying a property its elements must satisfy:

$$M = \{x : A(x)\} = \{x \mid A(x)\},$$

the set of all x that satisfy the condition/property $A(x)$.

Example.

$$\{x \mid x \text{ is an integer and } x^2 \leq 9\} = \{-3, -2, -1, 0, 1, 2, 3\}.$$

2.2.3 Empty Set

Definition (Empty set). The EMPTY SET is the set containing no elements. It is denoted by

$$\{\} = \emptyset.$$

2.2.4 Sets as Elements

Remark. A set is allowed to contain another set as one of its elements.

Example. Let $M = \{1, \text{elephant}, \{a, 5\}\}$. Then

$$\{a, 5\} \in M.$$

Here $\{a, 5\}$ is a set, but also an element of M .

Example.

$$M = \{\{\}\} = \{\emptyset\}$$

is a set containing exactly one element: the empty set.

2.2.5 Subsets

Definition (Subset). Let P and Q be sets.

- (1) $P \subseteq Q$ means that $\forall x \in P : x \in Q$ (equivalently, $x \in P \Rightarrow x \in Q$). We say P is a **SUBSET** of Q .
- (2) $P \subsetneq Q$ means $P \subseteq Q$ but $P \neq Q$. We say P is a **PROPER SUBSET** of Q .
- (3) $P \not\subseteq Q$ means $\neg(P \subseteq Q)$.

Example.

$$\begin{aligned} \{-1, 2, 5\} &\subseteq \{x \mid x \text{ is an integer and } x^2 \leq 30\} \\ \{-1, 2, 5\} &\not\subseteq \{x \mid x \text{ is an integer and } x^2 \leq 20\}. \end{aligned}$$

2.2.6 Operations on Sets

Definition (Set operations). Let P and Q be sets.

$$\begin{aligned} P \cap Q &:= \{x \mid x \in P \wedge x \in Q\} && \text{(intersection),} \\ P \cup Q &:= \{x \mid x \in P \vee x \in Q\} && \text{(union),} \\ P \setminus Q &:= \{x \mid x \in P \wedge x \notin Q\} && \text{(difference),} \\ P \Delta Q &:= (P \cup Q) \setminus (P \cap Q) && \text{(symmetric difference).} \end{aligned}$$

Definition (Complement). If all sets under consideration are subsets of one ambient set X (sometimes called the *Grundmenge*), then we write

$$P^c := X \setminus P,$$

the **COMPLEMENT** of P in X . Sometimes one also writes $\overline{P}$.

Definition (Union and intersection of families). Let $\mathcal{A}$ be a non-empty family (= set) of sets. We define

$$\begin{aligned} \bigcup_{A \in \mathcal{A}} A &= \{x \mid \exists A \in \mathcal{A} \text{ s.t. } x \in A\} && \text{(union),} \\ \bigcap_{A \in \mathcal{A}} A &= \{x \mid \forall A \in \mathcal{A} : x \in A\} && \text{(intersection).} \end{aligned}$$

Remark. In practice, $\mathcal{A}$ often just parametrises/indexes a family of sets.

Example. Let $\mathcal{A} = \{2, 3, \dots\}$. For each $n \in \mathcal{A}$, define

$$A_n = \{x \mid x \text{ is an integer, } x \geq 1, \text{ and } n^2 \text{ divides } x\}.$$

Then

$$A_2 = \{4, 8, 12, \dots\}, \quad A_3 = \{9, 18, \dots\}, \quad \dots$$

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid \exists k, r \in \mathbb{Z}, k, r \geq 1, \text{ s.t. } a = k^2 \cdot r\}, \quad \bigcap_{n \in \mathcal{A}} A_n = \emptyset.$$

2.2.7 Cartesian Product

Definition (Cartesian product). Let X and Y be sets. The **CARTESIAN PRODUCT** of X and Y , denoted $X \times Y$, is the set of all *ordered* pairs (x, y) with $x \in X, y \in Y$.

Example. Let $X = \{1, 2, \text{elephant}\}$ and $Y = \{4, \text{dog}\}$. Then

$$X \times Y = \{(1, 4), (1, \text{dog}), (2, 4), (2, \text{dog}), (\text{elephant}, 4), (\text{elephant}, \text{dog})\}.$$

So $(1, 4) \in X \times Y$, but $(4, 1) \notin X \times Y$.

Definition (n -fold Cartesian product). Let X be a set. We define its **n -FOLD CARTESIAN PRODUCT** as

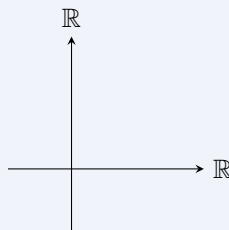
$$X^n = \underbrace{X \times \dots \times X}_{n \text{ times}} = \{(x_1, \dots, x_n) \mid x_1, \dots, x_n \in X\}.$$

The elements $(x_1, \dots, x_n)$ are called **n -TUPLES**.

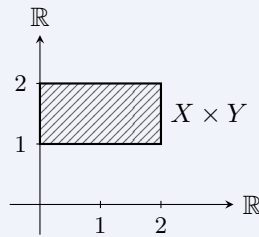
Remark. In particular,

$$X^2 = X \times X, \quad X^3 = X \times X \times X = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in X\}, \quad \text{etc.}$$

Example. Let $X = \mathbb{R}$, the real numbers. Then $X^2 = \mathbb{R}^2$ is the plane.



Let $X = [0, 2]$ and $Y = [1, 2]$. Then $X \times Y$ is the rectangle:



2.2.8 Power Sets

Definition (Power set). Let X be a set. The POWER SET of X is

$$\mathcal{P}(X) := \{ Q \mid Q \text{ is a subset of } X \}.$$

Sometimes written 2^X .

Example.

$$\mathcal{P}(\{1, 2, 3\}) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}.$$

This set has 8 elements.

3

2.2.9 Cardinality of a Set

Definition (Cardinality). Let X be a finite set, i.e. a set with finitely many elements; $X = \{x_1, \dots, x_n\}$. We say that the CARDINALITY of X is n (the number of elements in X) and write

$$|X| = n \quad \text{or} \quad \text{card}(X) = n.$$

Exercise

Let A be a finite set. Let $\mathcal{P}(A)$ be the power set of A . Show that

$$|\mathcal{P}(A)| = 2^{|A|}.$$

2.2.10 Russell's Paradox

Let $U_0 :=$ the set of all sets. Define

$$R := \{ P \mid P \in U_0 \text{ and } P \notin P \},$$

the set of all sets that do not contain themselves.

For example: $U_0 \in U_0$, so $U_0 \notin R$.

3

Claim. $R \in R \Leftrightarrow R \notin R$.

Proof. If $R \in R$, then by definition of R we have $R \notin R$. This shows $R \in R \Rightarrow R \notin R$.

If $R \notin R$, then by definition R does not contain itself, so $R \in R$. This shows $R \notin R \Rightarrow R \in R$. $\square$

Remark. We will not solve this paradox in this course. The solution comes from *axiomatic set theory* (in contrast to “naive set theory”), where one defines more precisely what a set is. It turns out that our U_0 and R cannot be regarded as sets.

Filip Jaskovic

Chapter 3

Foundations of linear algebra

Lecture 4: Grundlagen

26 Sep 2025

04 Maps: Sets and Functions

04.1 Cardinality

Definition (Cardinality). The CARDINALITY of a finite set X is the number of elements in X . For example, if $X = \{x_1, \dots, x_n\}$, then $|X| = n$.

Exercise

Let X be a finite set. Show that

$$|\mathcal{P}(X)| = 2^{|X|}.$$

04.2 Functions: basic notions

Definition (Functions). A function is written

$$f : X \longrightarrow Y,$$

where X is called the DOMAIN of f , and Y is called the TARGET of f .

Two functions

$$f : X \longrightarrow Y, \quad g : X' \longrightarrow Y'$$

are equal if and only if $X = X'$, $Y = Y'$, and for all $x \in X$ we have $f(x) = g(x)$.

Example. • $f(x) = x^2$.

• $X := \{x \in \mathbb{Z} \mid x \geq 0\}$.

- $Y := \{\text{all family names}\}$.
- $g : X \rightarrow Y, \quad g(x) := \text{family name of the student at ETH whose ID number is } x$.

Here g is *not well-defined*: e.g. $g(0) = ?$, $g(10^{10}) = ?$.

But if we extend the target to

$$Y' := \{\text{family names}\} \cup \{\text{Error}\},$$

then $g : X \rightarrow Y'$ becomes well-defined.

Definition (Image). Let $f : X \rightarrow Y$. The IMAGE of f is the set

$$\text{im } f := \{y \in Y \mid \exists x \in X \text{ s.t. } f(x) = y\}.$$

Equivalently,

$$\text{im}(f) = f(X) \subseteq Y.$$

Definition (Restriction). Let $f : X \rightarrow Y$ and let $A \subseteq X$. We define a new map $f|_A : A \rightarrow Y$, called the RESTRICTION of f to A by

$$(f|_A)(x) := f(x) \quad \text{for } x \in A.$$

Example. • The IDENTITY MAP: $\text{id} : X \rightarrow X, \text{id}(x) := x$ for all $x \in X$.

- A CONSTANT FUNCTION: given $y \in Y$, define $f : X \rightarrow Y, f(x) := y$ for all $x \in X$.
- The CHARACTERISTIC FUNCTION of $A \subseteq X$:

$$1_A : X \rightarrow \mathbb{R}, \quad 1_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

04.3 Injective, Surjective, Bijective Maps

Definition (Injective, Surjective, Bijective Maps). Let $f : X \rightarrow Y$.

- (1) f is called INJECTIVE (or an INJECTION) if

$$\forall x_1, x_2 \in X : f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

- (2) f is called SURJECTIVE (or an SURJECTION) if

$$\forall y \in Y \exists x \in X : f(x) = y.$$

Equivalently, $f(X) = Y$.

(3) f is called BIJECTIVE (or a BIJECTION) if it is both injective and surjective. Equivalently,

$$\forall y \in Y \exists! x \in X : f(x) = y.$$

Example. Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

This map is not injective and not surjective.

However,

$$f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, \quad f(x) = x^2,$$

is injective, but not surjective.

04.4 Inverse Map

Definition (Inverse Map). Let $f : X \rightarrow Y$ be a bijection. The INVERSE MAP of f is the map

$$f^{-1} : Y \rightarrow X,$$

which assigns to each $y \in Y$ the unique $x \in X$ such that $f(x) = y$.

4

04.5 Composition of Maps

Definition (Composition). Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$. The COMPOSITION $g \circ f : X \rightarrow Z$ is defined by

$$(g \circ f)(x) := g(f(x)).$$

Diagrammatically,

$$X \xrightarrow{f} Y \xrightarrow{g} Z, \quad X \xrightarrow{g \circ f} Z.$$

Example. We have

$$f \circ f^{-1} = \text{id}_Y, \quad f^{-1} \circ f = \text{id}_X.$$

The order of composition matters: $f \circ g$ and $g \circ f$ are not the same map. It can even happen that $g \circ f$ is defined but $f \circ g$ is not.

Example. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + 1$, and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$. Then

$$(g \circ f)(x) = g(x + 1) = (x + 1)^2 = x^2 + 2x + 1,$$

$$(f \circ g)(x) = f(x^2) = (x^2) + 1 = x^2 + 1.$$

Thus $g \circ f \neq f \circ g$.

Example. The map

$$f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, \quad f(x) = x^2,$$

is bijective.

Lemma. Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$.

- (1) If f and g are injective, then $g \circ f$ is injective.
- (2) If f and g are surjective, then $g \circ f$ is surjective.
- (3) If f and g are bijective, then $g \circ f$ is bijective.

Proof. (1) We need to prove that for all $x_1, x_2 \in X$, if

$$(g \circ f)(x_1) = (g \circ f)(x_2),$$

then $x_1 = x_2$. So let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since g is injective, this implies $f(x_1) = f(x_2)$. Since f is injective, we then get $x_1 = x_2$, as required.

(2) We need to show that for all $z \in Z$, there exists $x \in X$ with $(g \circ f)(x) = z$. So let $z \in Z$. Since g is surjective, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Then

$$(g \circ f)(x) = g(f(x)) = g(y) = z,$$

as desired.

(3) If both f and g are bijective, then they are both injective and surjective. By (1) and (2), it follows that $g \circ f$ is both injective and surjective, hence bijective. $\square$

04.6 Image and Inverse Image

Definition (Image of a Subset). Let $f : X \rightarrow Y$ and $A \subseteq X$. The IMAGE of A under f is

$$f(A) := \{ y \in Y \mid \exists x \in A \text{ s.t. } f(x) = y \}.$$

Equivalently,

$$f(A) = \{ f(x) \mid x \in A \}.$$

Definition (Inverse Image). Let $f : X \rightarrow Y$ and $B \subseteq Y$. The INVERSE IMAGE of B under f is

$$f^{-1}(B) := \{ x \in X \mid f(x) \in B \} \subseteq X.$$

4

Example. For $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$,

$$\begin{aligned} f^{-1}(\{4\}) &= \{-2, 2\}, & f^{-1}(\emptyset) &= \emptyset, \\ f^{-1}(\{1, 9\}) &= \{-3, -1, 1, 3\}, & f^{-1}(\{-1\}) &= \emptyset, \\ f^{-1}(\{0\}) &= \{0\}, & f^{-1}(\{-1, 0\}) &= \{0\}. \end{aligned}$$

Lecture 5: beginning of Linear Algebra

1 Oct 2025

04.7 Graph of a Function

04.8 Bases

Definition. Let $f: X \rightarrow Y$ be a map. The graph of f is the subset $\text{Graph}(f) \subseteq X \times Y$ defined by

$$\text{Graph}(f) := \{(x, y) \in X \times Y \mid y = f(x)\}.$$

05 Fields

Definition (Field). A field is a tuple $(K, +, \cdot, 0, 1)$ consisting of a set K with two maps

$$+: K \times K \rightarrow K, \quad (x, y) \mapsto x + y, \quad \cdot: K \times K \rightarrow K, \quad (x, y) \mapsto x \cdot y,$$

and distinguished elements $0, 1 \in K$, such that the following field axioms hold:

Additive structure

$$\forall x, y, z \in K: \quad x + (y + z) = (x + y) + z \quad (\text{associativity of addition}) \quad (\text{K1})$$

$$\forall x, y \in K: \quad x + y = y + x \quad (\text{commutativity of addition}) \quad (\text{K2})$$

$$\forall x \in K: \quad 0 + x = x \quad (\text{additive identity}) \quad (\text{K3})$$

$$\forall x \in K: \quad \exists (-x) \in K: x + (-x) = 0 \quad (\text{additive inverse}) \quad (\text{K4})$$

Multiplicative structure

$$\forall x, y, z \in K: \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z \quad (\text{associativity of multiplication}) \quad (\text{K5})$$

$$\forall x \in K: \quad 1 \cdot x = x \quad (\text{multiplicative identity}) \quad (\text{K6})$$

$$\forall x \in K \setminus \{0\}: \quad \exists x^{-1} \in K: x \cdot x^{-1} = 1 \quad (\text{multiplicative inverse}) \quad (\text{K7})$$

5

Linking laws and nontriviality

$$\forall x, y, z \in K : \quad x \cdot (y + z) = x \cdot y + x \cdot z, \quad (\text{distributivity}) \quad (\text{K8})$$

$$(y + z) \cdot x = y \cdot x + z \cdot x$$

$$1 \neq 0 \quad (\text{nontriviality}) \quad (\text{K9})$$

$$\forall x, y \in K : \quad x \cdot y = y \cdot x \quad (\text{commutativity of multiplication}) \quad (\text{K10})$$

Example. $\mathbb{Q} = \{a | a = \frac{m}{n}, m, n \in \mathbb{Z}, n \neq 0\}$

$1 = 1/1, \quad 0 = 0/1, \quad +, \times$ are the usual definitions. $\mathbb{R}$ = the real numbers.

Example. $\mathbb{C} = \{z | z = a + ib, a, b \in \mathbb{R}, i^2 = -1\}$.

Example. $\mathbb{F}_2 = \{0, 1\}$ with $1 + 1 = 0$.

Example. $\mathbb{F}_3 = \{0, 1, 2\}$ operations modulo 3.

Example. $\mathbb{Z} \rightarrow$ not a field.

Let $x = 2 \in \mathbb{Z}$. Does there exist $x' \in \mathbb{Z}$ such that $2x' = 1$? NO!

5

06 Linear equations Systems of Linear Equations

We start with the system

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases}$$

06.1 Row reduction

$$\begin{aligned} & \begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \xrightarrow{-2E_1 + E_2 \rightarrow E_2} \begin{cases} x + 3y + 4z = -1 \\ -7y - 7z = 7 \end{cases} \\ & \xrightarrow{-\frac{1}{7}E_2 \rightarrow E_2} \begin{cases} x + 3y + 4z = -1 \\ y + z = -1 \end{cases} \xrightarrow{-3E_2 + E_1 \rightarrow E_1} \begin{cases} x + z = 2 \\ y + z = -1 \end{cases} \end{aligned}$$

06.2 Parametric form of the solution

Set of all solutions (x, y, z) : choose z arbitrarily, $z = t \in \mathbb{R}$, $y = -1 - t$, $x = 2 - t$.

$$\{(x, y, z) \in \mathbb{R}^3 \mid z = t \in \mathbb{R}, y = -1 - t, x = 2 - t\}.$$

Definition (SYSTEMS OF LINEAR EQUATIONS AND MATRIX NOTATION). Fix a field F . A system of m equations in n unknowns over F :

Unknowns: $x_1, x_2, \dots, x_n$. Coefficients $a_{ij} \in F$, right-hand sides $b_i \in F$.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}.$$

a_{ij} — coefficient of x_j in equation i .

$1 \leq i \leq m, \quad 1 \leq j \leq n$.

b_i — right-hand side of equation i .

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right] \quad (\text{augmented}) \ m \times (n+1) \text{ matrix over } F.$$

Sometimes we write

$$A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n} \in M_{m \times n}(F).$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

$$S: \quad A\mathbf{x} = \mathbf{b}.$$

$$L(S) = \{\mathbf{x} \in F^n \mid A\mathbf{x} = \mathbf{b}\} = \left\{ (x_1, \dots, x_n) \in F^n \mid A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{b} \right\}.$$

Matrix–vector product via row sums

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^n a_{1j}x_j \\ \vdots \\ \sum_{j=1}^n a_{ij}x_j \\ \vdots \\ \sum_{j=1}^n a_{mj}x_j \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Component-wise formula

For $1 \leq i \leq m$,

$$(A\mathbf{x})_i = \sum_{j=1}^n a_{ij}x_j \implies A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

5

Example.

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \iff \begin{cases} x_1 + 3x_2 + 4x_3 = -1 \\ 2x_1 - x_2 + x_3 = 5 \end{cases}.$$

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & -1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$

Extended matrix: $[A \mid \mathbf{b}]$

$$\text{Exp. } \begin{cases} 2x_1 - x_2 + 3x_3 + 2x_4 = 0, \\ x_1 + 4x_2 - x_4 = 3, \\ 2x_1 + 6x_2 - x_3 + 5x_4 = 9. \end{cases}$$

06.3 Elementary row operations

- choose $0 \neq c \in F$; multiply row i by c ($c \times R_i \longrightarrow R_i$)
- choose $c \in F$, choose $1 \leq i, j \leq m$

Lecture 6

3 Oct 2025

06.4 Elementary Row Operations and Row Equivalence

- (1) choose $0 \neq c \in F$, multiply eq./row number i by c . ($c \times R_i \longrightarrow R_i$)
- (2) choose $c \in F$, choose $1 \leq i, j \leq m$, $i \neq j$, replace eq./row number j by eq./row number $j + c$ times eq./row number i . ($R_j + c \times R_i \longrightarrow R_j$)
- (3) Interchange eq./row number i and eq./row number j . ($R_i \longleftrightarrow R_j$)

Definition. Let C and C' be $r \times s$ matrices with entries in F . We say that C' is *row equivalent* to C if C' can be obtained from C by a finite sequence of elementary row operations.

$$C = C_0 \longrightarrow C_1 \longrightarrow C_2 \longrightarrow \dots \longrightarrow C_k = C'$$

Theorem. Let $S: A\mathbf{x} = \mathbf{b}$ and $S': A'\mathbf{x} = \mathbf{b}'$ be two systems of m linear equations in n unknowns over F . Suppose that the augmented matrices $[A \mid \mathbf{b}]$ and $[A' \mid \mathbf{b}']$ are row equivalent. Then $L(S') = L(S)$.

Proof. We begin with the following auxiliary statements.

Claim. If S' is obtained from S by one elementary row operation, then $L(S) \subseteq L(S')$.

Proof of claim. Let $(x_1, \dots, x_n) \in L(S)$. We need to show that $(x_1, \dots, x_n) \in L(S')$. Consider the three types of elementary operations:

(1) $c \times R_i \rightarrow R_i$ with $c \neq 0$. In this case, all equations of S' coincide with those of S except equation $\#i$. If

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i,$$

then multiplying both sides by c gives

$$ca_{i1}x_1 + ca_{i2}x_2 + \dots + ca_{in}x_n = cb_i.$$

Thus $(x_1, \dots, x_n)$ satisfies equation $\#i$ of S' as well (since $c \neq 0$).

(2) $R_j + c \times R_i \rightarrow R_j$ with $i \neq j$. Again, all equations coincide except equation $\#j$. From

$$a_{j1}x_1 + \dots + a_{jn}x_n = b_j$$

and $c(a_{i1}x_1 + \dots + a_{in}x_n) = cb_i$, we obtain

$$a_{j1}x_1 + \dots + a_{jn}x_n + c(a_{i1}x_1 + \dots + a_{in}x_n) = b_j + cb_i,$$

so $(x_1, \dots, x_n)$ satisfies equation $\#j$ of S' .

(3) $R_i \leftrightarrow R_j$. Trivially, any solution of S is a solution of S' .

Claim. If S' is obtained from S by one elementary row operation, then S can be obtained from S' by one elementary row operation.

Proof of claim. (1) $(1/c) \times R_i \rightarrow R_i$ transforms S to S' , then $(1/c) \times R_i \rightarrow R_i$ transforms S' back to S .

(2) If $R_j + c \times R_i \rightarrow R_j$ (with $i \neq j$) transforms S to S' , then $R_j - c \times R_i \rightarrow R_j$ transforms S' back to S (row $\#i$ is unchanged).

(3) If $R_i \leftrightarrow R_j$ transforms S to S' , then the same swap transforms S' back to S .

Claim. If S' is obtained from S by one elementary row operation, then $L(S) = L(S')$.

Proof of claim. By Claim 1, $L(S) \subseteq L(S')$. By Claim 2, S can be obtained from S' by one elementary row operation; applying Claim 1 with the roles of S and S' interchanged yields $L(S') \subseteq L(S)$. Hence $L(S) = L(S')$.

We are now in a position to prove the theorem. By assumption, there is a finite sequence of elementary row operations

$$S = S_0 \longrightarrow S_1 \longrightarrow \dots \longrightarrow S_k = S'.$$

By Claim 3, we have

$$L(S) = L(S_0) = L(S_1) = \dots = L(S_{k-1}) = L(S_k) = L(S').$$

□

Definition. A $m \times n$ matrix A is called ROW REDUCED if the following two conditions hold:

1. The first non-zero entry in each non-zero row of A is 1 (the leading entry or pivot).
2. Each column of A which contains the leading non-zero entry of some row has all its other entries equal to 0.

Examples.

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & -3 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is row reduced.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

is not row reduced, because the second column contains a leading 1 (in row 2) and also a nonzero entry (in row 3).

$$\begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \end{pmatrix}.$$

is now reduced.

Theorem. Every $m \times n$ matrix A is row equivalent to a row-reduced matrix.

Proof. Let $A = (a_{ij})$ with $1 \leq i \leq m$, $1 \leq j \leq n$, and entries in F . If all entries in the first row of A are zero, then that row already satisfies the pivot condition. Otherwise, let k be the smallest index with $a_{1k} \neq 0$. Multiply row #1 by $1/a_{1k}$; now the pivot in row #1 equals 1. For each $2 \leq i \leq m$, add $-a_{ik}$ times row #1 to row #i (that is, $R_i + (-a_{ik})R_1 \rightarrow R_i$). As a result, the entries in column k below the pivot become zero. Moreover, entries to the left of column k in each row remain unchanged because the pivot in row #1 is the leftmost nonzero entry.

We now turn to row #2. If every entry in row #2 is zero, we leave the row unchanged. Otherwise, let the pivot be in column k' with $k' \neq k$. Divide row #2 by its pivot so that the leading entry becomes 1, and then eliminate the other entries in column k' (except the pivot itself) by adding suitable multiples of row #2 to the other rows. $\square$

Lecture 7

8 Oct 2025

06.5 Row-Reduced Echelon Form

Definition. An $m \times n$ matrix A is said to be in row-reduced echelon form if the following holds:

1. A is row-reduced.
2. Every row of A that has only 0 entries appears below every row which has a non-zero entry.
3. If rows $1, \dots, r$ are the non-zero rows of A , and if the pivot of row i is in column k_i , $i = 1, \dots, r$, then $k_1 < k_2 < \dots < k_r$.

Theorem. Every $m \times n$ matrix A is row equivalent to a row-reduced echelon matrix.

Proof. Apply the previous theorem and, if necessary, perform permutations of the rows. $\square$

Remark. Useful for solving

$$Ax = b$$

We can replace $[A \mid b]$ by a row-reduced echelon matrix row equivalent to it:

$$[A' \mid b'].$$

06.6 Worked example

$$\left(\begin{array}{cccc|c} 0 & 1 & 3 & 4 & 19 \\ 1 & 4 & 0 & -1 & 5 \\ 6 & -1 & 5 & -5 & -5 \end{array} \right) \xrightarrow{-\frac{1}{3}R_1 \rightarrow R_1} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array} \right)$$

$$R_2 - 4R_1 \rightarrow R_2, \quad R_3 - 6R_1 \rightarrow R_3$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 2 & 0 & 1 & \frac{23}{3} & 1 \end{array} \right) \xrightarrow{R_3 - 2R_2 \rightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 0 & 0 & -\frac{5}{3} & \frac{55}{9} & -17 \end{array} \right)$$

$$\xrightarrow{-\frac{3}{5}R_3 \rightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array} \right)$$

$$\underbrace{R_2 - \frac{4}{3}R_3 \rightarrow R_2, \quad R_1 + \frac{1}{3}R_3 \rightarrow R_1}_{\text{eliminate the 3rd column above the pivot}} \left(\begin{array}{cccc|c} 0 & -\frac{5}{3} & \frac{12}{5} & \frac{17}{3} & -\frac{23}{5} \\ 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array} \right)$$

06.7 Another worked example

07 Vector spaces: Definition of Vector Spaces

Definition (Vector space). A vector space over a field K is a set V (the elements of K are called scalars, the elements of V are called vectors) endowed with two operations

$$+ : V \times V \rightarrow V, \quad (u, v) \mapsto u + v, \quad \cdot : K \times V \rightarrow V, \quad (\lambda, v) \mapsto \lambda v,$$

such that the following axioms hold:

- (V1) For all $v_1, v_2, v_3 \in V$, we have $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$ (associativity of vector addition).
- (V2) There exists an element $0 \in V$ such that for all $v \in V$, $0 + v = v$ (additive identity).
- (V3) For every $v \in V$, there exists $v' \in V$ such that $v + v' = 0$ (additive inverse).
- (V4) For all $v_1, v_2 \in V$, we have $v_1 + v_2 = v_2 + v_1$ (commutativity of vector addition).
- (V5) For all $a, b \in K$ and all $v \in V$, we have $(ab)v = a(bv)$ (associativity of scalar multiplication).
- (V6) For all $v \in V$, $1 \cdot v = v$ (multiplicative identity).
- (V7) For all $a \in K$ and all $v_1, v_2 \in V$, we have $a(v_1 + v_2) = av_1 + av_2$ (distributivity over vector addition).
- (V8) For all $a_1, a_2 \in K$ and all $v \in V$, we have $(a_1 + a_2)v = a_1v + a_2v$ (distributivity over field addition).

Example. Some Examples: Example 1.: The coordinate space K^n . We have

$$n \in \mathbb{N}, \quad K^n = \{ (a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n \}.$$

Equivalently,

$$\underbrace{K \times K \times \dots \times K}_{n \text{ times}} \cong K^n.$$

We turn $V = K^n$ into a vector space by defining, for

$$v = (a_1, \dots, a_n), \quad w = (b_1, \dots, b_n) \in K^n,$$

the operations

$$v + w = (a_1 + b_1, \dots, a_n + b_n) \in K^n, \quad a \in K, \quad a \cdot v := (a a_1, \dots, a a_n) \in K^n,$$

and the zero vector

$$0 := (0, \dots, 0) \in K^n.$$

EXC. Show that K^n with these operations is a vector space.

Lemma. Let V be a vector space over K . Then:

- (a) The zero vector is unique, i.e., if $0' \in V$ also satisfies $\forall v \in V : 0' + v = v$, then $0' = 0$.
- (b) Let $v \in V$. The element $v' \in V$ from (V3) is unique, i.e., if $\tilde{v} \in V$ satisfies $v + \tilde{v} = 0$, then $\tilde{v} = v'$.
- (c) $\forall v \in V : 0 \cdot v = 0$ (the zero vector).
- (d) $\forall v \in V, \forall a \in K : a \cdot 0 = 0$.
- (e) $\forall v \in V : (-1) \cdot v = -v$.
- (f) $\forall v \in V : -(-v) = v$.
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then $a = 0$ or $v = 0$.
- (h) Associativity for $+$ holds also for n elements, i.e., $v_1 + v_2 + \dots + v_n$ makes sense no matter where we put the parentheses; for example,

$$(v_1 + v_2) + ((v_3 + v_4) + v_5) = v_1 + (v_2 + (v_3 + (v_4 + v_5)))$$

or

$$((v_1 + v_2) + v_3) + (v_4 + v_5)$$

Proof. (a) Uniqueness of 0. Suppose $e \in V$ satisfies $\forall v \in V : e + v = v$. We need to prove that $e = 0$. Indeed, $e + 0_V = 0_V$ by assumption. At the same time $0_V + e = e$ by the definition of 0_V . From these two equations we get $e = 0_V$.

(b) Uniqueness of v' . We need to prove that $v = v'$ and $u \in V$ satisfy $v + u = 0$ and $v + v' = 0$. We have to show that $u = v'$. So:

$$u = u + 0 = (v + v') + u = v + (v' + u) = v' + (v + u) = v' + 0 = v'.$$

This proves statement (b).

(c) Let $v \in V$. We have to show that $0 \cdot v = 0$. Indeed,

$$0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v.$$

so we have $0 \cdot v = 0 \times 0 + 0 \times 0$

Add $-0 \times v$ to both sides.

$$\Rightarrow 0 = 0 \times v + 0 \times v - 0 \times v = 0 \times v + (0 \times v - 0 \times v) = 0 \times v + 0 = 0 \times v$$

(d) Let $a \in K, v \in V$ and assume that $a \cdot v = 0$. We need to prove that either $a = 0$ or $v = 0$. Indeed, assume that $a \neq 0$. By the axioms of a field, $\exists$ an element $a^{-1} \in K$ such that $a^{-1}a = 1$. Now: $v = 1 \cdot v = (a^{-1} \cdot a) \cdot v = a^{-1}(a \cdot v) = 0$ by assumption.

$$= a^{-1}(a \cdot v) = 0 \text{ statement d of the lemma.}$$

□

Definition. Let V be a vector space over a field K . A subset $W \subseteq V$ is called a *linear subspace* of V if the following holds:

LSS1: $W \neq \emptyset$

LSS2: $\forall w_1, w_2 \in W$, we have $w_1 + w_2 \in W$.

LSS3: $\forall a \in K, \forall w \in W$ we have $a \cdot w \in W$.

Lemma. Let V be a vector space over K , and let $W \subseteq V$ be a subspace. Then W is a vector space in its own right when endowed with the operations coming from V .

Proof. By (LSS2) and (LSS3) the operations $+$ and $\cdot$ from V indeed define corresponding operations on W . It remains to check the axioms (V1)–(V8).

Exercise: Do that!

We check axiom (V2) for W . Pick any element $u \in W$ (in particular $W \neq \emptyset$). Since $0_V = 0_K \cdot u$ and W is a subspace, we have $0_V \in W$ by (LSS3). Define $0_W := 0_V$.

Clearly, for all $w \in W$ we have $0_W + w = 0_V + w = w$. $\square$

Lemma (Criterion for a subset to be a subspace). Let V be a V.S. over a field K , and $W \subseteq V$ a subset. Then W is a subspace of V if the following holds:

1. $0 \in W$.

2. $\forall a_1, a_2 \in K, \forall w_1, w_2 \in W$ we have $a_1 w_1 + a_2 w_2 \in W$.

Proof. " $W \subseteq V$ is a subspace $\Rightarrow$ (1)+(2)": This follows at once from the fact that W is in itself a vect. sp..

(1)+(2) $\Rightarrow W$ is a subspace:

clearly $W \neq \emptyset$ b.c. $0 \in W$. (LSS1)

(LSS2): Let $w_1, w_2 \in W$. By (2), $1 \cdot w_1 + 1 \cdot w_2 \in W$. So (LSS2) holds.

(LSS3): By (2), if $a \in K, w \in W$, then $a \cdot w + 0 \cdot 0 \in W$. So $a \cdot w \in W$. $\square$

Lecture 8: Linear Subspaces

10 Oct 2025

07.1 Linear Subspaces

1) Let V be a v.s. over K . Then $\{0_v\} \subseteq V$ is a subspace called the 0-subspace of V or the trivial subspace of V .

2) Fix $b \in K$. Consider the subset

$$U_b := \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\} \subseteq K^3.$$

Claim. U_b is a subspace iff $b = 0$.

Proof. $\Rightarrow$: Suppose U_b is a linear subspace. If

$$(x_1, x_2, x_3), (y_1, y_2, y_3) \in U_b,$$

then

$$x_1 + x_2 + x_3 = b, \quad y_1 + y_2 + y_3 = b. \quad (*)$$

Since U_b is a subspace,

$$\begin{aligned} (x_1 + y_1, x_2 + y_2, x_3 + y_3) &\in U_b \\ \Rightarrow (x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) &= b. \end{aligned}$$

But from $(*)$ we get

$$(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = b + b = 2b,$$

so $b = 2b$, hence $b = 0$. $\square$

Exec. 1) Use the fact that $0 \in U_b$ to deduce that $b = 0$. 2) Prove " $\Leftarrow$ ".

Let A be an $m \times n$ matrix with entries in K .

$$(A \in \text{Mat}_{m \times n}(K) \text{ or } A \in M_{m \times n}(K)).$$

Fix $b \in K^m$, $b = (b_1, \dots, b_m)$ viewed as a column vector. Consider the set

$$L := \{x \in K^n \mid Ax = b\} \subseteq K^n.$$

Claim: L is a linear subspace iff $b = 0 \in K^m$. For the proof, we need preparations.

Lemma. $\forall u, v \in K^n \quad \forall a \in K$ we have: 1) $A(u + v) = Au + Av$ 2) $A(au) = a(Au)$

Proof. Write $A = (a_{ij})$, $u = (u_1, \dots, u_n)^T$, $v = (v_1, \dots, v_n)^T$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

$$Au + Av = \begin{pmatrix} a_{11}u_1 + a_{12}u_2 + \dots + a_{1n}u_n \\ \vdots \\ a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n \\ \vdots \\ a_{m1}u_1 + a_{m2}u_2 + \dots + a_{mn}u_n \end{pmatrix} + \begin{pmatrix} a_{11}v_1 + a_{12}v_2 + \dots + a_{1n}v_n \\ \vdots \\ a_{i1}v_1 + a_{i2}v_2 + \dots + a_{in}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \dots + a_{mn}v_n \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}(u_1 + v_1) + a_{12}(u_2 + v_2) + \dots + a_{1n}(u_n + v_n) \\ \vdots \\ a_{i1}(u_1 + v_1) + a_{i2}(u_2 + v_2) + \dots + a_{in}(u_n + v_n) \\ \vdots \\ a_{m1}(u_1 + v_1) + a_{m2}(u_2 + v_2) + \dots + a_{mn}(u_n + v_n) \end{pmatrix} = A(u + v).$$

The scalar multiplication property is shown similarly:

$$A(au) = \begin{pmatrix} a_{11}(au_1) + \dots + a_{1n}(au_n) \\ \vdots \\ a_{m1}(au_1) + \dots + a_{mn}(au_n) \end{pmatrix} = a \begin{pmatrix} a_{11}u_1 + \dots + a_{1n}u_n \\ \vdots \\ a_{m1}u_1 + \dots + a_{mn}u_n \end{pmatrix} = a(Au).$$

□

Proof of claim: $\Leftarrow$: Assume $b = 0$, show that L is a subspace. Indeed:

$$0 \in L \text{ b.c. } A0 = 0 = b.$$

Lecture 9

15.10.2025

Proof. Proof of the claim continued:

$\Leftarrow$: Assume $b = 0$, and we will prove that L is a subspace. Clearly $0 \in L$, because $A0 = 0 = b$. If $x = (x_1, \dots, x_n) \in L$ and $y = (y_1, \dots, y_n) \in L$, then

$$A(x + y) = A \cdot x + A \cdot y = 0 + 0 = 0,$$

hence $x + y \in L$. Let $x \in L$ and $a \in K$. Then

$$A(ax) = a(Ax) = a \cdot 0 = 0,$$

so $ax \in L$. It follows that $L \subseteq K^n$ is a subspace.

$\Rightarrow$: Suppose L is a subspace, and we will show that $b = 0$. Since $L \subseteq K^n$ is a subspace, $0 \in L$. Hence $A \cdot 0 = b$, which implies $b = 0$. □

07.2 More examples of vector spaces and subspaces

Example. Let K be a field.

Define $K^\infty := \{(a_1, a_2, a_3, \dots) \mid a_i \in K \forall i\}$. Addition and scalar multiplication on K^∞ are given by

$$(a_1, a_2, a_3, \dots) + (b_1, b_2, b_3, \dots) = (a_1 + b_1, a_2 + b_2, a_3 + b_3, \dots),$$

and for $a \in K$,

$$a \cdot (a_1, a_2, a_3, \dots) = (a \cdot a_1, a \cdot a_2, a \cdot a_3, \dots).$$

Exercise. K^∞ with $+$ and $\cdot$ as defined here is a vector space.

Example. Take $k = \mathbb{R}$.

$$\text{Fib} := \{(a_1, a_2, a_3, \dots) \mid a_k = a_{k-1} + a_{k-2} \ \forall k \geq 3\}.$$

Exercise. Show that

$$\text{Fib} \subseteq \mathbb{R}^\infty$$

is a linear subspace.

More generally, fix $\alpha, \beta \in \mathbb{R}$.

Then

$$U_{\alpha, \beta} = \{(a_1, a_2, \dots, a_k, \dots) \in \mathbb{R}^\infty \mid a_k = \alpha a_{k-1} + \beta a_{k-2} \ \forall k \geq 3\}.$$

Exercise. Show that $U_{\alpha, \beta} \subseteq \mathbb{R}^\infty$ is a subspace.

07.3 Examples of Vector Spaces

9

07.4 Spaces of Functions

Let K be a field, and let S be a non-empty set.

Define

$$K^S = \{\text{all functions } f : S \rightarrow K\}.$$

Define $+$ and $\cdot$ on K^S :

Let $f, g \in K^S$, $a \in K$.

$$\begin{aligned} (f + g) &: S \rightarrow K, \\ (af) &: S \rightarrow K \end{aligned}$$

Define $(f + g) : S \rightarrow K$ by

$$(f + g)(x) := f(x) + g(x) \quad \forall x \in S$$

(where the addition is in K).

Also define $af : S \rightarrow K$ by

$$(af)(x) := a f(x) \quad \forall x \in S.$$

Exercise. K^S with these operations is a vector space over K .

Example. Another example. Let $S = [0, 1] := \{x \in \mathbb{R} | 0 \leq x \leq 1\}$. Define $C(S) := \{f \in \mathbb{R}^S | f \text{ is continuous in } [0, 1]\}$

Claim. $C(S) \subseteq \mathbb{R}^S$ is a linear subspace. (exercise)

07.5 Polynomials

Let K be a field. Let x be a formal variable. A polynomial over K (or a polynomial with coefficients in K) in the variable x is an expression of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

where $n \in \mathbb{Z}_{\geq 0}, a_i \in K \forall 0 \leq i \leq n$.

The zero polynomial is the polynomial with all coefficients equal to zero.

$$f(x) = 0 \text{ i.e. } a_0 = 0, a_1 = 0, a_2 = 0, \dots, a_r = 0$$

Remark.

$$K^3 \leftrightarrow K^{\{1,2,3\}} \longrightarrow a : \{1, 2, 3\} \rightarrow K \longrightarrow (a(1), a(2), a(3)) = (a_1, a_2, a_3) \in K^3.$$

9 Back to polynomials. We allow to omit terms like $0 \cdot x^i$.

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_r x^r, \quad g(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots + b_m x^m.$$

If $r < m$, we set $a_{n+1} = a_{n+2} = \dots = a_m = 0$.

If $m < r$, we set $b_{m+1} = b_{m+2} = \dots = b_r = 0$.

Put $d = \max\{r, m\}$ and now we can write:

$$\begin{aligned} f(x) &= a_0 + a_1 x + a_2 x^2 + \dots + a_r x^r, \\ g(x) &= b_0 + b_1 x + b_2 x^2 + \dots + b_r x^r. \end{aligned}$$

Define

$$f(x) + g(x) := (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots + (a_r + b_r)x^r, .$$

If $c \in K, f(x)$ a polynomial:

$$c \cdot f(x) := (c \cdot a_0) + (c \cdot a_1)x + (c \cdot a_2)x^2 + \dots + (c \cdot a_r)x^r.$$

Notation: $K[x]$ is the set of all polynomials with coefficients in K . Claim. $K[x]$ together with the operations $+$ and $\cdot$ defined above is a vector space over K . (exercise)

Every $f(x) \in K[x]$ defines also a function $f : K \rightarrow K$. The subset of K^K which comes from polynomials is a linear subspace of K^K isomorphic to $K[x]$. (exercise)

But, from $\tilde{f}$ we cannot always recover f .

Example.

$$K = \mathbb{F}_2 = \{0, 1\}, \quad f(x) = x^2 + x + 1, \quad \tilde{f}(0) = 1, \quad \tilde{f}(1) = 1.$$

$$\begin{aligned} f(x) &= x \\ g(x) &= x^2 f(x) \neq g(x) \text{ or polynomials} \end{aligned}$$

but they give rise to the same functions $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ This is b.c. $\forall x \in \mathbb{F}_2, x^2 = x$.

Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be a polynomial with $a_n \neq 0$. We say that the degree of f is n and write $\deg(f) := n \in \mathbb{Z}_{\geq 0}$.

If $f(x) = 0$, i.e. $f(x)$ is the zero polynomial, we define $\deg(f(x)) := -\infty$.
(So $\deg f(x) = -\infty \iff f(x)$ is the 0-polynomial)

$$\deg f(x) := \max \{k \in \mathbb{Z}_{\geq 0} \mid a_k \neq 0\}$$

Fix $d \in \mathbb{Z}_{\geq 0} \cup \{-\infty\}$.

Define

$$K[x]_{\leq d} := \{f \in K[x] \mid \deg(f) \leq d\}$$

(We agree that $-\infty \leq k$ for all $k \in \mathbb{Z}$.)

Claim. $K[x]_{\leq d} \subseteq K[x]$ is a linear subspace (exercise)

Q. Is $\{f \in K[x] \mid \deg f = d\}$ a subspace of $K[x]$?

Q. Consider $\{p \in K[x]_5 \mid p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5, \\ a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 0\}$.

Is this subspace of $K[x]$?

07.6 Spaces of Matrices

Consider $M_{m \times n}(K)$, the set of all $m \times n$ matrices with entries in K . Define $+$ and $\cdot$:
 $A = (a_{ij}), B = (b_{ij}) \in M_{m \times n}(K)$.

$$A + B = (a_{ij} + b_{ij}), \quad \alpha A = (\alpha a_{ij}).$$

Claim. With these operations, $M_{m \times n}(K)$ is a vector space over K . (exercise)

$$M_{3 \times 2}(K) \longleftrightarrow K^6,$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \mapsto (a_{11}, a_{12}, a_{21}, a_{22}, a_{31}, a_{32}).$$

$$M_{m \times n}(K) \cong K^{mn},$$

$$A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n} \mapsto (a_{11}, a_{12}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn}).$$

9

Remark. The map above is a linear isomorphism (vector-space bijection) if K is a field.

Let V be a vector space. Let $S \subseteq V$ be a subspace Question: What is the smallest linear subspace of V that contains S' ?ad

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Chapter 4

Vector spaces and bases

Lecture 10: Span of a set

16.10.2025

08.a Span: Span of a Set

Continuation in the next lecture.

Lecture 11

16.10.2025

08.b Span: Span of a Set (Continued)

Sol. Take the col. vectors $v_1 \dots v_n$ and make a $m \times n$ matrix out of them:

$$A = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{pmatrix}.$$

If

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in K^n, \text{ then.}$$

$$A \cdot x = x \cdot v_1 + x_2 \cdot v_2 + \dots + x_n \cdot v_n = \sum_{i=1}^n x_i \cdot v_i.$$

Given $w \in K^n$, the question is whether or not the system of equations

$$Ax = w$$

has a solution.

If a solution exists, then $w \in \text{Sp}(v_1, \dots, v_n)$.

If a solution does **not** exist, then $w \notin \text{Sp}(v_1, \dots, v_n)$.

In the first case, any solution x will give a choice of coefficients $x_1, \dots, x_n$ such that $w = \sum_{i=1}^n x_i \cdot v_i$.

$$(A|w) = \left(\begin{array}{c|c|c|c|c} | & | & & | & | \\ v_1 & v_2 & \dots & v_n & w \\ | & | & & | & | \end{array} \right)_{x \neq 0}.$$

Examples of spaces and how to span them.

1. $V = K^n$. $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0) \dots e_n = (0, 0, \dots, 1)$ Then $\text{Sp}(e_1, e_2, \dots, e_n) = K^n$.
2. $W = K[x]_d$ (polynomials in x of degree $\leq d$ with coeff. in K). $W = \text{Sp}(1, x, x^2, \dots, x^d)$. (exercise)

2') $V = K[x]$ (all polynomials in x). $V = \text{Sp}(1, x, x^2, x^3, \dots)$ infinite set.

3. $V = M_{m \times n}(K)$. For every $1 \leq k \leq m, 1 \leq l \leq n$ let E_{kl} be the matrix which has all entries $= 0$, except entry in row k and column l which is $= 1$.

$$k \rightarrow \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} = E_{k,\ell} \quad a_{ij} = \begin{cases} 1, & \text{if } i = k \text{ and } j = \ell, \\ 0 & \text{otherwise.} \end{cases}$$

$\uparrow \ell$ $E_{k,\ell} = (a_{ij})$

Then $M = (\text{Sp}(\{E_{kl} \mid 1 \leq k \leq m, 1 \leq l \leq n\}))$. e.g.

$$M_{2 \times 2}(K) = \text{Sp} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Remark. $K^n, n \in \mathbb{Z}_{\geq 0}, M_{m \times n}, K[x]_d$ are all finite-dimensional vector spaces. Exc: Show that $K[x]$ is not finite-dimensional.

09 Linear independence: Linear Independence

09.1 Definition

Let V be a vector space over a field K .

Definition. Let $v_1, v_2, \dots, v_n \in V$ be a list of n vectors in V . We say that the vectors $v_1, v_2, \dots, v_n$ are *linearly independent* if the *only* linear combination of $v_1, v_2, \dots, v_n$ that equals $0 \in V$ is the one with 0 coefficients.

Alternatively, $v_1, v_2, \dots, v_n$ are linearly independent if $\forall a_1, a_2, \dots, a_n \in K$ for which $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ we must have that $a_1 = a_2 = \dots = a_n = 0$. Lets agree that the empty list is linearly independent.

If the vectors $v_1, v_2, \dots, v_n$ are *not* linearly independent, we say that it is *linearly dependent*.

Remark. 1) 0 can always be written as a linear combination of any list of vectors.
 $0 = v_1 \cdot 0 + v_2 \cdot 0 + \dots + v_n \cdot 0$.

We call this the trivial linear combination.

Remark. 2) Suppose that in our list $v_1, v_2, \dots, v_n$ there is a repetition, i.e. $v_i = v_\ell$ for $1 \leq k < \ell \leq n$. Then $v_1, \dots, v_n$ is *not* linearly independent $v_k + (-1) \cdot v_\ell = 0$

Remark. 3) The order of the elements in the list $v_1, \dots, v_n$ does not matter for linear independence.

Remark. 4) If 0 is one of the elements in the list $v_1, \dots, v_n$, then this list cannot be linearly independent. $1 \cdot 0 + 0 \cdot v_2 + \dots + 0 \cdot v_n = 0$

09.2 Generalization to families

Definition. Let $F = \{v_i\}_{i \in I}$ be a family of vectors in a vector space V over a field K . We say that F is linearly independent if $\forall n \in \mathbb{Z}_{\geq 1}$ and any sequence of *distinct* indices $i_1, i_2, \dots, i_n \in I$, the list of vectors $v_{i_1}, v_{i_2}, \dots, v_{i_n}$ is linearly independent.

Definition. continued Let $\emptyset \neq S \subseteq V$ be a subset. We say that S is lin. independent if every finite list of distinct vectors $v_1, \dots, v_n \in S$ is linearly independent ($n \in \mathbb{Z}_{\geq 1}$).

Lemma. If $S \subseteq V$ is linearly independent, then every subset $S' \subseteq S$ is also linearly independent.

Example. 1) $e_1, \dots, e_n \in K^n$ is linearly independent.

Proof. Suppose that $a_1 e_1 + a_2 e_2 + \dots + a_n e_n = 0$.

$$a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

□

Example. 2 The set $e_1, e_2, \dots, e_n, \dots$ in K^∞ .

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \quad \dots e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \end{pmatrix} \quad \dots$$

is linearly independent. (exercise)

Example. 3 The set $\{1, x, x^2, x^3, \dots, x^d\} \subseteq K[x]_d$ is linearly independent. $\{1, x, x^2, x^3, \dots, x^d\} \subseteq K[x]$ is linearly independent. (exercise)

Example. 4 Let V be a vector space. Let $v \in V$. Show that $\{v\}$ is linearly independent iff $v \neq 0$.

10.a Basis

Definition. A subset $S \subseteq V$ is called a basis of V if the following holds:

- 1) S is linearly independent.
- 2) S spans V . ($\text{Sp}(S) = V$).

Lecture 12

26.10.2025

Proposition. A subset $S \subseteq V$ is a basis of V iff every $v \in V$ can be written in a unique way as a linear combination of vectors from S (up to order of summation).
 Meaning of "in a unique way..." The meaning is: suppose $v \in V$ is written as
 $v = \text{linear combination \#1 (with vectors from } S)$
 $v = \text{linear combination \#2 (with vectors from } S)$

Define

$$C_1 = \left\{ \text{all the vectors that appear in lin. combin. \#1 with non-zero coeff.} \right\}$$

$$C_2 = \left\{ \text{same as above but for lin. combin. \#2} \right\},$$

then $C_1 = C_2$ and $\forall u \in C_1 = C_2$ the coeff. of u in lin. combin. \#1 is the same as

the coeff. of u in lin. combin. #2.

In case S is a finite set, say $S = \{v_1, v_2, \dots, v_n\}$. The uniqueness means that if $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$ and $v = b_1v_1 + b_2v_2 + \dots + b_nv_n$, with $a_i, b_i \in K$, then $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$.

Proof. For simplicity, assume $S = \text{finite}$. ($S = \text{infinite} \leftarrow \text{exc.}$)

$\Leftarrow$: Assume every $v \in V$ can be written in a unique way as a linear combination of vectors from S .

$\Rightarrow sp(S) = V$.

We will show now that S is linearly independent. Write $S = \{v_1, v_2, \dots, v_n\}$. Consider the $0 \in V$. Clearly $0 = 0 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n$. But, by assumption there is a unique linear combination of $v_1, \dots, v_n$ that gives 0. It follows that if $a_1, a_2, \dots, a_n = 0$ then $a_1 = 0, \dots, a_n = 0$. $\Rightarrow S$ is linearly independent.

$\Rightarrow$: Assume that S is a basis.

$sp(S) = V \Rightarrow$ every $v \in V$ can be written as a linear combination of S .

It remains to show uniqueness. Assume by contradiction that $\exists v \in V$ that can be written as $v = a_1v_1 + \dots + a_nv_n$ and $v = b_1v_1 + \dots + b_nv_n$ and $a_k \neq b_k$ for some $1 \leq k \leq n$. $\Rightarrow 0 = (a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \dots + (a_k - b_k)v_k + \dots + (a_n - b_n)v_n$. $\Rightarrow 0$ can be written as a non-trivial linear combination of $v_1, \dots, v_n$.

$\Rightarrow S$ is NOT linearly independent. Contradiction to the assumption that S is a basis. $\square$

12

Exps. $\mathbb{K}^n$, $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, \dots, 0, 1)$.

$S = \{e_1, \dots, e_n\}$ is a basis for $\mathbb{K}^n$ (a.k.a. the standard basis).

1) $V = K[x]_{\leq d}$, $S = \{1, x, x^2, \dots, x^d\}$ is a basis.

2') $V = K[x]$, $S = \{1, x, x^2, \dots, x^n, \dots\}$ is a basis.

$$3) \quad V = M_{m \times n}(K), \quad E_{ij} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}$$

where the 1 is in row i and column j .

Then

$$S = \{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

is a basis for $M_{m \times n}(K)$.

Lemma (A). Let $v_1, \dots, v_m \in V$ be a linearly dependent list of vectors. Then there exists an index j with $1 \leq j \leq m$ such that:

- 1) $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$
- 2) $\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \text{Sp}(v_1, \dots, v_m)$

Proof. Since the list is linearly dependent, $\exists a_1, \dots, a_m \in K$, not all of them 0, s.t.

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0.$$

Define $j := \max\{i \mid a_i \neq 0\}$. (i.e. $a_j \neq 0$ and $a_{j+1} = a_{j+2} = \dots = a_m = 0$).

$$\Rightarrow a_1 v_1 + a_2 v_2 + \dots + a_j v_j = 0.$$

$$\Rightarrow v_j = -\frac{a_1}{a_j} v_1 - \frac{a_2}{a_j} v_2 - \dots - \frac{a_{j-1}}{a_j} v_{j-1}.$$

This is possible because $a_j \neq 0$. To prove 2): let $v \in \text{Sp}(v_1, \dots, v_m)$.

$$\Rightarrow v = c_1 v_1 + c_2 v_2 + \dots + c_j v_j + \dots + c_m v_m.$$

for some $c_1, \dots, c_m \in K$. Substitute (*) in to (v) and we obtain that v is a linear combination of $v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m$.

$$\Rightarrow v \in \text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m).$$

$$\Rightarrow \text{Sp}(v_1, \dots, v_m) \subseteq \text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m).$$

But clearly we also have

$$\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) \subseteq \text{Sp}(v_1, \dots, v_m).$$

$$\Rightarrow (2) \text{ holds.}$$

□

Lemma (B). Under the same assumptions as in Lemma A, assume in addition that the first k items $v_1, \dots, v_k$ in the list $v_1, \dots, v_n$ are linearly independent, where $1 \leq k < m$. Then the index j from Lemma A satisfies $k < j$.

Proof. Assume by contradiction that $j \leq k$. We have:

$$v_j = a_1 v_1 + \dots + a_{j-1} v_{j-1}$$

for some $a_1, \dots, a_{j-1} \in K$.

$$\Rightarrow 0 = a_1 v_1 + \dots + a_{j-1} v_{j-1} - 1 \cdot v_j.$$

This is a non-trivial linear combination of $v_1, \dots, v_j$ that gives 0. But $v_1, \dots, v_k$ are linearly independent and $j \leq k$. $\Rightarrow v_1, \dots, v_j$ are linearly independent. Contradiction. □

Lemma (C). Let $w_1, \dots, w_n \in V$ such that $\text{Sp}(w_1, \dots, w_n) = V$, and let $v \in V$. Then the list $v, w_1, \dots, w_n$ is linearly dependent.

Proof. Write $v = a_1 w_1 + a_2 w_2 + \dots + a_n w_n$ for some $a_1, \dots, a_n \in K$.

$$\Rightarrow (-1) \cdot v + a_1 w_1 + a_2 w_2 + \dots + a_n w_n = 0.$$

This is a non-trivial linear combination that gives 0, proving dependence. $\square$

Lemma (D). Let $v_1, \dots, v_n \in V$ s.t. $\text{Sp}(v_1, \dots, v_n) = V$. Let $v_1, \dots, v_m$ be a list of linearly independent vectors in V . Then $m \leq n$.

Proof. The proof will have m steps.

Step $\ell = 1$: We will replace u_1 by one of the vectors $v_1, \dots, v_n$. How? Consider $u_1, v_1, \dots, v_n$. By Lemma C, this list is linearly dependent. By Lemma A one of the vectors in this list is in the span of the vectors that appear before it in the list and if we drop this vector from the list the overall span of the list is still V . By Lemma B that vector that we can drop cannot be u_1 (because u_1 standalone is linearly independent.) We now drop that vector and we now get a list of n vectors that still spans V and the first vector in this list is v_1 .

Step $2 \leq \ell \leq m$:

From step $\ell - 1$ we should have a list of n vectors that spans V and the first $\ell - 1$ vectors in this list are $u_1, \dots, u_{\ell-1}$ and the remaining $n - (\ell - 1)$ vectors are taken from the list $v_1, \dots, v_n$. Let's write this new list of n vectors as $u_1, \dots, u_{\ell-1}, w_1, \dots, w_{n-(\ell-1)}$. Consider now the list $u_1, \dots, u_{\ell-1}, u_\ell, w_1, \dots, w_{n-(\ell-1)}$.

This list $u_1, \dots, u_{\ell-1}, u_\ell$ spans V , and by Lemma C the larger list $u_1, \dots, u_{\ell-1}, u_\ell, w_1, \dots, w_{n-(\ell-1)}$ must be linearly dependent. By Lemma A, we can omit one of the vectors in this list and still span V and by Lemma B, the vector that we omit can be taken from the $w_1, \dots, w_{n-(\ell-1)}$ and not from the u 's. Moreover the vector we omit must be in the span of those lying before it. The new list (for step ℓ) is:

$$u_1, \dots, u_\ell, \boxed{n - \ell \text{ vectors all taken from } v_1, \dots, v_n}. \leftarrow \text{list of length } n$$

Assume now by contradiction that $m > n$. Then we can perform step $\ell = 1, 2, \dots, n$ and obtain a list $\hat{t}$ that looks like $u_1, u_2, \dots, u_n$ and this list spans V . Since $m > n$, we still have the vectors $u_{n+1}, \dots, u_m$. Consider $u_1, u_2, \dots, u_n, u_{n+1}$. By Lemma C this list is linearly dependent. Contradiction! (because we assumed $u_1, \dots, u_m$ is linearly independent). $\square$

Lecture 13

29.10.2025

11.b Basis: Dimension

In fact we proved a bit more than Lemma D, namely: we can remove m vectors from the list $v_1, \dots, v_n$, s.t. the remaining $n - m$ vectors $v_{i_1}, v_{i_2}, \dots, v_{i_{n-m}}$ ($1 \leq i_1 < i_2 < \dots < i_{n-m} \leq n$) when put together with the list $u_1, \dots, u_m$ spans together V .

In particular (means has a special case a less general case), this shows if $m = n$ (i.e. the u -list and the v -list have the same size) then $\text{Sp}(u_1, \dots, u_n) = V$.

Theorem. Let V be a finite-dimensional vector space over a field K . Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

For the proof, we need: *Lemma E*

Lemma (E). Let $w_1, \dots, w_\ell \in V$ and assume $\forall 1 \leq i \leq \ell, w_i \notin \text{Sp}(w_1, \dots, w_{i-1})$. Then $w_1, \dots, w_\ell$ are linearly independent.

Proof. If $w_1, \dots, w_\ell$ were linearly dependent then by Lemma A, there would exist an index j with $1 \leq j \leq \ell$ such that $w_j \in \text{Sp}(w_1, \dots, w_{j-1})$, contradicting our assumption. $\square$

Lemma (F). Assume $\text{Sp}(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $v_1, \dots, v_n$ which is a basis of V .

Proof. We will have n steps, and in each step we will consider one of the vectors from the list $v_1, \dots, v_n$, and decide whether or not to drop it from the list.

Step 1: if $v_1 = 0$, we drop it.

If $v_1 \neq 0$, we keep it.

Step $2 \leq j \leq n$: If $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$, we drop it, otherwise we keep it.

After performing n steps as above, we get a possibly shorter list $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ with $1 \leq i_1 < i_2 < \dots < i_m \leq n$. We claim that $\text{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_m}) = V$. Indeed, in any of the steps $1 \leq j \leq n$ we dropped the vector v_j only if $v_j \in \text{Sp}(v_1, \dots, v_{j-1}) \Rightarrow \text{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_m}) = V$.

We now claim that $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ is linearly independent. Indeed, $\forall 1 \leq k \leq m$, we have

$$v_{i_k} \notin \text{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_{k-1}}). \text{ In particular}$$

$$v_{i_k} \notin \text{Sp}(v_1, v_2, \dots, v_{i_k-1}).$$

By Lemma E, applied to the list $v_{i_1}, v_{i_2}, \dots, v_{i_m}$, we deduce that $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ are linearly independent. $\Rightarrow v_{i_1}, v_{i_2}, \dots, v_{i_m}$ both spans V and is linearly independent. $\square$

Proof of the Theorem. [Proof of the Theorem] We need to prove:

1. V has a basis with finitely many elements.
2. Every basis of V has finitely many elements.
3. For every two bases A, B of V , we have $|A| = |B|$.

1.: Take any finite set $S \subseteq V$ s.t. $\text{Sp}(S) = V$ (possible because V is finite-dimensional). By Lemma F, $\exists S' \subseteq S$ which is a basis of V .

2.: Let $\mathcal{C}$ be any basis for V . We will show that $\mathcal{C}$ is a finite set.

Suppose by contradiction that $\mathcal{C}$ is not finite. Also, take any finite basis, say S , for V (exists by 1.), say

$$S = \{v_1, v_2, \dots, v_n\}.$$

Choose $n + 2025$ vectors $u_1, u_2, \dots, u_{n+2025} \in \mathcal{C}$. $\text{Sp}(v_1, v_2, \dots, v_n) = V$. and $u_1, u_2, \dots, u_{n+2025}$ are linearly independent. By Lemma D, we have that $n + 2025 \leq n$. Contradiction.

3.: By statement 2., A and B are both finite. Now $\text{Sp}(B) = V$ and A is linearly independent. By Lemma D, $|A| \leq |B|$. We also know that $\text{Sp}(A) = V$ and B is linearly independent. By Lemma D again, $|B| \leq |A|$. Thus, $|A| = |B|$. (applause) $\square$

13

Note. From every good theorem we get a definition:

12 Dimension

Definition. Let V be a finite-dimensional vector space over a field K . We define the dimension of V to be the number $n \in \mathbb{Z}_{\geq 0}$ of elements in (any!) a basis of V . We write $\dim(V) = n$.

$$\dim_K(V) = n.$$

(*) $\dim V$ is well defined. !

$$\dim\{0\} = 0 \quad \text{because a basis for } \{0\} \text{ is the empty set } \emptyset,$$

$$\text{and } |\emptyset| = 0.$$

Example (Examples). 1) $\dim K^n = n$ because the standard basis has n elements.
 2) $\dim K[x]_d = d + 1$
 3) $\dim M_{m \times n}(K) = m \cdot n$ 4) Let S be a set. $K^S = \{f : S \rightarrow K \text{ is a function}\}$. This is a vector space over K .

EXERCISE: If S is a finite set, say $S = \{s_1, s_2, \dots, s_n\}$. Then $\dim(K^S) = |S|$. Find a basis.

13

Summary: Let V be a finite vector space. Then

- 1) Every finite list of vectors that spans V contains a sublist which is a basis for V .
- 2) Every linearly independent list of vectors $u_1, \dots, u_m$ can be extended to a basis of V .
 - 1) $v_1, \dots, v_n \supset v_{i_1}, \dots, v_{i_k}$ basis.
 - 2) $u_1, \dots, u_m \supset u_1, \dots, u_m, w_1, \dots, w_{d-m}$ basis.

Lecture 14

31.10.2025

Theorem (1). Let V be a finite-dimensional vector space of dimension $n := \dim V$. The following statements are equivalent:

1. $v_1, \dots, v_n$ are linearly independent.
2. $v_1, \dots, v_n$ spans V .
3. $v_1, \dots, v_n$ form a basis for V .

Proof. $1 \Rightarrow 3$. By what was done last time, we can extend the list $v_1, \dots, v_n$ and get a basis for V . But this list already has n ($= \dim V$) vectors, $\Rightarrow$ there is no extension possible, hence $v_1, \dots, v_n$ is already a basis for V .

$3 \Rightarrow 1 \& 2$: By definition of "basis".

$2 \Rightarrow 3$: By what we did last time we can possibly drop some of the vectors from the list $v_1, \dots, v_n$ and get a basis. But the length of the list $v_1, \dots, v_n$ is already n ($= \dim V$), so dropping is impossible. $\Rightarrow v_1, \dots, v_n$ was a basis from the beginning. $\square$

14

Theorem (2). Suppose V is finite-dimensional and $n := \dim V$. Let $v_1, \dots, v_k \in V$ be a list. Then

1. If $k < n$, then $v_1, \dots, v_k$ do *not* span V .
2. If $k > n$, then $v_1, \dots, v_k$ are *cannot* be linearly independent.

Proof. 1) Assume $k < n$. Suppose by contradiction that $\text{Sp}(v_1, \dots, v_k) = V$. By point 1 on the very left backboard :) we can thin this list to a sublist $v_{i_1}, \dots, v_{i_k}$ which is a basis for V . But a basis for V has n elements, so $k = n$. ($\Rightarrow d \leq k < n$). Contradiction.

2) Assume $k > n$. Suppose by contradiction that $v_1, \dots, v_k$ are linearly independent. By point 2 on the famous very left backboard ;) we can extend this list to a basis of V . ($\Rightarrow \dim V \geq k > n$). Quod est absurdum. $\square$

12.1 Dimension of subspaces

Proposition. Let V be a finite-dimensional vector space. Then every subspace $U \subseteq V$ is also finite-dimensional and we have $\dim(U) \leq \dim(V)$. Moreover we have equality iff $U = V$.

Note (Lemma E). Let $w_1, \dots, w_\ell \in V$ and assume that $\forall 1 \leq j \leq \ell, w_j \notin \text{Sp}(w_1, \dots, w_{j-1})$. Then $w_1, \dots, w_\ell$ are linearly independent. This was proved last time.

Proof of Proposition. [Proof of Proposition]

Put $n := \dim(V)$. We will show first that $\mathcal{U}$ has a finite basis. If $\mathcal{U} = \{0\}$, then $\emptyset$ is a basis for $\mathcal{U}$. So assume $\mathcal{U} \neq \{0\}$. Take $u_1 \in \mathcal{U}, u_1 \neq 0$. If $\mathcal{U} = \text{Sp}(u_1)$ the u_1 is a basis for $\mathcal{U}$ and we are done. If $\mathcal{U} \neq \text{Sp}(u_1)$, choose $u_2 \in \mathcal{U} \setminus \text{Sp}(u_1)$. ($\subsetneq$) By Lemma E, u_1, u_2 are linearly independent. We continue this process and after j steps we obtain a list $u_1, u_2, \dots, u_j \in \mathcal{U}$ that is linearly independent. If $\text{Sp}(u_1, \dots, u_j) = \mathcal{U}$ we are done. Otherwise, choose $u_{j+1} \in \mathcal{U} \setminus \text{Sp}(u_1, \dots, u_j)$. By Lemma E, $u_1, \dots, u_j, u_{j+1}$ are linearly independent.

However, in V a list of linearly independent vectors can have at most length $n := \dim(V)$. $\Rightarrow$ the procedure we are describing must end (i.e. $\text{Sp}(u_1, \dots, u_j) = \mathcal{U}$) after not more than n steps. Say that $k \leq n$ steps. So we obtain a list $u_1, \dots, u_k$ ($k \leq n$) in $\mathcal{U}$ which is linearly independent and spans $\mathcal{U}$. In other words, $u_1, \dots, u_k$ is a basis for $\mathcal{U}$.

$\Rightarrow \dim(\mathcal{U}) = k \leq n = \dim(V)$.

Equality ($\dim(\mathcal{U}) = \dim(V)$) holds iff $\mathcal{U} = V$: obviously if $\mathcal{U} = V$ then $\dim \mathcal{U} = \dim V$.

Conversely, suppose $k = n$. $\Rightarrow$ the list of vectors $u_1, \dots, u_k$ is a basis for $\mathcal{U}$, hence these are linearly independent vectors and they continue to be linearly independent in V . By Theorem 4.1 (applied to $u_1, \dots, u_k \in V$), $u_1, \dots, u_n$ is a basis for V . $\Rightarrow \mathcal{U} = \text{Sp}(u_1, \dots, u_n) = V$. $\square$

Example. Consider K^m . Let $v_1, \dots, v_n \in K^m$.

Q1. Let $w \in K^m$. How to determine whether or not $w \in \text{Sp}(v_1, \dots, v_n)$?

Q2. Describe $\text{Sp}(v_1, \dots, v_n)$.

Q3. How to determine if $v_1, \dots, v_n$ are linearly independent?

A1. $w \in \text{Sp}(v_1, \dots, v_n) \Leftrightarrow \exists x_1, \dots, x_n \in K$ s.t. $w = x_1 v_1 + \dots + x_n v_n$.

$$\Leftrightarrow \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = w$$

So the question is whether or not the linear system with unknowns $x_1, \dots, x_n$ has a solution.

A2. Apply A1 to a general w . Perform elimination on $(A \mid w)$ until we obtain a reduced row echelon matrix $(A' \mid w')$. The equation $A'x = w'$ has a solution if and only if $Ax = w$ has a solution.

If A' has no zero rows (i.e. each row of A' contains a pivot), then $A'x = w'$ always has a solution. Hence

$$\text{Sp}(v_1, \dots, v_n) = K^m.$$

If A' has some zero rows, say the last $m - r$ rows are zero, then:

Does $A'x = w'$ have a solution?

This system has a solution iff the last entries satisfy

$$w'_{r+1} = \dots = w'_m = 0.$$

Conclusion:

$$\text{Sp}(v_1, \dots, v_n) = \{ b \in K^m \mid b_{r+1} = \dots = b'_m = 0 \}.$$

14

Example.

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \in \mathbb{R}^3. \quad \text{Find } \text{Sp}(v_1, v_2).$$

$$(A|b) = \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 1 & 0 & b_2 \\ 1 & -1 & b_3 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} 1 & 0 & b_2 \\ 0 & 1 & b_1 - b_2 \\ 0 & -2 & b_3 - b_1 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_1 - b_2 \\ 0 & 0 & b_1 - 2b_2 + b_3 \end{array} \right).$$

$$\Rightarrow \text{Sp}(v_1, v_2) = \left\{ \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3 \mid b_1 - 2b_2 + b_3 = 0 \right\}.$$

Example (A3). $v_1, \dots, v_n \in K^m$ are linearly independent iff the system of equations

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

14

has only the trivial solution $\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$. Take

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix}.$$

Do elimination to get A' . Then $v_1, \dots, v_n$ are linearly independent iff $A'x = 0$ has only the trivial solution. This system of equations has only the trivial solution $\Leftrightarrow$ the reduced row echelon matrix A' has precisely n pivots

$$\text{i.e. } A' = \begin{pmatrix} 1 & 0 & \dots & 0 & * & * & \dots & * \\ 0 & 1 & \dots & 0 & * & * & \dots & * \\ \vdots & \vdots & \text{ots} & \vdots & \vdots & \vdots & \text{ots} & \vdots \\ 0 & 0 & \dots & 1 & * & * & \dots & * \end{pmatrix} \quad (\Leftrightarrow \text{ no free variables}).$$

Lecture 15

05.11.2025

13 Row and col spaces

Theorem (2 revisited). Take a list $v_1, \dots, v_k \in K^m$ where $n < m$.

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{pmatrix} \quad \text{we can have } \leq n \text{ pivots but } m > n.$$

When trying to solve $A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$ we'll get

$$\left(\begin{array}{ccc|c} 1 & \dots & 0 & b' \\ & \ddots & \vdots & \vdots \\ 0 & \dots & 1 & \\ \hline 0 & \dots & 0 & b'' \\ \vdots & & \vdots & \vdots \\ 0 & \dots & 0 & \end{array} \right)$$

after elimination. Each entry in b' and in b'' is of the type $\alpha_1 b_1 + \dots + \alpha_m b_m$ with $\alpha_i \in K$.

$$\text{Sp}(v_1, \dots, v_n) = \{b \in K^m \mid b'' = 0\}'.$$

2) Assume now $v_1, \dots, v_k \in K^m$ and $n > m$. ($m = \dim K^m$). Why is $v_1, \dots, v_k$ linearly

15

dependent? Well, we need to check if the system

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

has non-trivial solutions. After elimination we get

$$A' = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

and A' can have at most m pivots (b.c. every pivot is in a different row). But # of variables $< n > m \Rightarrow$ we must have a free variable $\Rightarrow \exists$ non-trivial solutions.

13.1 Row and Column Spaces

Definition. Let $A \in M_{m \times n}(K)$, and denote by $u_1, \dots, u_m \in K^n$ the rows of A , and by $v_1, \dots, v_n \in K^m$ the columns of A .

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v & \dots & v_n \\ | & & | \end{pmatrix}.$$

Define two spaces:

- $\text{RowS}(A) := \text{Sp}(u_1, \dots, u_m) \subseteq K^n$
- $\text{ColS}(A) := \text{Sp}(v_1, \dots, v_n) \subseteq K^m$ Column Space of A

Lemma. If $A, B \in M_{m \times n}(K)$ are row equivalent, then $\text{RowS}(A) = \text{RowS}(B)$.

Proof. Let $M \in M_{m \times m}(K)$.

1. If we exchange two rows of M , the span of the rows does not change. So $\text{RowS}(A)$ does not change under this operation.
2. The operation $\lambda R_i \rightarrow R_i$, $\lambda \in K \setminus \{0\}$. Here too, the span of the rows does NOT change.

$$\alpha_1 R_1 + \dots + \alpha_i (\lambda R_i) + \dots + \alpha_m R_m = \alpha_1 R_1 + \dots + (\alpha_i \lambda) R_i + \dots + \alpha_m R_m,$$

$$\beta_1 R_1 + \dots + \beta_m R_m = \beta_1 R_1 + \dots + \left(\frac{\beta_i}{\lambda} \right) (\lambda R_i) + \dots + \beta_m R_m.$$

3. The operation $R_i + \lambda R_j \rightarrow R_i$, $i \neq j$. We need to show

$$\text{Sp}(R_1, \dots, R_m) = \text{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m).$$

Indeed,

$$\alpha_1 R_1 + \dots + \alpha_i (R_i + \lambda R_j) + \dots + \alpha_m R_m = \alpha_1 R_1 + \dots + \alpha_i R_i + \dots + \alpha_m R_m + (\alpha_i \lambda) R_j,$$

so the span is contained in $\text{Sp}(R_1, \dots, R_m)$.

Conversely,

$$R_i = (R_i + \lambda R_j) - \lambda R_j,$$

so $\text{Sp}(R_1, \dots, R_m) \subseteq \text{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m)$.

Summary: $\text{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m) = \text{Sp}(R_1, \dots, R_m)$.

Conclusion: After one elementary row operation on M , the $\text{RowS}(M)$ is UNchanged.

If A and B are row equivalent, then $\exists$ a finite sequence $A = A_0, A_1, \dots, A_r = B$ such that A_{i+1} is obtained from A_i by one elementary row operation. By what we just proved,

$$\text{RowS}(A) = \text{RowS}(A_0) = \dots = \text{RowS}(A_r) = \text{RowS}(B).$$

□

Remark. It is NOT true that $\text{ColS}(A) = \text{ColS}(B)$ (exc. find an example).

Definition. Let $A \in M_{m \times n}(K)$. Define

$$\text{row-rank}(A) := \dim(\text{RowS}(A))$$

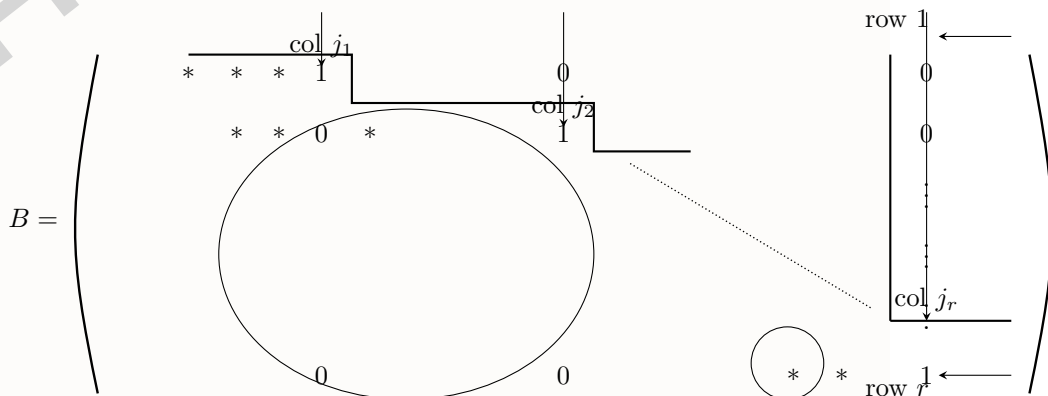
$$\text{col-rank}(A) := \dim(\text{ColS}(A)).$$

15

Lemma. Let $B \in M_{m \times n}(K)$ be in row reduced echelon form. Then the rows of B that are not totally 0 form a basis of $\text{RowS}(B)$. In particular, $\text{row-rank}(B)$ equals the number of pivots in B . Also the pivot columns of B form a basis of $\text{ColS}(B)$. In particular, for matrices B that are in row reduced echelon form, we have

$$\text{row-rank}(B) = \text{col-rank}(B).$$

Proof.



Denote by $j_1 < j_2 < \dots < j_r$ the column numbers of the pivots. Denote by the $u_1, \dots, u_r$ the non-zero rows of B . We claim that $u_1, \dots, u_r$ are linearly independent. Indeed, assume that $\lambda_1 u_1 + \dots + \lambda_r u_r = 0$ for some $\lambda_1, \dots, \lambda_r \in K$. This is a vector

in K^n . Entry number j_k in this vector is precisely λ_k i.e.

$$\lambda_1 u_1 + \dots + \dots + \lambda_r u_r = (\dots, \lambda_1, \dots, \lambda_2, \dots, \lambda_k, \dots, \lambda_r, \dots).$$

But if we assume that this vector is 0, then $\lambda_1 = \lambda_2 = \dots = \lambda_r = 0$. This shows our claim (that $u_1, \dots, u_r$ are linearly independent).

But $\text{RowS}(B) = \text{Sp}(u_1, \dots, u_r)$ By Thm 1, $u_1, \dots, u_r$ is a basis of $\text{RowS}(B)$. Also we get that $\text{row-rank}(B) = r$.

ColS(B): clearly $\text{ColS}(B) \subseteq \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \in K^m \mid x_{r+1} = \dots = x_m = 0 \right\} = K^r \times \{0\} \leftarrow 0\text{-vector of } K^{m-r}.$

At the same time, the pivot columns of B are of the form:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_r = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad (\text{the } i\text{th entry is } 1).$$

So $e_1, \dots, e_r$ form a basis for $K^r \times \{0\}$.

$$\Rightarrow K^r \times \{0\} = \text{span}(e_1, \dots, e_r) \subseteq \text{ColS}(B) \subseteq K^r \times \{0\},$$

$$\Rightarrow \text{ColS}(B) = K^r \times \{0\} \quad \text{and} \quad e_1, \dots, e_r \text{ form a basis for it.}$$

(the pivot columns)

□

Lecture 16: Sums of vector spaces

12.11.2025

14 Sums of vector spaces

Last time we established the meaning of the pivots in the row reduced matrix. In particular, we found that if r is the amount of pivots, then $\text{row-rank}(B) = \text{col-rank}(B) = r$.

Let's quickly discuss an algorithm to find a basis for the span of a list of vectors

$u_1, \dots, u_m \in K^n$. Just put them in a matrix

$$A = \begin{pmatrix} - & u_1 & - \\ - & u_2 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix}$$

Apply elimination to get a matrix B in RREF. The non-zero rows of B give a basis for the span of $\text{Sp}(u_1, \dots, u_m)$, literally just citing the last Lemma.

Remark. The number of pivots one gets after elimination does *not* depend on the specific elimination process. Why? Because the number of pivots equal $\text{row-rank}(A)$, which is invariant.

Definition (Transpose of a matrix). Let $A = (a_{ij}) \in M_{m \times n}(K)$. Define the transposed matrix of A

$$A^\top := (b_{ij}) \in M_{n \times m}(K),$$

where $b_{ij} = a_{ji}$.

We have a few simple properties of the transpose. $\forall A, B \in M_{m \times n}(K) : \forall \lambda \in K$:

$$(A + B)^\top = A^\top + B^\top$$

$$(\lambda A)^\top = \lambda A^\top$$

$$(A^\top)^\top = A$$

$$\text{RowS}(A^\top) = \text{ColS}(A)$$

$$\text{ColS}(A^\top) = \text{RowS}(A)$$

14.1 Sums of Vector Spaces

Let's maybe look at what happens when you sum vector spaces.

Definition (Sum of subspaces). Let V be a VS and $U, W \subseteq V$ linear subspaces of V . We define the sum

$$U + W = \{u + w \mid u \in U, w \in W\}.$$

You can extend this to more than just two subspaces. For $U_1, \dots, U_r \subseteq V$, we define

$$U_1 + \dots + U_r = \{u_1 + \dots + u_r \mid u_i \in U_i \forall 1 \leq i \leq r\}.$$

Proposition. Let $U, W \subseteq V$ be subspaces of V . Then,

- 1) $U + W = \text{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is therefore a subspace.
- 2) If U, W are finite dimensional, then $U + W$ is also finite dimensional. One can

write a basis for $U + W$ as follows:

- Choose a basis $p_1, \dots, p_k$ for $U \cap W$ ($k := \dim(U \cap W)$).
- Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, u_1, \dots, u_{l-k}$ of U with $l := \dim U$.
- Extend $p_1, \dots, p_k$ to a basis of W , $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ with $m = \dim W$.

Then, $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ is a basis for $U + W$.

3) In particular, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Proof. 1) Clearly, $U + W \subseteq \text{Sp}(U \cup W)$. We'll show also that $\text{Sp}(U \cup W) \subseteq U + W$. Indeed, let $v \in \text{Sp}(U \cup W)$. Then,

$$v = \sum_{i=1}^s a_i u_i + \sum_{j=1}^r b_j w_j = \bar{u} + \bar{w},$$

because those sums remain in the subspaces they originated from hence $\bar{u} \in U$ and $\bar{w} \in W$. This proves the equality.

2) We first show that $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ are linearly independent. Indeed, assume for some $a_i, b_i, c_i \in K$

$$a_1 p_1 + \dots + a_k p_k + b_1 u_1 + \dots + b_{l-k} u_{l-k} + c_1 w_1 + \dots + c_{m-k} w_{m-k} = 0.$$

Write

$$v = b_1 u_1 + \dots + b_{l-k} u_{l-k} + c_1 w_1 + \dots + c_{m-k} w_{m-k} \in W.$$

Note that from the way we defined v , we have

$$v = (a_1 p_1 + \dots + a_k p_k + b_1 u_1 + \dots + b_{l-k} u_{l-k}) \in U.$$

Hence $v \in U \cap W$. Since $p_1, \dots, p_k$ form a basis for $U \cap W$, $\exists$ coefficients $\alpha_1, \dots, \alpha_k \in K$, s.t.

$$v = \alpha_1 p_1 + \dots + \alpha_k p_k.$$

But

$$v = c_1 w_1 + \dots + c_{m-k} w_{m-k}$$

and

$$\alpha_1 p_1 + \dots + \alpha_k p_k + (-c_1) w_1 + \dots + (-c_{m-k}) w_{m-k} = 0.$$

But $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ are a basis for W , hence $\alpha_1 = \dots = \alpha_k = 0$ and $c_1 = \dots = c_{m-k} = 0$, hence $v = 0$. But this means

$$0 = v = -(a_1 p_1 + \dots + a_k p_k + b_1 u_1 + \dots + b_{l-k} u_k).$$

But $p_1, \dots, p_k, u_1, \dots, u_k$ are a basis (for U), hence $a_1 = \dots = a_k = 0$ and $b_1 = \dots = b_{l-k} = 0$. This shows that $p_1, \dots, p_k, u_1, \dots, u_{l-k}, \dots, w_1, \dots, w_{m-k}$ are linearly independent.

It remains to show that this list of vectors span $U + W$. Indeed, let $z \in U + W$. Write $z = u + w$ for some $u \in U, w \in W$. We can write

$$u = \alpha_1 p_1 + \dots + \alpha_k p_k + a_1 u_1 + \dots + a_{l-k} u_{l-k}$$

for some $\alpha_i, a_j \in K$. Then write

$$w = \beta_1 p_1 + \dots + \beta_k p_k + b_1 u_1 + \dots + b_{m-k} w_k$$

for some $\beta_i, b_j \in K$. Then,

$$z = (\alpha_1 + \beta_1)p_1 + \dots + (\alpha_k + \beta_k)p_k + a_1 u_1 + \dots + a_{l-k} u_{l-k} + b_1 w_1 + \dots + b_{m-k} w_{m-k} \in \text{Sp}(U \cup W).$$

3) We have the length of the basis we just constructed given by

$$\dim(U + W) = k + (l - k) + (m - k) = j + m - k = \dim U + \dim W - \dim(U \cap W).$$

□

Corollary. Let $U, W \subseteq V$ be finite dimensional subspaces. Then the following five statements are equivalent:

1. $\dim(U + W) = \dim U + \dim W$
2. $\dim(U \cap W) = 0$.
3. $U \cap W = \{0\}$.
4. Every $v \in U + W$ can be written in a *unique* way as $v = u + w$ with $u \in U, w \in W$.
5. If $u + w = 0$, where $u \in U, w \in W \Rightarrow u = w = 0$.

Proof. $1 \Leftrightarrow 2$: Follows by the previous proposition (3), the dimension formula.

$2 \Leftrightarrow 3$: This is immediate by the fact that there is only one space which has $\dim = 0$, the trivial space $\{0\}$.

$3 \Rightarrow 4$: By definition, $\forall v \in U + W$ can be written as $v = u + w$ with $u \in U, w \in W$. We need to show the uniqueness. Assume $v = u + w = u' + w'$ where $u, u' \in U$ and $w, w' \in W$. So $u - u' = w' - w$, so $u - u', w' - w \in U \cap W = \{0\}$, hence $u' = u, w' = w$.

$4 \Rightarrow 5$: $0 = 0 + 0$, so if $0 = u + w$ and 4 holds, then $u = 0, w = 0$.

$5 \Rightarrow 3$: Let $v \in U \cap W$. Clearly $0 = v + (-v)$, with $v \in U$ and $-v \in W$. If 5 holds, then $v = 0$, hence $U \cap W = \{0\}$. □

Definition (Complement of a subspace). Let V be a vector space and $U \subseteq V$ a subspace. A subspace $W \subseteq V$ is called a complement to U if $U + W = V$ and

16

$$U \cap W = \{0\}.$$

Lecture 17

14.11.2025

Proposition (14.5). Let V be a finite dimensional vector space and $U \subseteq V$ a subspace. Then there exists a complement $W \subseteq V$ which is a complement of U .

Proof. Choose a basis $u_1, \dots, u_\ell$ of U ($\ell := \dim U$). Extend this basis to a basis of V : $u_1, \dots, u_\ell, w_1, \dots, w_m$ ($\ell + m = \dim V$). Take $W := \text{Sp}(w_1, \dots, w_m)$. Clearly $U + W = V$, b.c. $U + W = \text{Sp}(U \cup W) \subseteq \text{Sp}(u_1, \dots, u_\ell, w_1, \dots, w_m) = V$. Also $U \cap W = \{0\}$, b.c. if

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^m b_j w_j, \text{ then}$$

$$\sum_{i=1}^{\ell} a_i u_i + \sum_{j=1}^m (-b_j) w_j = 0.$$

But $u_1, \dots, u_\ell, w_1, \dots, w_m$ are linearly independent, so all coefficients are 0 $\Rightarrow a_i = 0 \forall i$ and $b_j = 0 \forall j$. Alternatively:

$$\dim(U \cap W) = \dim(U + W) - \dim U - \dim W = (\ell + m) - \ell - m = 0.$$

□

17

Chapter 5

Linear maps and matrices

15.a Linear maps: Homomorphisms of Vector Spaces

Maps bring interest into vector spaces. A vector space is a set with some structure, and we are gonna study maps that respect this structure. Linear maps are also called linear transformations and homomorphisms of vector spaces.

Definition (15.a.1). Let V, W be vector spaces over the same field K . A map $T : V \rightarrow W$ is called linear if:

1. $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$ (additivity)
2. $\forall a \in K, \forall v \in V : T(av) = aT(v)$ (homogeneity)

Sometimes $Tv = T(v)$ is used instead of $T(v)$.

The set of all linear maps $T : V \rightarrow W$ is denoted by $\text{Hom}_K(V, W)$ or simply $\text{Hom}(V, W)$.

as sometimes $\text{hom}(V, W)$

Exercise: – Show that $T : V \rightarrow W$ is linear $\Leftrightarrow T(a_1v_1 + a_2v_2) = a_1T(v_1) + a_2T(v_2) \forall v_1, v_2 \in V, \forall a_1, a_2 \in K$

Example (1).

$V =$ vector space, $id : V \rightarrow V$, is a linear map.

Example (2).

$V, W =$ vector spaces, $O : V \rightarrow W, v \in V \mapsto 0 \in W$

the zero map is linear.

Example (3). $D : K[x] \rightarrow K[x]$, the (formal) derivative.

$$D(a_0 + a_1x + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1}.$$

$$\sum_{i=0}^n a_i x^i \mapsto \sum_{i=1}^n i a_i x^{i-1}.$$

This is a linear map.

$$D(ap(x) + bq(x)) = a \cdot p'(x) + b \cdot q'(x) = aD(p(x)) + bD(q(x)).$$

$$a, b \in K \quad p(x), q(x) \in K[x]$$

Example (4).

$$C([a, b]) := \text{contin. functions } f : [a, b] \rightarrow \mathbb{R},$$

$$T : C([a, b]) \rightarrow \mathbb{R},$$

$$T(f) := \int_a^b f(x) dx.$$

Example (5).

$$S : K^\infty \rightarrow K^\infty$$

$$S(a_1, a_2, \dots) := (a_2, a_3, \dots, a_n, \dots)$$

The shift map, it is linear.

Example (6). Important example. Let $A \in M_{m \times n}(K)$.

Define

$$T_A : K_{\text{col}}^n \rightarrow K_{\text{col}}^m$$

$$T_A \left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right) := A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K_{\text{col}}^m, \quad A \in K^{m \times n}.$$

We showed that $A \cdot (a \cdot x) = a \cdot (Ax) \quad \forall a \in K, x \in K_{\text{col}}^n$

and

$$A(x + y) = Ax + Ay, \quad \forall x, y \in K_{\text{col}}^n.$$

$\Rightarrow T_A$ is a linear map.

Proposition (15.a.2). Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map. Then:

$$1. \forall n \in \mathbb{Z}_{\geq 0}, \forall v_1, \dots, v_n \in V, \forall a_1, \dots, a_n \in K :$$

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

$$2. T(0_V) = 0_W$$

Proof. 1. Follows from the definition of linear maps (15.a.1) by induction on n .
 2. $T(0_W) = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V) \Rightarrow T(0_V) = 0_W$.

□

Theorem. Let V be a finite dimensional VS and $v_1, \dots, v_n \in V$ a basis for V . Let W be any vector space and $w_1, \dots, w_n \in W$ a list of arbitrary vectors. Then $\exists! T : V \rightarrow W$ linear s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n$$

Proof. Existence of T : Let $v \in V$. We want to define $T(v) \in W$. Since $v_1, \dots, v_n$ form a basis of V , $\exists a_1, \dots, a_n \in K$ s.t. $v = \sum_{i=1}^n a_i v_i$. Define $T(v) := a_1 w_1 + \dots + a_n w_n = \sum_{i=1}^n a_i w_i \in W$. First let that T is well-defined, because $\forall v \in V$ the coefficients $a_1, \dots, a_n \in K$ s.t. $v = a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i$ are unique. (because $v_1, \dots, v_n$ is a basis of V).

We claim that T is linear: Indeed, let $v', v'' \in V \exists a'_1, \dots, a'_n \in K, a''_1, \dots, a''_n \in K$ s.t.
 $v' = \sum_{i=1}^n a'_i v_i = a'_1 v_1 + \dots + a'_n v_n, v'' = \sum_{i=1}^n a''_i v_i = a''_1 v_1 + \dots + a''_n v_n$.

$$v' + v'' = \sum_{i=1}^n (a'_i + a''_i) v_i = (a'_1 + a''_1) v_1 + \dots + (a'_n + a''_n) v_n$$

$$\begin{aligned} \text{By our definition. (*)}, T(v' + v'') &= \sum_{i=1}^n (a'_i + a''_i) w_i = (a'_1 + a''_1) w_1 + \dots + (a'_n + a''_n) w_n \\ &= \sum_{i=1}^n a'_i w_i + \sum_{i=1}^n a''_i w_i = a'_1 w_1 + \dots + a'_n w_n + a''_1 w_1 + \dots + a''_n w_n \\ &= T(v') + T(v''). \end{aligned}$$

Let $\alpha \in K, v \in V$. Write $v = a_1 v_1 + \dots + a_n v_n$ ($a_i \in K$).

$$T(\alpha v) = T(\alpha a_1 v_1 + \dots + \alpha a_n v_n) \stackrel{(*)}{=} \alpha a_1 w_1 + \dots + \alpha a_n w_n = \alpha (a_1 w_1 + \dots + a_n w_n) \stackrel{(*)}{=} \alpha T(v).$$

This proves that T is linear. Note that $T(v_i) = w_i$, because $v_i = 0 \cdot v_1 + \dots + 1 \cdot v_i + \dots + 0 \cdot v_n$.

Uniqueness of T . Suppose $T, S : V \rightarrow W$ are two linear maps with $T(v_i) = w_i = S(v_i)$ for all $i \leq n$. We'll show $T = S$. Let $v \in V$. We will show that $T(v) = S(v)$. Write $v = a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i$ ($a_i \in K \forall i$).

$$\begin{aligned} T(v) &= T(a_1 v_1 + \dots + a_n v_n) = T\left(\sum_{i=1}^n a_i v_i\right) = a_1 T(v_1) + \dots + a_n T(v_n) \\ &= \sum_{i=1}^n a_i T(v_i) = a_1 w_1 + \dots + a_n w_n = \sum_{i=1}^n a_i w_i \\ &= S(a_1 v_1 + \dots + a_n v_n) = S\left(\sum_{i=1}^n a_i v_i\right) = S(v). \end{aligned}$$

□

Example (Important example 15.a.4). Consider K^n with the standard basis $e_1, \dots, e_n$. Let $w_1, \dots, w_n \in K^m$ be n arbitrary vectors. By the previous theorem (15.a.3), $\exists!$ linear map $T : K^n \rightarrow K^m$ s.t. $T(e_i) = w_i$. We can describe T as $T = T_A$, where

$$A = \begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix} \in K^{m \times n}.$$

Recall:

$T_A : K^n \rightarrow K^m$, is linear.

$$T_A(v) = A \cdot v.$$

$$T_A(e_i) = \begin{pmatrix} | \\ w_i \\ | \end{pmatrix} \quad \forall i \quad e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (1 \text{ at the } i\text{-th position}).$$

Lemma. For every linear map $T : K^n \rightarrow K^m$, there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $T_A = T$. That matrix is

$$A = \begin{pmatrix} | & & | \\ T e_1 & \dots & T e_n \\ | & & | \end{pmatrix}.$$

Proof. This is immediate, as demonstrated in the last example. □

Lecture 18

19.11.2025

15.b Linear maps: Kernel and Image of a linear map

Definition (Prop. + Def. 15.b.1). Let V, W be vector spaces and $T : V \rightarrow W$ a linear map.

1. The set

$$\ker(T) := \{v \in V \mid T(v) = 0_W\} \subseteq V$$

is a subspace of V . It is called the *kernel* of T .

2. The set

$$\operatorname{Im}(T) := \{T(v) \mid v \in V\} \subseteq W$$

is a subspace of W . It is called the *image* of T .

Hence,

$$\ker(T) = T^{-1}(\{0_W\}), \quad \operatorname{Im}(T) = T(V).$$

Proof.

1. $T(0_V) = 0_W$, hence $0_V \in \ker(T)$. Let $v_1, v_2 \in \ker(T)$ and $a, b \in K$. We have

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = 0 \Rightarrow av_1 + bv_2 \in \ker(T).$$

This shows that $\ker(T)$ is a subspace of V .

2. $T(0_V) = 0_W \Rightarrow 0_W \in \operatorname{Im}(T)$. Let $w_1, w_2 \in \operatorname{Im}(T)$ and $a, b \in K$. Then $\exists v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. We have

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = aw_1 + bw_2.$$

so

$$aw_1 + bw_2 = T(\text{something}) \Rightarrow aw_1 + bw_2 \in \operatorname{Im}(T).$$

□

Example. Let $A \in M_{m \times n}(K)$. Consider $T_A : K^n \rightarrow K^m$. ($T_A(x) := Ax$)

$$\ker(T_A) = \{x \in K^n \mid Ax = 0\}.$$

Proposition (15.b.2). Let $T : V \rightarrow W$ be a linear map.

1. T is injective $\Leftrightarrow \ker(T) = \{0_V\}$.
2. T is surjective $\Leftrightarrow \operatorname{Im}(T) = W$.

Proof. (2) is a triviality.

(1) $\Rightarrow$: Clearly $\{0_V\} \subseteq \ker(T)$ because $T(0_V) = 0_W$. Since T is injective, $T^{-1}(\{0_W\})$

consists of at most one element. But we know that $0_V \in T^{-1}(\{0_W\})$. So

$$\ker(T) = T^{-1}(\{0_W\}) = \{0_V\}.$$

$\Leftarrow$: Assume $\ker(T) = \{0_V\}$. Let $v_1, v_2 \in V$ and assume $T(v_1) = T(v_2)$. We need to show that $v_1 = v_2$. Indeed,

$$T(v_1) - T(v_2) = 0_W,$$

$$T(v_1 - v_2) = 0_W.$$

So,

$$v_1 - v_2 \in \ker(T) \Rightarrow v_1 - v_2 = 0_V$$

$$\Rightarrow v_1 = v_2.$$

□

Example. Let $T : V \rightarrow W$ be a linear map.

1. If $V' \subseteq V$ is a linear subspace, then $T(V') \subseteq W$ is a linear subspace.
2. If $W' \subseteq W$ is a linear subspace, then $T^{-1}(W') \subseteq V$ is a linear subspace.

$$f : A \rightarrow B \cup B', \quad B' \subseteq B$$

$$f^{-1}(B') = \{a \in A \mid f(a) \in B'\}.$$

18

Definition (Invertible maps). A map $f : X \rightarrow Y$ is called bijective if it is both injective & surjective. In this case $\exists!$ map $g : Y \rightarrow X$ s.t. $g \circ f = id_X$ and $f \circ g = id_Y$. This map g is called the inverse of f . It's denoted by f^{-1} .

Definition (Isomorphism). A linear map $T : V \rightarrow W$ is called an isomorphism if $\exists$ a linear map $S : W \rightarrow V$ s.t. $S \circ T = id_V$ and $T \circ S = id_W$. If such a T exists, we say that V & W are *isomorphic* or that V is isomorphic to W . We write $V \cong W$ (sometimes $V \simeq W$).

Remark. If T is unique, then T is bijective (b.c. it has an inverse S).

Lemma (15.b.4). Let $T : V \rightarrow W$ be a linear map. Assume T is bijective. Then its inverse map $T^{-1} : W \rightarrow V$ must be linear.

Proof. Let $w_1, w_2 \in W$ and $a, b \in K$. We need to show that $T^{-1}(aw_1 + bw_2) = aT^{-1}(w_1) + bT^{-1}(w_2)$.

Put $v_1 := T^{-1}(w_1)$ and $v_2 := T^{-1}(w_2)$.

$$T(v_1) = w_1, \quad T(v_2) = w_2.$$

Since T is linear,

$$T^{-1}(aw_1 + bw_2) = T^{-1}(T(av_1 + bv_2)) = av_1 + bv_2 = aT^{-1}(w_1) + bT^{-1}(w_2).$$

$$\Rightarrow T^{-1}(aw_1 + bw_2) = av_1 + bv_2 = aT^{-1}(w_1) + bT^{-1}(w_2).$$

□

Lemma (15.b.5). Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps. Then the composition $S \circ T : V \rightarrow U$ is also a linear.

Proof. Let $v_1, v_2 \in V$ and $a, b \in K$.

$$S \circ T(av_1 + bv_2) = S(aT(v_1) + bT(v_2)) = aS(T(v_1)) + bS(T(v_2)) = a(S \circ T)(v_1) + b(S \circ T)(v_2).$$

□

EXERCISE — Show that " $\cong$ " (being isomorphic) is an equivalence relation on the class of all vector spaces over a field K .

18

Definition. A linear map $T : V \rightarrow V$ is called an *endomorphism* of V . We denote the set of all endomorphisms $T : V \rightarrow V$ by

$$\text{End}_K(V) \quad (= \text{Hom}_K(V, V)).$$

Definition. A linear map $T : V \rightarrow V$ which is also an isomorphism is called an *automorphism* of V . The set of autos (automorphisms) of V is denoted by

$$\text{Aut}(V).$$

Definition (More jargon). Let $T : V \rightarrow W$ be a linear map.

1. If T is injective, we also say that T is a *monomorphism*.
2. If T is surjective, we also say that T is a *epimorphism*.

Lemma (15.b.8). Let $T : V \rightarrow W$ be a linear map. Let $v_1, \dots, v_n \in V$ be a list that spans V . Then

$$\text{Im}(T) = \text{Sp}(T(v_1), \dots, T(v_n)).$$

Proof. We'll first show $\text{Im}(T) \subseteq \text{Sp}(T(v_1), \dots, T(v_n))$. Let $a_1, \dots, a_n \in K$.
 Let $\sum_{i=1}^n a_i T v_i \in \text{Sp}(T(v_1), \dots, T(v_n))$.

$$\sum_{i=1}^n a_i T v_i = T \left(\sum_{i=1}^n a_i v_i \right) \in \text{Im}(T).$$

We'll show now that $\text{Im}(T) \supseteq \text{Sp}(T(v_1), \dots, T(v_n))$. Let $w \in \text{Im}(T)$. By definition $\exists v \in V$ s.t. $T(v) = w$. Since $v_1, \dots, v_n$ span V , $\exists a_1, \dots, a_n \in K$ s.t.

$$v = \sum_{i=1}^n a_i v_i.$$

$$\Rightarrow w = T(v) = T \left(\sum_{i=1}^n a_i v_i \right) = \sum_{i=1}^n a_i T(v_i) \in \text{Sp}(T(v_1), \dots, T(v_n)).$$

This shows $\text{Im}(T) \supseteq \text{Sp}(T(v_1), \dots, T(v_n))$. □

Theorem (15.b.9: The Rank Theorem). Let $T : V \rightarrow W$ be a linear map, where V is finite dimensional. Then $\dim \ker(T) + \dim \text{Im}(T) = \dim V$.

Proof. Write $n := \dim V$. Let $v_1, \dots, v_k$ be a basis for $\ker(T)$ ($k = \dim \ker(T)$). (Note that $\ker(T)$ is finite dimensional, being a subspace of V which is finite dimensional). Extend this basis to a basis

$$u_1, \dots, u_k, v_1, \dots, v_{n-k} \text{ of } V.$$

By Lemma 15.b.8, $\text{Im}(T) = \text{Sp}(T(u_1), \dots, T(u_k), T(v_1), \dots, T(v_{n-k}))$.

....

We'll show that $T(v_1), \dots, T(v_{n-k})$ is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T(v_i) = 0_W (*).$$

Since T is linear,

$$(*) = T \left(\sum_{i=1}^{n-k} a_i v_i \right).$$

So,

$$T \left(\sum_{i=1}^{n-k} a_i v_i \right) = 0_W \Rightarrow \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

18

Now, $u_1, \dots, u_k$ form a basis for $\ker(T)$, $\Rightarrow \exists b_1, \dots, b_k \in K$ s.t.

$$\sum_{i=1}^{n-k} a_i v_i = b_1 u_1 + \dots + b_k u_k \Rightarrow b_1 u_1 + \dots + b_k u_k + (-a_1) v_1 + \dots + (-a_{n-k}) v_{n-k} = 0_V.$$

Since $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ form a basis for V , we have:

$$b_1 = \dots = b_k = 0, \quad a_1 = \dots = a_{n-k} = 0.$$

This shows that $T(v_1), \dots, T(v_{n-k})$ are linearly independent. We have already seen that those span $\text{Im}(T)$, so they actually form a basis of $\text{Im}(T)$.

$$\begin{aligned} &\Rightarrow \dim \text{Im}(T) = n - k = \\ &= \dim V - \dim \ker(T). \Rightarrow \dim \ker(T) + \dim \text{Im}(T) = \dim V. \end{aligned}$$

□

Lecture 19

21.11.2025

19

Corollary. Let $T : V \rightarrow W$ be a linear map, where V & W are finite dimensional. Then:

1. If $\dim W < \dim V$, then T is *not* injective.
2. If $\dim W < \dim V$, then T is *not* surjective.
3. If $\dim V = \dim W$, then T is bijective $\Leftrightarrow T$ is injective $\Leftrightarrow T$ is surjective.

Proof.

1. Since $\text{Im}(T)$ is a subspace of W , we have:

$$\dim \text{Im}(T) \leq \dim W < \dim V.$$

By the rank theorem (15.b.9)

$$\dim \ker(T) = \dim V - \dim \text{Im}(T) > 0 \Rightarrow \ker(T) \neq \{0_V\}.$$

By proposition 15.b.2, T is *not* injective.

2. $\dim \text{Im}(T) = \dim V - \dim \ker(T) \leq \dim V < \dim W.$

$$\Rightarrow \text{Im}(T) \subsetneq W \Rightarrow T \text{ is not surjective..}$$

3. T is injective $\Leftrightarrow \ker(T) = \{0_V\} \Leftrightarrow \dim \ker(T) = 0 \Leftrightarrow \dim \text{Im}(T) = \dim V \Leftrightarrow \dim \text{Im}(T) = \dim W$ (b.c. $\dim V = \dim W$) $\Leftrightarrow \text{Im}(T) = W \Leftrightarrow T$ is surjective.

□

Corollary. Two finite dimensional vector spaces V and W are isomorphic iff $\dim V = \dim W$.

Proof. " $\Rightarrow$ ": Suppose $T : V \rightarrow W$ is an isomorphism. By proposition 15.b.2, $\ker(T) = \{0_V\}$ and $\text{Im}(T) = W$. By the rank theorem (15.b.9) we have:

$$\dim W = \dim \text{Im}(T) = \dim V - \underbrace{\dim \ker(T)}_{=0} = \dim V.$$

" $\Leftarrow$ ": Assume $\dim V = \dim W$. Denote by $n := \dim V = \dim W$. Let $v_1, \dots, v_n$ be a basis of V . Let $w_1, \dots, w_n$ be a basis of W . Define a linear map $T : V \rightarrow W$ by $T(v_i) := w_i \forall i \in \{1, \dots, n\}$. By Theorem 15.a.3, $\exists$ a linear map T with these (*) properties. By Lemma 15.b.8, $\text{Im}(T) = \text{Sp}(T(v_1), \dots, T(v_n)) = \text{Sp}(w_1, \dots, w_n) = W$. So T is surjective. By corollary 15.b.10 we have that T is bijective. $\square$

Theorem. Let $T : V \rightarrow W$ be an isomorphism between two vector spaces V & W . Let S be a set of vectors from V . Write

$$T(S) := \{T(v) \mid v \in S\}.$$

Then:

1. S is linearly independent iff $T(S)$ is linearly independent.
2. S spans V iff $T(S)$ spans W .
3. S is a basis of V iff $T(S)$ is a basis of W .

19

$$\begin{aligned} V &\cong W \\ \exists T : V &\rightarrow W \\ \&\exists R : W &\rightarrow V \\ \text{linear s.t. } R \circ T &= id_V \\ \text{and } R \circ T &= id_W \end{aligned}$$

$$\boxed{V \cong K^n}$$

$$K[x]_3 \cong K^4.$$

Definition (15.b.13). Let $T : V \rightarrow W$ be a linear map. Define the rank of T , to be $\text{rank}(T) := \dim \text{Im}(T)$.

EXERCISE — Let $T : V \rightarrow W, S : W \rightarrow U$ be linear maps, where U, V, W are finite dimensional. Show that:

1. $\text{rank}(S \circ T) \leq \min\{\text{rank}(T), \text{rank}(S)\}$.
2. If S is injective, then $\text{rank}(S \circ T) = \text{rank}(T)$.
3. If T is surjective, then $\text{rank}(S \circ T) = \text{rank}(S)$.

Hint: Just use the definitions.

16 Linear maps and bases: Writing linear maps using bases/coordinates

In this discussion all vector spaces are finite dimensional. We'll write a basis as an *ordered* tuple $B = (v_1, \dots, v_n)$.

Definition (16.1). Let V be a vector space over K and $B = (v_1, \dots, v_n)$ a basis for V . Let $v \in V$. Define the *coordinate vector* of v with respect to B as

$$[v]_B := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique list of scalars in K s.t. $v = a_1 v_1 + \dots + a_n v_n$. Define a map $\Phi_B : V \rightarrow K^n$ by

$$\Phi_B(v) := [v]_B \in K^n \quad \forall v \in V.$$

Proposition (16.2). Φ_B is linear and is an isomorphism.

Proof. Linearity of Φ_B . Let $v, w \in V$, $a, b \in K$. Write v & w a linear combination of the elements of B .

$$v = a_1 v_1 + \dots + a_n v_n$$

$$w = b_1 v_1 + \dots + b_n v_n$$

$$\Rightarrow [v]_B = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad [w]_B = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

$$\Rightarrow av + bw = (aa_1 + bb_1)v_1 + \dots + (aa_n + bb_n)v_n.$$

$$\Rightarrow [av + bw]_B = \begin{pmatrix} aa_1 + bb_1 \\ \vdots \\ aa_n + bb_n \end{pmatrix} = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

$$\Phi_B(av + bw) = [av + bw]_B = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a[v]_B + b[w]_B = a\Phi_B(v) + b\Phi_B(w).$$

This proves that Φ_B is linear. We'll show now that Φ_B is an isomorphism. We'll

show that $\text{Ker}(\Phi_B) = \{0_V\}$. Indeed, if $v \in \text{ker}(\Phi_B)$, then $[v]_B = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \in K^n$.

19

$\Rightarrow v = 0v_1 + \dots + 0v_n = 0$. So $\ker(\Phi_B) = \{0\}$. Since $\dim V = \dim K^n$, Φ_B is an isomorphism. (Alternatively: show Φ_B is surjective). $\square$

Lecture 20

26.11.2025

Remark. The inverse Φ_B^{-1} of Φ_B can be written as:

$$\Phi_B^{-1} : K^n \rightarrow V,$$

$$\Phi_B^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \dots + a_n v_n.$$

Example. Consider $V = \mathbb{R}^2$ and let $\mathcal{E} = ((1, 0), (0, 1))$ be the standard basis. Let $B = ((1, 1), (1, -1))$ be another basis (check it's a basis). Let $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

$$\left[\begin{pmatrix} x \\ y \end{pmatrix} \right]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Why? Because $x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$. What is $\left[\begin{pmatrix} x \\ y \end{pmatrix} \right]_B$? Consider first $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$$\text{Write } \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]_B = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}.$$

$$a_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$\begin{cases} a_1 + a_2 = 1, \\ a_1 - a_2 = 0, \end{cases} \Rightarrow a_1 = a_2 = \frac{1}{2}.$$

So

$$\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]_B = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}.$$

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]_B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix},$$

$$b_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

20

$$\begin{cases} b_1 + b_2 = 0 \\ b_1 - b_2 = 1 \end{cases} \Rightarrow b_1 = \frac{1}{2}, b_2 = -\frac{1}{2}.$$

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]_B = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Using linearity of Φ_B ,

$$\begin{aligned} \left[\begin{pmatrix} x \\ y \end{pmatrix} \right]_B &= x \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]_B + y \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]_B \\ &= \begin{pmatrix} \frac{x}{2} \\ \frac{x}{2} \end{pmatrix} + \begin{pmatrix} \frac{y}{2} \\ -\frac{y}{2} \end{pmatrix} = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{pmatrix}. \end{aligned}$$

Let V, W be vector spaces (finite dimensional) $n := \dim V$, $m := \dim W$. Let T be a linear map $T : V \rightarrow W$.

$T : V \rightarrow W$. Fix a basis $\mathcal{B}$ for V , and $\mathcal{C}$ for W .

$$T : V \rightarrow W$$

Fix a basis B for V , and a basis C for W .

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_B^{-1} \uparrow & \circlearrowleft & \downarrow \Phi_C \\ K^n & \xrightarrow{\Phi_C \circ T \circ \Phi_B^{-1}} & K^m \end{array} \quad \text{diag. commutes}$$

Recall that composition of linear maps is linear, hence

$$\Phi_C \circ T \circ \Phi_B^{-1} : K^n \rightarrow K^m$$

is linear.

By previous knowledge (15,a,4), $\exists! A \in M_{m \times n}(K)$ such that

$$\Phi_C \circ T \circ \Phi_B^{-1} = T_A$$

(where $T_A : K^n \rightarrow K^m$ is $T_A x := Ax$).

The matrix A is called the representation of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. It is denoted

$$[T]_{\mathcal{B}}^{\mathcal{C}} := A, \quad [T]_{\mathcal{B}}^{\mathcal{C}} \in M_{m \times n}(K).$$

How to calculate $[T]_{\mathcal{C}}^{\mathcal{B}}$?

Put $A := [T]_{\mathcal{C}}^{\mathcal{B}}$.

$$\begin{array}{ccc}
 v_i \in V & \xrightarrow{T} & W \in T v_i \\
 \uparrow \Phi_B^{-1} & \odot & \downarrow \Phi_C \\
 e_i \in K^u & \xrightarrow{T_A} & K^w \in [T v_i]_C
 \end{array}$$

$$\begin{aligned}
 T_A(e_i) &= \Phi_C \circ T \circ \Phi_B^{-1}(e_i) \\
 &= \Phi_C(T(v_i)) \\
 &= [T v_i]_C
 \end{aligned}$$

In other words

$$A = [T]_C^B = \left(\begin{array}{c|ccc|c} & & \text{col \#} j & & \\ & | & & | & \\ [T v_1]_C & \cdots & [T v_j]_C & \cdots & [T v_n]_C \\ & | & & | & \end{array} \right) \in M_{m \times n}(K) \longrightarrow \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

Another way to describe $[T]_C^B$:

Write $[T]_C^B = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$. The entries a_{ij} are uniquely defined by:

$$T v_j = a_{1j} w_1 + \cdots + a_{mj} w_m = \sum_{i=1}^m a_{ij} w_i$$

$\forall j$ (Here $C = (w_1, \dots, w_m)$).

Proposition ((16.4)). Let $T : V \rightarrow W$ be a linear map between two finite dimensional vector spaces. Let B, C be bases for V, W respectively. Put $A = [T]_C^B$. Then $\forall v \in V, [T v]_C = A[v]_B$. Equivalently:

$$[T v]_C = [T]_C^B \cdot [v]_B.$$

Proof.

$$\begin{array}{ccccc}
 & & T & & \\
 \omega & \in & V \xrightarrow{T} W & \ni & T\omega \\
 \downarrow & & \Phi_B \downarrow \odot \downarrow \Phi_C & & \downarrow \\
 [\omega]_B & \in & K^n \xrightarrow{T_A} K^m & \ni & [T\omega]_C
 \end{array}$$

$$\text{So } [T\omega]_C = T_A([\omega]_B) = A[\omega]_B. \quad \square$$

□

Example.

1. Let

$$V = K^n, \quad W = K^m,$$

and let

$$T : K^n \rightarrow K^m \text{ be } T = T_Q,$$

where

$$Q \in M_{m \times n}(K).$$

So $Tv = Qv$. Take $\mathcal{B} = (e_1, \dots, e_n)$, $\mathcal{E} = (e_1, \dots, e_m)$. Then

$$[T]_{\mathcal{E}}^{\mathcal{B}} = Q.$$

2.

$$V = K[x]_{\leq 3}, \quad W = K[x]_{\leq 2}.$$

$$D : V \rightarrow W, \quad D(p(x)) := p'(x).$$

D is linear. Take $\mathcal{B} = (1, x, x^2, x^3)$ and $\mathcal{C} = (1, x, x^2)$. Find $[D]_{\mathcal{C}}^{\mathcal{B}} \in M_{3 \times 4}(K)$.

$$\begin{array}{lcl} D(1) & = & 0 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ D(x) & = & 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \end{array} \quad \begin{array}{lcl} D(x^2) & = & 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2 \\ D(x^3) & = & 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2 \end{array}$$

$$[D]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

2'. Another variation of the problem. View D as a map $D : K[x]_3 \rightarrow K[x]_3$. Take $\mathcal{C} = \mathcal{B}$. Then

$$[D]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is just as valid and even makes sense because you will never get as a function value from D a polynomial of degree 3.

Let $T : V \rightarrow W$ be linear maps. Let

$$\mathcal{A} = (v_1, \dots, v_n) \text{ be a basis for } V,$$

$$\mathcal{B} = (w_1, \dots, w_m) \text{ be a basis for } W.$$

and $e = (u_1, \dots, u_n)$ be a basis for U . Consider $S \circ T : V \rightarrow U$. *Question:* What is the relation between $[S \circ T]_e^{\mathcal{A}}$ and $[T]_{\mathcal{B}}^{\mathcal{A}}, [S]_e^{\mathcal{B}}$? Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} \in M_{m \times n}$,

$$[S]_e^{\mathcal{B}} = (b_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq m}} \in M_{p \times m}$$

$$[S \circ T]_e^{\mathcal{A}} = (c_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}} \in M_{p \times n}.$$

We know that

$$Tv_j = a_{1j}w_1 + \dots + a_{mj}w_m = \sum_{i=1}^m a_{ij}w_i \forall 1 \leq j \leq n$$

$$Sw_j = b_{1j}u_1 + \dots + b_{pj}u_p = \sum_{i=1}^p b_{ij}u_i \forall 1 \leq j \leq m$$

$$(S \circ T)v_j = c_{1j}u_1 + \dots + c_{pj}u_p = \sum_{i=1}^p c_{ij}u_i \forall 1 \leq j \leq n.$$

$$\begin{aligned} \text{But } (S \circ T)v_j &= S(Tv_j) = S\left(\sum_{i=1}^m a_{ij}w_i\right) = \sum_{i=1}^m a_{ij}S(w_i) = \sum_{i=1}^m a_{ij}\left(\sum_{l=1}^p b_{il}u_l\right) = \\ &= \sum_{k=1}^m \sum_{i=1}^p a_{ij}b_{ik}u_i = \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik}a_{kj}\right) u_i. \end{aligned}$$

$$\text{So } c_{ij} = \sum_{k=1}^m b_{ik}a_{kj}.$$

Hence we find the matrix (is cols = M????)

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = \begin{pmatrix} c_{11} & \dots & c_{1j} & \dots & c_{1m} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \dots & \dots & \dots & c_{im} \\ \vdots & & \vdots & & \vdots \\ p^1 & \dots & c_{pj} & \dots & c_{pm} \end{pmatrix}.$$

$$\begin{array}{ccccc} V & \xrightarrow{T} & W & \xrightarrow{S} & U \\ \uparrow \Phi_A & & \uparrow \Phi_B & & \uparrow \Phi_C \\ K^n & \xrightarrow{T_{[T]_{\mathcal{B}}^{\mathcal{A}}}} & K^m & \xrightarrow{T_{[S]_{\mathcal{C}}^{\mathcal{B}}}} & K^p \\ & \searrow T_{[T]_{\mathcal{B}}^{\mathcal{A}} \cdot [S]_{\mathcal{C}}^{\mathcal{B}}} & & & \end{array}$$

We are looking for the unique matrix representing the linear map $T_{[T]_{\mathcal{B}}^{\mathcal{A}} \cdot [S]_{\mathcal{C}}^{\mathcal{B}}} := T_{[T]_{\mathcal{B}}^{\mathcal{A}}} \circ T_{[S]_{\mathcal{C}}^{\mathcal{B}}}$, where T_X just maps $v \rightarrow Xv$, via vector-matrix multiplication. We found them, they are given by

$$c_{ij} = \sum_{k=1}^m b_{ik}a_{kj},$$

Lecture 21

28.11.2025

17.a Matrices: Matrix multiplication

So let's try to see what happens in total entry notation

$$\begin{pmatrix} \vdots & & \vdots \\ b_{i1} & \dots & b_{ik} & \dots & b_{im} \\ \vdots & & \vdots \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{kj} \\ \vdots \\ a_{mj} \end{pmatrix} = \begin{pmatrix} \vdots \\ \dots & c_{ij} & \dots \\ \vdots \end{pmatrix}$$

Notice that we take the vector formed by the i -th row of the first matrix and multiply element-wise and sum with the j -th column of the second matrix, giving you a number which you put into the entry i, j of the resulting matrix. Let's look at this purely from the numerical-matrix-entry view.

Definition (Matrix multiplication (17.a.1)). Let $A \in M_{m \times n}(K)$ and $B \in M_{n \times p}(K)$. Their entries will be

$$A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}, \quad B = (b_{ij})_{1 \leq j \leq n, 1 \leq j \leq p}.$$

Define a new matrix $C \in M_{m \times p}(K)$, with entries $(c_{ij})_{1 \leq i \leq m, 1 \leq j \leq p}$, where

$$c_{ij} := a_{i1}b_{1j} + \dots + a_{ik}b_{kj} + \dots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj},$$

where the index k is highlighted for mnemonic purposes. We denote this new matrix C by $A \cdot B$ and call it the *multiplication* of A and B or the product of A and B .

Remark. Finally, we can truly understand why it makes sense to define matrix multiplication exactly this way. After all, if you had only seen the index by index definition of the product, there would be no real information hidden in there. But because we motivated this definition by looking at the coordinate representation of linear operators between two *arbitrary* finite dimensional vector spaces, we can say that we know where this comes from.

IMPORTANT — $A \cdot B$ is defined only when $\# \text{columns } A = \# \text{rows } B$.

Another way to describe $A \cdot B$:

$$(A) \begin{pmatrix} \left| \begin{array}{c} z_1 \\ \vdots \end{array} \right| & \left| \begin{array}{c} \vdots \\ \vdots \end{array} \right| & \left| \begin{array}{c} z_j \\ \vdots \end{array} \right| & \dots & \left| \begin{array}{c} z_p \\ \vdots \end{array} \right| \end{pmatrix} = \begin{pmatrix} \left| \begin{array}{c} Az_1 \\ \vdots \end{array} \right| & \left| \begin{array}{c} \vdots \\ \vdots \end{array} \right| & \left| \begin{array}{c} Az_j \\ \vdots \end{array} \right| & \dots & \left| \begin{array}{c} Az_p \\ \vdots \end{array} \right| \end{pmatrix}.$$

Example. $\begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} =$

$$= \begin{pmatrix} 3 \cdot 1 + 2 \cdot 0 + 1 \cdot 4 & 3 \cdot 2 + 2 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + 2 \cdot 4 & 1 \cdot 2 + 0 \cdot 1 + 2 \cdot 0 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 9 & 2 \end{pmatrix} = A \cdot B.$$

Example. $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} =$

$$= \begin{pmatrix} 1 \cdot 3 + 2 \cdot 1 & 1 \cdot 2 + 2 \cdot 0 & 1 \cdot 1 + 2 \cdot 2 \\ 0 \cdot 3 + 1 \cdot 1 & 0 \cdot 2 + 1 \cdot 0 & 0 \cdot 1 + 1 \cdot 2 \\ 4 \cdot 3 + 0 \cdot 1 & 4 \cdot 2 + 0 \cdot 0 & 4 \cdot 1 + 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 5 \\ 1 & 0 & 2 \\ 12 & 8 & 4 \end{pmatrix} = B \cdot A$$

Special Cases

1. Let $A \in M_{m \times n}$, $v \in K_{col}^n$. We defined (long time ago)

$$A \cdot v \in K_{col}^m.$$

We can view v as an $n \times 1$ matrix. Then

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \cdot \begin{pmatrix} | \\ v \\ | \end{pmatrix} = A \cdot v.$$

2. Let $B \in M_{n \times p}$, $v \in K_{row}^n$. We can define $v \cdot B \in M_{1 \times p} = K_{row}^p$.

$$\begin{pmatrix} - & v & - \end{pmatrix} \begin{pmatrix} B \\ \end{pmatrix} = \begin{pmatrix} - \end{pmatrix}.$$

So matrix multiplication can also be described like:

$$\text{If } \begin{pmatrix} - & v_1 & - \\ & \vdots & \\ - & v_n & - \end{pmatrix} \begin{pmatrix} B \\ \end{pmatrix} = \begin{pmatrix} - & v_1 \cdot B & - \\ & \vdots & \\ - & v_n \cdot B & - \end{pmatrix}.$$

3. Let $v = (a_1, \dots, a_n) \in K_{row}^n = M_{1 \times n}$ and $w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K_{col}^n = M_{n \times 1}$. Then

$$v \cdot w \in M_{1 \times 1} = K:$$

$$(a_1 \ \dots \ a_n) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a_1 \cdot b_1 + \dots + a_n \cdot b_n. \text{ scalar product}$$

CONCLUSION — $[S \circ T]_e^A = [S]_e^B \cdot [T]_B^A$.

NOTATION — Let I be an index set. For all $i, j \in I$, define

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j \end{cases}$$

(Kronecker delta).

Definition (Identity matrix). Define the identity matrix $I_n \in M_{n \times n}(K)$ as

$$I_n := (\delta_{ij})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Remark. Let V be a finite dimensional vector space, and $n = \dim V$. Let $\mathcal{B}$ be any basis for V . Then $[\text{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n$. ← check!

Proposition ((17.a)).

1. $\forall A \in M_{m \times n}, B \in M_{n \times p}, C \in M_{p \times q}$ we have:

$$(A \cdot B) \cdot C = A \cdot (B \cdot C) \quad (\text{associativity}).$$

Hence, there is a meaning to write $A \cdot B \cdot C$ without parentheses.

2. $\forall A, B \in M_{n \times n}, C \in M_{n \times p}, C' \in M_{q \times m}$ we have:

$$(A + B) \cdot C = A \cdot C + B \cdot C \quad (\text{left distributivity})$$

$$C' \cdot (A + B) = C' \cdot A + C' \cdot B \quad (\text{right distributivity}).$$

3. $\forall A \in M_{m \times n}$ we have:

$$I_m \cdot A = A, \quad I_n \cdot A = A \quad (\text{identity element}).$$

4. $\forall \alpha \in K, A \in M_{m \times n}(K), B \in M_{n \times p}(K)$ then

$$\alpha \cdot (A \cdot B) = (\alpha \cdot A) \cdot B = A \cdot (\alpha \cdot B) \quad (\text{compatibility with scalar multiplication}).$$

Proof. We need to cook!

1. Let $T_A : K^n \rightarrow K^m, T_B : K^p \rightarrow K^n, T_C : K^q \rightarrow K^p$ be the linear maps defined using the matrices A, B, C respectively. (i.e. $T_A(v) := A \cdot v$ etc.) $\forall \ell$, denote by $\mathcal{E}_\ell$ the standard basis for K^ℓ .

$$[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} = A, [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} = B, [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = C.$$

We have:

$$\begin{array}{ccccccc}
 (T_A \circ T_B) \circ T_C & = & T_A \circ (T_B \circ T_C) \\
 \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 (K^p \rightarrow K^m) & & (K^q \rightarrow K^p) & & (K^n \rightarrow K^m) & & (K^q \rightarrow K^n)
 \end{array}$$

Composition is $K^q \rightarrow K^m$.

$$\Rightarrow [(T_A \circ T_B) \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} = [T_A \circ (T_B \circ T_C)]_{\mathcal{E}_q}^{\mathcal{E}_m}.$$

By what we have done earlier:

$$\begin{aligned}
 [T_A \circ T_B]_{\mathcal{E}_p}^{\mathcal{E}_m} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} &= [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_n} \\
 ([T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n}) \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} &= [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot ([T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p}).
 \end{aligned}$$

2. If $T, S : V \rightarrow W$ are linear, B is a basis for V , and $\mathcal{E}$ is a basis for W , then

$$[T + S]_{\mathcal{E}}^B = [T]_{\mathcal{E}}^B + [S]_{\mathcal{E}}^B.$$

$$\Rightarrow (A + B) \cdot C = A \cdot (B \cdot C),$$

$$\begin{aligned}
 2) \quad \underbrace{[(T_A + T_B) \circ T_C]_{\mathcal{E}_M}^{\mathcal{E}_P}}_{(A+B)C} &= [T_A \circ T_C]_{\mathcal{E}_M}^{\mathcal{E}_P} + [T_B \circ T_C]_{\mathcal{E}_M}^{\mathcal{E}_P} \\
 &= [T_A]_{\mathcal{E}_M}^{\mathcal{E}_P} \cdot [T_C]_{\mathcal{E}_M}^{\mathcal{E}_P} + [T_B]_{\mathcal{E}_M}^{\mathcal{E}_P} \cdot [T_C]_{\mathcal{E}_M}^{\mathcal{E}_P} \\
 &= (A \cdot C) + (B \cdot C).
 \end{aligned}$$

The first identity in 2) – exc.

3.

$$\begin{aligned}
 I_m &= [\text{id}_{K^m}]_{\mathcal{E}_m}^{\mathcal{E}_m}, \\
 \text{id}_{K^m} T_A = T_A &\Rightarrow \underbrace{[\text{id}]_{\mathcal{E}_m}^{\mathcal{E}_m}}_{=I_m} \cdot A = A.
 \end{aligned}$$

□

17.b Matrices: Invertible matrix

Last time, we defined matrix multiplication as a way to compose linear maps in matrix form w.r.t. certain bases we picked. Then we defined the special identity matrix and started proving computation rules for matrices. Today we ask how we can invert the matrix, in the similar sense we can invert a map.

Definition (Invertible matrix). An $n \times n$ matrix $A \in M_{n \times n}$ is called invertible if there exists $B \in M_{n \times n}$ such that

$$A \cdot B = B \cdot A = I_n.$$

Remark. If $A \in M_{n \times n}$ is invertible, the inverse B is unique. Indeed, assume that $B, C \in M_{n \times n}$ and

$$A \cdot B = B \cdot A = I_n, \quad A \cdot C = C \cdot A = I_n,$$

then

$$B = B \cdot I_n = B \cdot (A \cdot C) = (B \cdot A) \cdot C = I_n \cdot C = C,$$

hence $B = C$ and the inverse is unique and we denote it $A^{-1} \in M_{n \times n}$.

Definition (General linear group). We denote the very special set

$$\text{GL}_n(K) := \{A \in M_{n \times n}(K) \mid A \text{ is invertible}\}.$$

Remark. Very clearly, $\text{GL}_n(K) \neq \emptyset$, because $I_n \in \text{GL}_n(K)$ and additionally, $0 \notin \text{GL}_n(K)$, hence $\text{GL}_n(K)$ is a proper subset of $M_{n \times n}(K)$. It is also immediately *so far from a subspace that you shouldn't even think about it*, as zero is not in there. It is a group though, which will be studied later in other subjects, not really linear algebra.

Proposition (Invertibility rules). Let $A, B \in \text{GL}_n(K)$. Then, $A \cdot B \in \text{GL}_n(K)$ and also $A^{-1} \in \text{GL}_n(K)$ and moreover

$$(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}, \quad (A^{-1})^{-1} = A.$$

Proof. We just need to verify this, which is pretty easy. Consider that

$$(B^{-1} \cdot A^{-1}) \cdot (A \cdot B) = B^{-1} \cdot (A^{-1} \cdot A) \cdot B = B^{-1} \cdot I_n \cdot B = B^{-1} \cdot B = I_n$$

and from the other side

$$(A \cdot B) \cdot (B^{-1} \cdot A^{-1}) = A \cdot (B \cdot B^{-1}) \cdot A^{-1} = A \cdot I_n \cdot A^{-1} = A \cdot A^{-1} = I_n,$$

hence the product is in $\text{GL}_n(K)$ and it is as claimed.

For the inverse,

$$A \cdot A^{-1} = I_n, \quad A^{-1} \cdot A = I_n \Rightarrow (A^{-1})^{-1} = A,$$

by definition of inverse applied to A^{-1} . □

Let's see how we can apply this to solve LSEs.

Corollary. Let $A \in \text{GL}_n(K)$. Then $\forall b \in K^n$, the linear system of equations $A \cdot x = b$ has a unique solution, which is $x = A^{-1}b$.

Proof. If $Ax = b$, then $A^{-1}Ax = A^{-1}b$, so $I_n x = x = A^{-1}b$. Vice versa, if $x = A^{-1}b$, then $Ax = b$. $\square$

Proposition (Criterion for invertibility). Let $A \in \text{M}_{n \times n}(K)$. A is invertible if and only if

$$T_A : K^n \rightarrow K^n : v \mapsto Av$$

is an isomorphism. Moreover

$$(T_A)^{-1} = T_{A^{-1}}$$

Proof. Assume first that A is invertible. We claim that

$$T_{A^{-1}} \circ T_A = \text{id}_{K^n}$$

and the other way around

$$T_A \circ T_{A^{-1}} = \text{id}_{K^n}.$$

This will show that T_A is an isomorphism and that $(T_A)^{-1} = T_{A^{-1}}$. Indeed,

$$\forall v \in K^n : (T_{A^{-1}} \circ T_A)(v) = T_{A^{-1}}(Av) = A^{-1} \cdot (Av) = (A^{-1} \cdot A)v = v,$$

similarly for the other direction.

Conversely, assume that $T_A : K^n \rightarrow K^n$ is an isomorphism. Denote

$$S := (T_A)^{-1}.$$

Now S is a linear map $K^n \rightarrow K^n$. Recall that for any such linear map $S : K^n \rightarrow K^n$, there exists a unique matrix $M \in \text{M}_{n \times n}(K)$ such that

$$S(v) = Mv \Rightarrow S = T_M.$$

So, $\exists B \in \text{M}_{n \times n}$ such that $S = T_B$.

Now, $\forall v \in K^n$, we have

$$v = (S \circ T_A)(v) = (T_B \circ T_A)(v) = BAv \Rightarrow (BA)v = v.$$

Apply this to the standard basis $v = e_1, v = e_2, \dots, v = e_n$ and we deduce that $B \cdot A = I_n$. In a similar way, one shows $A \cdot B = I_n$. It follows that A is invertible. $\square$

Proposition. Let $A \in \text{M}_{m \times n}$ and $B \in \text{M}_{n \times p}$. Then,

$$(AB)^\top = B^\top A^\top.$$

Moreover, if $A \in \text{GL}_n(K)$, then $A^\top \in \text{GL}_n(K)$ and

$$(A^\top)^{-1} = (A^{-1})^\top.$$

Proof. This is left as a very simple exercise. Use the component definition and verify the candidate for the inverse using the first rule. $\square$

Definition (Triangular and diagonal matrices). A matrix $A \in M_{n \times n}$ with $A = (a_{ij})_{1 \leq i, j \leq n}$ is called *upper triangular* if

$$\forall i > j : a_{ij} = 0.$$

The matrix A is called *lower triangular* if

$$\forall i < j : a_{ij} = 0.$$

The matrix A is called *diagonal* if

$$\forall i \neq j : a_{ij} = 0.$$

Lemma. Let $A, B \in M_{n \times n}$ be two upper triangular matrices (resp. lower triangular or diagonal) matrices, then the product $AB \in M_{n \times n}$ is the same.

Proof. Done in Problem Sheet 11 $\square$

22

17.b.1 Change of basis

This is allegedly a very important process in the application of applied math (*aka physics*) and other applications. What we want to study is: *How does $[T]_{\mathcal{C}}^{\mathcal{B}}$ depend on the choices of $\mathcal{B}$ and $\mathcal{C}$?*

Corollary. Let V, W be finite dimensional vector spaces. Let $n = \dim V, m = \dim W$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V and $\mathcal{C}, \mathcal{C}'$ two bases for W . Let $T : V \rightarrow W$ be a linear map. Then

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\text{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Furthermore,

$$[\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \in \text{GL}_n(K), [\text{id}_W]_{\mathcal{C}}^{\mathcal{C}'} \in \text{GL}_m(K)$$

and

$$[\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = \left([\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'}\right)^{-1} \text{ and } [\text{id}_W]_{\mathcal{C}'}^{\mathcal{C}} = \left([\text{id}_W]_{\mathcal{C}}^{\mathcal{C}'}\right)^{-1}$$

22

Proof. So, the map T is trivially

$$T = \text{id}_W \circ T \circ \text{id}_V = \text{id}_W \circ (T \circ \text{id}_V).$$

Hence, because of definitions,

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\text{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T \circ \text{id}_V]_{\mathcal{C}}^{\mathcal{B}'} = [\text{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

For the second identity, consider that $\text{id}_V = \text{id}_V \circ \text{id}_V$. Hence,

$$[\text{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \cdot [\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}},$$

the same for id_W . □

Lecture 23: Equivalence classes of matrices

Last time, at the end, we started inspecting what happens to the matrix of linear map when you change the bases. What we found was that there are two matrices generated by the identity map that are completely independent of the map T itself, that are only dependent on the bases themselves.

Definition (Transition matrix). Let V be a finite dimensional vector space and $n = \dim V$. Let $\mathcal{B}, \mathcal{B}'$ be two bases in V . Then the $n \times n$ matrix

$$[\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \in \text{GL}_n(K)$$

is called the change of basis matrix, or transition matrix, between $\mathcal{B}$ and $\mathcal{B}'$.

If we write $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$, then by definition,

$$[\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \dots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

This makes it especially easy to find these matrices in for example $\mathbb{R}^n$.

Proposition. Let $T : V \rightarrow W$ be an isomorphism between two finite dimensional vector spaces and let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W , respectively (this forces $n := \dim V = \dim W$). Then

$$[T]_{\mathcal{C}}^{\mathcal{B}} \in \text{GL}_n(K)$$

and moreover,

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}.$$

23

Proof. We have very simply that

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} = [T^{-1} \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n,$$

and similarly,

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = [T \circ T^{-1}]_{\mathcal{C}}^{\mathcal{C}} = [\text{id}_V]_{\mathcal{C}}^{\mathcal{C}} = I_n.$$

□

Let us now consider some endomorphism $T : V \rightarrow V$, with $\mathcal{B}, \mathcal{B}'$ two bases for V . We want to ask what is the relation between $[T]_{\mathcal{B}'}^{\mathcal{B}'}$ and $[T]_{\mathcal{B}}^{\mathcal{B}}$. It turns out to be pretty easy.

Corollary. So let $T : V \rightarrow V$ be an endomorphism and $\mathcal{B}, \mathcal{B}'$ two bases for V . We have

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = ([\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

Proof. This is an immediate consequence of expanding $T = \text{id}_V \circ T \circ \text{id}_V$ and applying the last theorem. □

17.c Matrices: Equivalent and similar matrices

We want to be able to compare different matrices and hope to discover some sort of classification for them.

Definition (Equivalent and similar matrices). Let $A, B \in M_{m \times n}(K)$. We say that A and B are *equivalent* if $\exists P \in \text{GL}_m(K)$ and $\exists Q \in \text{GL}_n(K)$ such that

$$B = PAQ.$$

This time let $A, B \in M_{n \times n}(K)$, then we say that A and B are *similar*, if $\exists P \in \text{GL}_n(K)$ such that

$$B = P^{-1}AP.$$

Remark. Equivalence between matrices as defined here is an equivalence relation on $M_{m \times n}(K)$. Furthermore, $A, B \in M_{m \times n}(K)$ are equivalent if and only if $\exists$ vector spaces V and W with $\dim V = n, \dim W = m$ and two bases $\mathcal{B}, \mathcal{B}'$ for V and two bases $\mathcal{C}, \mathcal{C}'$ for W and a linear map $T : V \rightarrow W$ such that

$$A = [T]_{\mathcal{C}}^{\mathcal{B}}, \quad B = [T]_{\mathcal{C}'}^{\mathcal{B}'}.$$

Similarity is also an equivalence relation on $M_{n \times n}(K)$. Again, $M, N \in M_{n \times n}(K)$ are equivalent if and only if $\exists$ n -dimensional vector space V and $T : V \rightarrow V$ and

two bases $\mathcal{B}$ and $\mathcal{B}'$ for V such that

$$M = [T]_{\mathcal{B}}, \quad N = [T]_{\mathcal{B}'}.$$

The proofs are left as an exercise, they should follow from the proofs we already did.

Proposition (Identity-like representation of a map). Let V, W be finite dimensional vector spaces over K of dimension $n = \dim V$ and $m = \dim W$. Let $T : V \rightarrow W$ be a linear map such that $\text{rank}(T) = \dim \text{Im}(T) = r$. Then there exist a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W such that

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Proof. So let $\ell = \dim \text{Ker}(T)$. By the rank theorem $r = n - \ell$. Let $v_1, \dots, v_\ell$ be a basis of $\text{Ker}(T)$. Extend this to a basis $v_1, \dots, v_\ell, v_{\ell+1}, \dots, v_n$ of V . We'll take $\mathcal{B}$ to be

$$\mathcal{B} = (v_{\ell+1}, \dots, v_n, v_1, \dots, v_\ell).$$

We have seen in the proof of the rank theorem (and in a problem sheet, probably) that

$$Tv_{\ell+1}, \dots, Tv_n$$

form a basis for $\text{Im}(T)$. Each v_i works obvious. Then we define

$$w_i = Tv_{\ell+i} \quad 1 \leq i \leq r.$$

Note that $w_1 = Tv_{\ell+1}, \dots, w_r = Tv_n$. Now extend $w_1, \dots, w_r, w_{r+1}, \dots, w_m$ to a basis of W ,

$$\mathcal{C} = (w_1, \dots, w_r, w_{r+1}, \dots, w_m).$$

But now

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | & & | \\ [Tv_{\ell+1}]_{\mathcal{C}} & \dots & [Tv_n]_{\mathcal{C}} & [Tv_1]_{\mathcal{C}} & \dots & [Tv_\ell]_{\mathcal{C}} \\ | & & | & & | \end{pmatrix}$$

We know that $Tv_1 = \dots = Tv_\ell = 0$ and $Tv_{\ell+i} = w_i$ for all $1 \leq i \leq r$. The matrix hence takes on the desired form. $\square$

Corollary (Equivalent to identity-like matrix). Take a matrix $A \in M_{m \times n}(K)$. Then $\exists P \in \text{GL}_m(K)$ and $\exists Q \in \text{GL}_n(K)$ such that

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}$$

where $r = \text{col-rank}(A)$. In particular, $\text{col-rank}(A) \leq \min \{m, n\}$.

Proof. Consider the map $T_A : K^n \rightarrow K^m : v \rightarrow Av$. Then we have by definition of the column space that

$$\text{ColS}(A) = \text{Im}(T_A),$$

so $r = \text{rank}(T_A)$. By the above proposition, there exist bases $\mathcal{B}$ and $\mathcal{C}$ for K^n and K^m respectively, such that

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

But we also know that

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = [\text{id}_{K^m}]_{\mathcal{C}}^{\mathcal{E}_m} \cdot [T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot [\text{id}_W]_{\mathcal{E}_n}^{\mathcal{B}},$$

so we arrived at the desired decomposition, as $[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} = A$, if we let $\mathcal{E}_m$ and $\mathcal{E}_n$ be the respective standard bases. The inequality holds because $\text{col-rank}(A) = \dim \text{Im}(T_A) = n - \dim \text{Ker}(T) \leq n$ and $\text{col-rank}(A) \leq \dim K^m = m$. $\square$

Corollary (Column rank equality is equivalence). Let $A, B \in M_{m \times n}(K)$. Then $A \sim B$ (equivalent) if and only if $\text{col-rank}(A) = \text{col-rank}(B)$.

Proof. Assume $r := \text{col-rank}(A) = \text{col-rank}(B)$. Then

$$A \sim \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

but also

$$B \sim \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

Hence by transitivity, $A \sim B$.

Now assume that $A \sim B$. Then by definition,

$$\exists P \in \text{GL}_m(K), Q \in \text{GL}_n(K) \text{ s.t. } B = PAQ.$$

This however implies that

$$T_B = T_P \circ T_A \circ T_Q.$$

But therefore,

$$\text{Im}(T_B) = T_B(K^n) = T_P(T_A(T_Q(K^n))) = T_P(T_A(K^m)),$$

since Q is invertible, hence T_Q is an isomorphism, so we have

$$T_P(T_A(K^m)) = T_P(\text{Im}(T_A)),$$

23

which has the same dimension as $\text{Im}(T_A)$, as T_P is an isomorphism as well. So in total,

$$\dim \text{Im}(T_B) = \dim \text{Im}(T_A) \Rightarrow \text{col-rank}(B) = \text{col-rank}(A).$$

□

Lecture 24

10.12.2025

Preparations for the proof

Lemma (1 (1.7.c.2)). Let $T : V \rightarrow W$ be a lin. map, and let $S : W \xrightarrow{\cong} U$ be an iso, and $L : P \xrightarrow{\cong} V$ also an iso. Then:

1. $\text{rank}(S \circ T) = \text{rank}(T)$.
2. $\text{rank}(T \circ L) = \text{rank}(T)$.

Remind. — $T : V \rightarrow W$, $\text{rank}(T) := \dim(\text{Im}(T))$.

Proof. 1. $\text{rank}(S \circ T) = \dim S(T(V))$, (*)
Note that $S|_{T(V)} : T(V) \rightarrow S(T(V))$ is a linear map which is also an isomorphism. So (*) = $\dim T(V) = \text{rank}(T)$.

2. $\text{rank}(T \circ L) = \dim T(L(P)) = \dim T(V) = \text{rank}(T)$. (because $L(P) = V$)

exercise — generalize to S injective, L is surjective. □

24

Lemma (2 (17.c.3)). Let $A \in M_{n \times n}$. Then A is invertible iff $T_A : K^n \rightarrow K^n$ is an isomorphism. Moreover,

$$(T_A)^{-1} = T_{A^{-1}},$$

where

$$T_A(v) := Av.$$

Lemma (3 (17.c.4)). Let $A, B \in M_{m \times n}(K)$. Let $P \in GL_m(K)$, $Q \in GL_n(K)$. Then:

1. $\text{col-rank}(A) = \text{col-rank}(PAQ)$
2. $\text{row-rank}(A) = \text{row-rank}(PAQ)$

In other words, if $B \sim A$,

$$\text{col-rank}(A) = \text{col-rank}(B), \quad \text{row-rank}(A) = \text{row-rank}(B).$$

Proof. 1) Let $B := PAQ$. This means

$$T_B = T_P \circ T_A \circ T_Q.$$

But since T_P and T_Q are isomorphisms by Lemma 2, we know that

$$\text{col-rank}(B) = \text{rank}(T_B) = \text{rank}(T_A) = \text{col-rank}(A).$$

2) We have $B^\top = Q^\top A^\top P^\top$. Since P and Q are invertible, so are $Q^\top$ and $P^\top$, hence by 1,

$$\text{row-rank}(B) = \text{col-rank}(B^\top) = \text{col-rank}(A^\top) = \text{row-rank}(A)$$

finishing the Lemma. $\square$

Theorem. Let $A \in M_{m \times n}(K)$. Then $\text{col-rank}(A) = \text{row-rank}(A)$. We define with this just the $\text{rank}(A)$.

Proof. We learned last time that there exist $P \in GL_m(K)$, $Q \in GL_n(K)$ such that

$$PAQ = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix} := B$$

for some $r \geq 0$ and denote by B this matrix

$$\begin{array}{c} \begin{array}{c} \uparrow r \\ \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & 1 & \vdots \\ \hline 0 & \vdots & 0 \end{pmatrix} \end{array} \end{array} \begin{array}{c} \begin{array}{c} \leftarrow r \\ \begin{array}{c|c} 0 & 0 \\ \hline 0 & 0 \end{array} \end{array} \end{array} =: B$$

Clearly

$$\text{row-rank}(B) = \text{col-rank}(B) = r.$$

But $\text{row-rank}(A) = \text{row-rank}(B) = r$ Lemma 17.c.4 $\text{col-rank}(A) = \text{col-rank}(B) = r$

$$\Rightarrow \text{row-rank}(A) = \text{col-rank}(A).$$

$\square$

From now on: $\text{rank}(A) := \text{row-rank}(A) = \text{col-rank}(A)$.

Remark. Suppose A' is a row-reduced echelon matrix which is row-equivalent to A . We know that $\text{RowS}(A) = \text{RowS}(A')$. So $\text{rank}(A) = \text{rank}(A')$. But $\text{rank}(A') = \#$ pivots in A .

17.d Matrices: Back to linear equations

Let $A \in M_{m \times n}(K)$, $b \in K^m$. Consider the system of linear equations $A \cdot x = b$.

Denote by $\text{Sol}(A, b) := \{x \in K^n : A \cdot x = b\}$.

set of solutions. We know that for $b = 0$, $\text{Sol}(A, 0)$ is a vector space (a subspace of K^n).

Claim. — $\dim \text{Sol}(A, 0) = n - \text{rank}(A)$. This is b.c.:

$\dim \text{Sol}(A, 0) = \dim \ker(T_A) = n - \dim \text{Im}(T_A) = n - \text{rank}(A)$ (by rank theorem).

$A \in M_{m \times n}, T_A : K^n \rightarrow K^m$.

$\text{Im}(T_A) = \text{ColS}(A)$.

$\text{rank}(T_A) = \text{rank}(A)$.

Let A' be a row-reduced echelon matrix which is row-equivalent to A .

$$\underbrace{\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & \dots & * \\ & & & & 1 & * & \dots & * \\ & & & & & 1 & \dots & * \\ & 0 & & & & & \ddots & \\ & & & & & & & 1 & \dots & * \end{pmatrix}}_{A'} \begin{pmatrix} x_1 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{pmatrix}$$

$\# \text{ pivots}(A') = \text{rank}(A') = \text{rank}(A)$.

$n - \text{rank}(A') = \# \text{ free variables} = \dim \text{Sol}(A', 0) = \dim \text{Sol}(A, 0)$.

$\text{Sol}(A, 0) = \text{Sol}(A', 0)$.

Denote by $x_{i_1}, \dots, x_{i_\ell}$ the free variables $1 \leq i_1 < \dots < i_\ell \leq n, \ell = n - \text{rank}(A)$. Denote by $x_{j_1}, \dots, x_{j_r}, 1 \leq j_1 < \dots < j_r \leq n, r = \text{rank}(A)$ the pivot variables. For each choice of values for the free variables $x_{i_1}, \dots, x_{i_\ell}$, the pivot variables are uniquely determined.

In other words we have a map

$$\Phi : K^\ell \rightarrow \text{Sol}(A, 0).$$

$\Phi(\lambda_1, \dots, \lambda_\ell) :=$ the unique solution of $A \cdot x = 0$ or $A' \cdot x = 0$ with

$$x_{i_1} = \lambda_1, \dots, x_{i_\ell} = \lambda_\ell.$$

Proposition (17.d.1). The map Φ is linear, and moreover it is an isomorphism.

Proof. Linearity of Φ . Consider row $\#k$ of A' and of $A' \cdot x = 0$. If $1 \leq k \leq r$, it looks like

$$x_{j_k} + (\text{linear combin. of free variables that come after } j_k) = 0$$

$$\text{So, } \underset{\substack{\uparrow \\ \text{pivot} \\ \text{var } \#k}}{x_{j_k}} = - \sum_{\{q | 1 \leq q \leq l, i_q > j_k\}} \underset{\substack{\uparrow \\ \text{free} \\ \text{vars,}}}{a'_{k,i_q}} \cdot \underset{\substack{\uparrow \\ \text{free} \\ \text{vars,}}}{x_{i_q}} \left(\begin{array}{c} \text{the *} \\ \text{coeffs} \\ \text{in row} \\ \#k \text{ of } A'. \end{array} \right)$$

$\Rightarrow x_{j_1}, \dots, x_{j_r}$ (the pivots) depend linearly on the free variables $x_{i_1}, \dots, x_{i_\ell}$. This shows Φ is linear. We will show that Φ is an isomorphism. We will 1st show $\ker(\Phi) = \{0\}$.

If $\Phi(\lambda_1, \dots, \lambda_\ell) = 0$, then by definition $\lambda_1 = \dots = \lambda_\ell = 0$.

But $\Phi : K^\ell \rightarrow \text{Sol}(A, 0)$, and $\dim \text{Sol}(A, 0) = n - \text{rank}(A) = n - \# \text{pivots}(A') = n - \# \text{free variables} = \ell$. Since Φ is injective, Φ is an isomorphism.

$$\Phi^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}.$$

$\text{Sol}(A, 0) =$ the values of the free variables. □

24

Corollary (17.d.2). Let $b \in K^m$, $A \in M_{m \times n}$. Then:

1. $\text{Sol}(A, b) \neq \emptyset$ iff $b \in \text{Col}(A)$
2. If $\text{Sol}(A, b) \neq \emptyset$, and $y \in \text{Sol}(A, b)$, then

$$\text{Sol}(A, b) = y + \text{Sol}(A, 0) := \{y + x : x \in \text{Sol}(A, 0)\}.$$

exercise with hints. 1. If $A = \left(\begin{array}{c|cc} & \dots & \\ v_1 & \dots & v_n \\ & \dots & \end{array} \right)$, $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, then $A \cdot x = x_1 \cdot$

$$\begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \dots + x_n \cdot \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}.$$

2. If $x \in \text{Sol}(A, 0)$, then

$$A(y + x) = Ay + Ax = b.$$

If $z \in \text{Sol}(A, b)$, then $x := z - y \in \text{Sol}(A, 0)$. □

Proposition ((17.d.3)). Let $A \in M_{m \times n}$, $b \in k^m$. Then the following statements are equiv.:

1. $\text{rank}(A \mid b) = \text{rank}(A)$
2. The syst. of eq. $Ax = b$ has a solut.

Proof. Write $A = (v_1 \ \dots \ v_n)$.

$$\begin{aligned}
 (1) &\Leftrightarrow \text{col-rank}(A \mid b) = \text{col-rank}(A) \\
 &\Leftrightarrow \text{Sp}(v_1, \dots, v_n, b) = \text{Sp}(v_1, \dots, v_n) \\
 &\Leftrightarrow b \in \text{Sp}(v_1, \dots, v_n) \Leftrightarrow b \in \text{colS}(A) \\
 &\Leftrightarrow \exists \text{ solut. to } Ax = b.
 \end{aligned}$$

□

24

Proposition (17.d.4). Let $A \in M_{m \times n}$ and $b \in K^m$. The th following are equivalent:

1. $\text{rank}(A) = \text{rank}(A \mid b) = n$.
2. The system $Ax = b$ has a unique solution.

Proof.

(Outline of proof.

$$A = \left(\begin{array}{c|c|c|c} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{array} \right).$$

$$\text{rank}(A) = n \iff \dim \text{Sp}(v_1, \dots, v_n) = n \iff v_1, \dots, v_n \text{ form a basis for } \text{Cols}(A).$$

□

Lecture 25

12.12.2025

17.e Matrices: Elementary row operations

25

Main point. These operations can be written by matrix multiplication. *Notation.* Let $n \in \mathbb{Z}_{\geq 1}$, $1 \leq i \leq n$, $1 \leq j \leq n$. Denote by $E_{i,j} \in M_{n \times n}(K)$ the matrix that has all entries 0, except (i,j)-where the entry is 1.

$$E_{i,j} = \left(\begin{array}{ccc|ccc} 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array} \right),$$

We will define now 3 other matrices: **Type I.** Let $1 \leq i \leq n$, $1 \leq j \leq n$, $i \neq j$, and let $\alpha \in K$. Define $Q_{i,j}(\alpha)$ to be the matrix obtained from I_n after applying $R_i + \alpha R_j \rightarrow R_i$.

We have $Q_{i,j}(\alpha) = I_n + \alpha E_{i,j}$.

$$\begin{aligned} Q_{i,j}(\alpha) &:= I_n + \alpha E_{i,j} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} + \alpha \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \alpha & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \end{aligned}$$

Type II. Let $1 \leq i \leq n$, $1 \leq j \leq n$. Define $P_{i,j}$ to be the matrix obtained from I_n by exchanging row $\#i$ and row $\#j$. ($R_i \longleftrightarrow R_j$)

$$P_{i,j} = \left(\begin{array}{ccc|ccc} \ddots & & & \vdots & \vdots & \\ & \ddots & & \vdots & \vdots & \\ & & \ddots & \vdots & \vdots & \\ & & & \vdots & \vdots & \\ \hline \cdots & \cdots & \cdots & 0 & 1 & \cdots & \cdots \\ \hline & & & \vdots & \vdots & \ddots & \\ \cdots & \cdots & \cdots & 1 & 0 & \cdots & \cdots \\ \hline & & & \vdots & \vdots & & \ddots & \\ & & & \vdots & \vdots & & & \ddots \end{array} \right), \quad (i \neq j).$$

Type III. Let $1 \leq i \leq n$, $0 \neq \alpha \in K$. Denote by $S_i(\alpha)$ the matrix obtained from I_n by

applying $\alpha R_i \rightarrow R_i$.

$$S_i(\alpha) = \left(\begin{array}{ccc|ccc} & & & \text{col } i & & \\ 1 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ 0 & \ddots & & \vdots & & & \vdots \\ \vdots & & 1 & 0 & & & 0 \\ 0 & \cdots & 0 & \alpha & 0 & \cdots & 0 \\ \hline 0 & & & 0 & 1 & & \vdots \\ \vdots & & & \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 & 1 \end{array} \right) \xleftarrow{\text{row } i}$$

Example. - In $M_{4 \times 4}(K)$ $Q_{1,3}(\alpha) = \begin{pmatrix} 1 & 0 & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $Q_{3,1}(\beta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$P_{1,2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $P_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$ $S_{3,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

25

Lemma (17.e.1). Let $A \in M_{n \times p}(K)$.

1. Multiplying A from the left by $Q_{i,j}(\alpha)$ results in applying the row operation $R_i + \alpha R_j \rightarrow R_i$ to A .
2. Multiplying A from the left by $P_{i,j}$ gives the application of $R_i \leftrightarrow R_j$ on A .
3. Multiplying A from the left by $S_i(\alpha)$ results in applying $\alpha R_i \rightarrow R_i$ to A .

outline Can be done by explicit straightforward calculation.

$$q_{ij}(\alpha) = \begin{pmatrix} \ddots & & & \\ & 1 & \alpha & \\ & 0 & 1 & \\ & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix}$$

$$Q_{ij}(\alpha) = \underline{I}_n + \alpha E_{ij}$$

$$Q_{ij}(\alpha) \cdot A = (\underline{I}_n + \alpha E_{ij}) \cdot A$$

$$= A + \alpha E_{ij} \cdot A$$

$$E_{ij}A = \begin{pmatrix} 0 \\ \vdots \\ \text{row } j \text{ of } A \\ \vdots \\ 0 \end{pmatrix} \quad (\text{this appears in row } i)$$

$\Rightarrow Q_{ij}(\alpha)A$ is A but with row $i \leftarrow \text{row } i + \alpha \cdot (\text{row } j \text{ of } A)$.

||| check details...

□

Elementary column operations.

1. ' Multiplying A from the right by $Q_{i,j}(\alpha)$ results the column operation $A \xleftarrow{C_j + \alpha C_i \rightarrow C_j} A \cdot Q_{i,j}(\alpha)$.
2. ' Multiplying A from the right by $P_{i,j}$ results in $C_i \leftrightarrow C_j$.

$$A \xleftarrow{C_i \leftrightarrow C_j} A \cdot P_{i,j}.$$

3. ' Multiplying A from the right by $S_i(\alpha)$ results in $\alpha C_i \rightarrow C_i$.

$$A \xleftarrow{\alpha C_i \rightarrow C_i} A \cdot S_i(\alpha).$$

To prove that 1'-3' just the transposed matrices.

$$\text{e.g. } (A \cdot Q_{i,j}(\alpha))^T = (*)Q_{i,j}(\alpha)^T \cdot A^T.$$

$$, \text{ but } Q_{i,j}(\alpha)^T = Q_{j,i}(\alpha).$$

$$\text{So } (*) = Q_{j,i}(\alpha) \cdot A^T$$

$$= \text{the result of applying } R_j + \alpha R_i \rightarrow R_j \text{ to } A^T.$$

$$\Rightarrow A \cdot Q_{i,j}(\alpha) = (*)^T \text{ is the result of applying } C_j + \alpha C_i \rightarrow C_j.$$

ect. (exc. complete the proof).

Lemma (17.e.2). Every elementary matrix is invertible and the inverse of each of them is also an elementary matrix. In fact:

$$(Q_{i,j}(\alpha))^{-1} = Q_{i,j}(-\alpha), \quad (P_{i,j})^{-1} = P_{i,j}, \quad (S_i(\alpha))^{-1} = S_i(\alpha^{-1}).$$

outline of the proof. $Q_{i,j}(-\alpha) \cdot Q_{i,j}(\alpha) =$ applying $R_i + (-\alpha)R_j$ to the matrix $Q_{i,j}(\alpha) \leftarrow$ (apply $R_i + \alpha R_j \rightarrow R_i$ to I_n) = Applying ist $(R_i + \alpha R_j \rightarrow R_i)$ to I_n and then $R_i + (-\alpha)R_j \rightarrow R_i$ to the previous result = do nothing to $I_n = I_n$.

Similarly one shows $Q_{i,j}(\alpha) \cdot Q_{i,j}(-\alpha) = I_n$. *exercise* — complete the proof (i.e. check for $P_{i,j}$ and $S_i(\alpha)$). $\square$

Theorem (17.e.3). $\forall A \in M_{m \times n}(K)$, $\exists k \geq 1$, elementary matrices $T_1, \dots, T_k$ such that

$$T_k \cdots T_{k-1} \cdots T_1 \cdot A = I_n.$$

Furthermore we have $A = T_1^{-1} \cdots T_k^{-1}$, and $A^{-1} = T_k \cdots T_1$.

Proof. We proved that $\forall A \in M_{m \times n}(K)$ can be brought to row-reduced echelon matrix A' by applying a sequence of elementary row operations. Since A is invertible, $\text{rank}(A) = n$. Now $\text{rank}(A') = \text{rank}(A) = n$.

$\Rightarrow A'$ has n pivots.

$\Rightarrow A' = I_n$.

But A' is obtained from A by applying a sequence of elementary row operations, hence $\exists$ elementary matrices $T_1, \dots, T_k$ such that

$$T_k (\cdots (T_2 (T_1 A)) \cdots) = A' = I_n.$$

The rest follows by straightforward calculation. $\square$

Theorem (17.e.4). Let $A \in M_{n \times n}(K)$. Place A together with I_n on the right to A ($A \mid I_n$). Then A is invertible iff the matrix $(A \mid I_n)$ can be brought by a sequence of elementary row operations to the form $(I_n \mid B)$ for some $B \in M_{n \times n}(K)$. In this case, $B = A^{-1}$.

Lemma (17.e.5). Let B, C, D be matrices $n \times n$. Then

$$B \cdot (C \mid D) = (B \cdot C \mid B \cdot D).$$

Proof — Exercise (by direct calculation.)

of Theorem 17.e.4. Assume A is invertible. By Theorem 17.e.3, $\exists$ elementary matrices $T_1, \dots, T_k$ such that

$$T_k \cdots T_{k-1} \cdots T_1 \cdot A = I_n.$$

By Lemma 17.e.5,

$$T_k \cdots T_{k-1} \cdots T_1 \cdot (A \mid I_n) = (I_n \mid \underbrace{T_k \cdots T_1}_{\text{denote by } B}).$$

By Theorem 17.e.3, $A^{-1} = B$. But $T_k \cdots T_1 (A \mid I_n)$ is precisely the matrix obtained from $(A \mid I_n)$ after applying a sequence of k elementary row operations. Conversely, assume $(A \mid I_n)$ can be brought to $(I_n \mid B)$ by a sequence of elementary row

operations. $\Rightarrow A$ itself can be brought to I_n by the same sequence of elementary row operations.

$\Rightarrow A$ is row-equivalent to I_n

$\Rightarrow A$ is invertible

Exercise — show that $B = A^{-1}$. □

25

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_{2 \times 2}(\mathbb{Q})$.

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{R_2 - 3R_1 \rightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right).$$

$$\xrightarrow{-\frac{1}{2}R_2 \rightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right) \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right).$$

So $A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

V, W vector spaces $\Rightarrow \text{Hom}(V, W)$ is also a vector space. If V & W are finite-dimensional, then $\text{Hom}(V, W)$ is also finite-dimensional and $\dim \text{Hom}(V, W) = \dim V \cdot \dim W$.

Lecture 26

17.12.2025

18.a New vector spaces out of old ones: $\text{Hom}(V, W)$

We want to learn about new vector spaces that naturally arise from other vector spaces we have studied this far. We start with $\text{Hom}(V, W)$ for vector spaces V and W over a common field K .

26

Proposition (Hom is a vector space). Given V, W vector spaces over K , $\text{Hom}(V, W)$ has the structure of a vector space over K as follows:

- $T_1, T_2 \in \text{Hom}(V, W)$, we define $(T_1 + T_2) : V \rightarrow W$ as $(T_1 + T_2)(v) = T_1(v) + T_2(v) \forall v \in V$.
- $\forall T \in \text{Hom}(V, W), \forall \alpha \in K$, we define $(\alpha T)(v) = \alpha(T(v)) \forall v \in V$.

Proof. We 1st claim that $T_1 + T_2$ is a linear map. Indeed, $\forall a \in K, v \in V$ we have

$$\begin{aligned}(T_1 + T_2)(av) &= T_1(av) + T_2(av) \\ &= aT_1(v) + aT_2(v) \\ &= a(T_1(v) + T_2(v)) \\ &= a(T_1 + T_2)(v).\end{aligned}$$

Let $v_1, v_2 \in V$. We have $(T_1 + T_2)(v_1 + v_2) = T_1(v_1 + v_2) + T_2(v_1 + v_2) = T_1(v_1) + T_1(v_2) + T_2(v_1) + T_2(v_2) = (T_1 + T_2)(v_1) + (T_1 + T_2)(v_2)$. This shows $T_1 + T_2 : V \rightarrow W$ is linear. Next one shows that αT is also a linear map (exc.) Now one needs to check that the operations

$$(T_1, T_2) \mapsto T_1 + T_2, \quad (\alpha, T) \mapsto \alpha \cdot T$$

satisfy all the axioms of a vector space.

The neutral element, $0 \in \text{Hom}(V, W)$, is the zero map

$$0 : V \rightarrow W, \quad v \mapsto 0.$$

We denote it by 0 .

One needs to verify that

$$0 + T = T, \quad \alpha \cdot 0 = 0,$$

etc.

This completes the proof. $\square$

Theorem. Let V, W be finite dimensional vector spaces and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W . Define

$$\Psi_{\mathcal{C}}^{\mathcal{B}} : \text{Hom}(V, W) \rightarrow M_{m \times n}(K)$$

by the value

$$\begin{aligned}\Psi_{\mathcal{C}}^{\mathcal{B}}(T) &:= [T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K). \\ &\begin{pmatrix} \left| \begin{array}{c} T(v_1)_{\mathcal{C}} \\ T(v_2)_{\mathcal{C}} \\ \vdots \end{array} \right| & \left| \begin{array}{c} T(v_2)_{\mathcal{C}} \\ \vdots \end{array} \right| & \cdots & \left| \begin{array}{c} T(v_n)_{\mathcal{C}} \\ \vdots \end{array} \right| \end{pmatrix}.\end{aligned}$$

Then, the map $\Phi_{\mathcal{C}}^{\mathcal{B}}$ is linear and an isomorphism.

Proof. Let $T \in \text{Hom}(V, W)$, and

$$M := [T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K).$$

Recall: we have

$$T_M = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}.$$

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_B^{-1} \uparrow & & \downarrow \Phi_C \\ K^n & \xrightarrow{T_M} & K^m \end{array}$$

where $\Phi_B(v) := [v]_B \quad \forall v \in V$,

$\Phi_C(w) := [w]_C \quad \forall w \in W$.

$$\Phi_B^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \cdots + a_n v_n,$$

$$\Phi_C^{-1} \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \cdots + b_m w_m.$$

We first claim that Ψ_C^B is bijective. To do this, define

$$\Theta : M_{m \times n}(K) \rightarrow \text{Hom}(V, W).$$

For all $A \in M_{m \times n}(K)$,

$$\Theta(A) = \Phi_C^{-1} \circ T_A \circ \Phi_B.$$

$$\begin{array}{ccc} V & \xrightarrow{\Theta(A)} & W \\ \Phi_B \downarrow & & \uparrow \Phi_C^{-1} \\ K^n & \xrightarrow{T_A} & K^m \end{array}$$

Consider $\Psi_C^B \circ \Theta : M_{m \times n}(K) \rightarrow M_{m \times n}(K)$. Let $A \in M_{m \times n}(K)$. Then

$$\begin{aligned} (\Psi_C^B \circ \Theta)(A) &= \Psi_C^B(\Phi_C^{-1} \circ T_A \circ \Phi_B) \\ &= \begin{pmatrix} | & & | \\ [(\Phi_C^{-1} \circ T_A \circ \Phi_B)(v_1)]_C & \cdots & [(\Phi_C^{-1} \circ T_A \circ \Phi_B)(v_n)]_C \\ | & & | \end{pmatrix}. \end{aligned}$$

□

Proof. Let $A \in M_{m \times n}(K)$. Then

$$\begin{aligned} \begin{pmatrix} | & & | \\ T_A \circ \Phi_B(v_1) & \cdots & T_A \circ \Phi_B(v_n) \\ | & & | \end{pmatrix} &= \begin{pmatrix} | & & | \\ T_A e_1 & \cdots & T_A e_n \\ | & & | \end{pmatrix} \quad (\text{since } \Phi_B(v_j) = e_j, j = 1, \dots, n) \\ &= A. \end{aligned}$$

Therefore,

$$\Rightarrow \Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta = \text{id}_{M_{m \times n}(K)}.$$

Now let $T \in \text{Hom}(V, W)$ and set $M := \Psi_{\mathcal{C}}^{\mathcal{B}}(T)$. Consider $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}}(T) = \Theta(M)$:

$$\Theta(M) = \Phi_{\mathcal{C}}^{-1} \circ T_M \circ \Phi_{\mathcal{B}} = T \quad (\text{by the 1st diagram we had earlier}).$$

Hence,

$$\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}} = \text{id}_{\text{Hom}(V, W)}.$$

It remains to show that $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is linear. Indeed, let $\alpha \in K$ and $T \in \text{Hom}(V, W)$. Then

$$\begin{aligned} \Psi_{\mathcal{C}}^{\mathcal{B}}(\alpha T) &= \begin{pmatrix} [(\alpha T)(v_1)]_{\mathcal{C}} & \cdots & [(\alpha T)(v_n)]_{\mathcal{C}} \\ | & & | \\ \alpha [T(v_1)]_{\mathcal{C}} & \cdots & \alpha [T(v_n)]_{\mathcal{C}} \\ | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \\ \vdots & & \vdots \end{pmatrix} \\ &= \begin{pmatrix} \alpha [T(v_1)]_{\mathcal{C}} & \cdots & \alpha [T(v_n)]_{\mathcal{C}} \\ | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \\ \vdots & & \vdots \end{pmatrix} \\ &= \alpha \begin{pmatrix} [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \\ \vdots & & \vdots \end{pmatrix} \\ &= \alpha \Psi_{\mathcal{C}}^{\mathcal{B}}(T). \end{aligned}$$

Similarly, $\Psi_{\mathcal{C}}^{\mathcal{B}}(T_1 + T_2) = \Psi_{\mathcal{C}}^{\mathcal{B}}(T_1) + \Psi_{\mathcal{C}}^{\mathcal{B}}(T_2)$. (exc.) □

26

Corollary (Dimension of $\text{Hom}(V, W)$). If V, W are finite dimensional then so is $\text{Hom}(V, W)$ and

$$\dim \text{Hom}(V, W) = \dim V \cdot \dim W.$$

Proof. Let $n = \dim V, m = \dim W$. By the previous theorem,

$$\text{Hom}(V, W) \cong M_{m \times n}(K).$$

Since $M_{m \times n}(K)$ is finite dimensional, so is $\text{Hom}(V, W)$ and $\dim \text{Hom}(V, W) = M_{m \times n}(K) = m \cdot n$. □

Definition (Ring). A ring $(R, +, \cdot)$ is defined similarly to a field, but we do *not* require that $a \cdot b = b \cdot a$, and we do not require that $\forall a \in R \setminus \{0\} \exists a^{-1}$. The element 1 is called the unit or the unity of R .

Example.

1. Every field K is a commutative ring.
2. $R = \mathbb{Z}$ is a commutative ring.
3. $M_{n \times n}(K)$ is a ring. Multiplication: $A, B \in M_{n \times n}(K)$, then $A \cdot B$ is the

multiplication of A & B as matrices. I_n is the unity of this ring. We get a non-commutative ring.

$$A \cdot (B + C) = A \cdot B + A \cdot C,$$

$$A \cdot (B + C) = (A \cdot B) \cdot C..$$

Definition (Homomorphism of rings). Let R, S be rings. A *homomorphism of rings* is a map $\Psi : R \rightarrow S$ such that $\forall r_1, r_2 \in R$,

$$\Psi(r_1 + r_2) = \Psi(r_1) + \Psi(r_2),$$

$$\Psi(r_1 \cdot r_2) = \Psi(r_1) \cdot \Psi(r_2),$$

$$\Psi(1) = 1,$$

$$\Psi(0) = 0.$$

Corollary (18.a.4). Let V be a vector space over a field K . Then $\text{End}(V) := \text{Hom}(V, V)$ is a ring (with a unit), in general non-commutative with the following operations:

+ operation: — the same as before for $\text{Hom}(V, V)$

· operation: $T_1 \cdot T_2 := T_1 \circ T_2 \forall T_1, T_2 \in \text{End}(V)$. The unity is id_V . Moreover, if V is finite-dimensional, and $B = (v_1, \dots, v_n)$ is a basis of V , then $\Phi_B^B : \text{End}(V) \rightarrow M_{n \times n}(K)$ is an isomorphism of rings.

Proof. $\forall T_1, T_2, T_3 \in \text{Hom}(V, V)$,

$$T_1 \circ (T_2 \circ T_3) = (T_1 \circ T_2) \circ T_3 \Rightarrow (\text{multiplication in } \text{End}(V) \text{ is associative}).$$

$$T \circ \text{id}_V = \text{id}_V \circ T = T \forall T \in \text{End}(V) \Rightarrow \text{id}_V \text{ is the unity.}$$

Also $\forall T, T_1, T_2$

$$T \cdot (T_1 + T_2) = T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2 = T \cdot T_1 + T \cdot T_2.$$

etc... (check all the other axioms of a ring)

For the second part, assume that V is finite dimensional. Let $T_1, T_2 \in \text{End}(V)$. We already know that

$$\Psi_B^B(T_1 + T_2) = \Psi_B^B(T_1) + \Psi_B^B(T_2)$$

and that Ψ_B^B is bijective. For the second requirement, notice that

$$\Psi_B^B(T_1 \circ T_2) = [T_1 \circ T_2]_B^B = [T_1]_B^B \cdot [T_2]_B^B = \Psi_B^B(T_1) \cdot \Psi_B^B(T_2),$$

hence this is a ring isomorphism. $\square$

Lecture 27: Last lecture of Linear Algebra 1

19.12.2025

18.a.1 Dual Space

Special case of $\text{Hom}(V, W)$ when $W = K$.

Definition. Let V be a vector space over the field K . The *dual space* of V is defined

$$V^* := \text{Hom}(V, K).$$

The elements of V^* are called *linear maps* $\ell : V \rightarrow K$ and are sometimes called *functionals* on V .

Corollary. Let V be a finite-dimensional vector space K . Then V^* is also finite-dimensional and $\dim V^* = \dim V$.

Proof. $\dim V^* = \dim \text{Hom}(V, K) = \dim V \cdot \dim K = \dim V$ □

Example. Take $V = K_{col}^n$. We will identify V^* with K_{row}^n . If $\ell \in K_{row}^n$, then ℓ defines a functional $f_\ell : K_{col}^n \rightarrow K$ by

$$f_\ell(v) = \ell \cdot v \in M_{1 \times n}(K) \cong K.$$

The map

$$D : K_{row}^n \rightarrow (K_{col}^n)^*, \quad D(\ell) := f_\ell.$$

Exercise — D is linear and is an isomorphism.

Let V be a finite-dimensional vector space over K , and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V . Define n functionals $v_1^*, \dots, v_n^* \in V^*$ as follows: $v_i^* : V \rightarrow K$ is the unique linear map such that

$$v_i^*(v_j) = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

In other words, $v_i^*(v_i) = 1$ and $v_i^*(v_j) = 0 \forall j \neq i$. Since $\mathcal{B}$ is a basis for V , this determines a functional $v_i^* : V \rightarrow K$ uniquely.

Lemma (18.a.7). The elements $v_1^*, \dots, v_n^* \in V^*$ form a basis for V^* . We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$. It is called the *dual basis* of $\mathcal{B}$. Moreover, $\forall \ell \in V^*$,

$$\ell = \sum_{i=1}^n \ell(v_i) v_i^*.$$

$$\text{i.e. } \forall v \in V, \ell(v) = \sum_{i=1}^n \ell(v_i) v_i^*(v).$$

Proof. Let $\ell \in V^*$. We claim that

$$\ell = \sum_{i=1}^n \ell(v_i) v_i^*, \quad (1)$$

id est

$$\forall v \in V, \quad \ell(v) = \sum_{i=1}^n \ell(v_i) v_i^*(v). \quad (2)$$

LHS of (1) is a linear map $V \rightarrow K$, and so is also RHS of (1). So to show LHS = RHS it is enough to check (2) on the elements of the basis of V , exempli gratia with the basis $\mathcal{B} = (v_1, \dots, v_n)$. So we will check (2) for $v = v_1, \dots, v = v_n$.

Let $1 \leq k \leq n$, and substitute $v = v_k$ in (2):

$$\text{RHS} = \sum_{i=1}^n \ell(v_i) \underbrace{v_i^*(v_k)}_{=\delta_{ik}} = \ell(v_k) \cdot \underbrace{\delta_{kk}}_{=1} = \ell(v_k).$$

So (2) holds for $v = v_k$.

The formula we have just proved, shows also that $v_1^*, \dots, v_n^*$ span V^* . But $\dim V^* = n$, hence $v_1^*, \dots, v_n^*$ form a basis of V^* . $\square$

27

Question — Let V be a finite-dimensional vector space over K . Let $\mathcal{B}, \mathcal{C}$ be two bases of V . Then we also get two bases $\mathcal{B}^*, \mathcal{C}^*$ for V^* . What is the relationship between $[\text{id}_V]_{\mathcal{C}^*}^{\mathcal{B}^*}$ and $[\text{id}_V]_{\mathcal{B}}^{\mathcal{C}}$?

Proposition (18.a.6).

$$\begin{aligned} [\text{id}_V]_{\mathcal{B}^*}^{\mathcal{C}^*} &= ([\text{id}_V]_{\mathcal{C}}^{\mathcal{B}})^{\top}, \\ [\text{id}_V]_{\mathcal{C}^*}^{\mathcal{B}^*} &= \left(([\text{id}_V]_{\mathcal{B}}^{\mathcal{C}})^{\top} \right)^{-1}. \end{aligned}$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_n)$. Then

$$A := [\text{id}_V]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} | & & | & & | \\ [v_1]_{\mathcal{C}} & \cdots & [v_i]_{\mathcal{C}} & \cdots & [v_n]_{\mathcal{C}} \\ | & & | & & | \end{pmatrix}.$$

Write also $A = (a_{ij})$. We have

$$v_j = \sum_{k=1}^n a_{kj} w_k. \quad (*)$$

Let $\mathcal{C}^* = (w_1^*, \dots, w_n^*)$ be the dual basis of $\mathcal{C}$. By the previous lemma (18.a.7), if

we take $\ell = w_j^*$, then

$$w_j^* = \sum_{i=1}^n w_j^*(v_i) v_i^* \underset{\text{by } (*)}{=} \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k \right) v_i^* \underset{\substack{\text{the only term that is } \neq 0 \\ \text{in } w_j^* \left(\sum_{k=1}^n a_{ki} w_k \right) \text{ is when } k=j}}{=} \sum_{i=1}^n a_{ji} v_i^*.$$

$$\Rightarrow [\text{id}_V]^*_{\mathcal{B}^*} = (a_{ji}) = A^\top.$$

□

18.a.2 The dual map

Definition (Lemma and Definition 18.a.9). Let V, W be vector spaces over a field K , and let $T : V \rightarrow W$ be a linear map. Define a new map $T^* : W^* \rightarrow V^*$ by

$$T^*(\ell) = \ell \circ T, \quad \forall \ell \in W^*.$$

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow \ell \circ T & \downarrow \ell \\ & & K \end{array}$$

The map $T^* : W^* \rightarrow V^*$ is called the *dual map* to T . Note that $\ell \circ T$ is indeed in V^* , since $\ell \circ T$ is linear (because T and ℓ are linear). Thus $T^* : W^* \rightarrow V^*$ is a linear map.

Proof. Let $\ell \in W^*$ and $\alpha \in V^*$. We have that $T^*(\alpha\ell)$ is the map $(\alpha\ell) \circ T$. So, for all $v \in V$,

$$\begin{aligned} T^*(\alpha\ell)(v) &= (\alpha\ell)(T(v)) \\ &= \alpha \cdot \ell(T(v)) \\ &= \alpha(\ell \circ T)(v) \\ &= \alpha T^*(\ell)(v). \end{aligned}$$

$$\Rightarrow T^*(\alpha\ell) = \alpha T^*(\ell).$$

Let $\ell_1, \ell_2 \in W^*$, and let $v \in V$.

$$\begin{aligned} T^*(\ell_1 + \ell_2)(v) &= (\ell_1 + \ell_2)(T(v)) \\ &= \ell_1(T(v)) + \ell_2(T(v)) \\ &= T^*(\ell_1)(v) + T^*(\ell_2)(v) \\ &= (T^*(\ell_1) + T^*(\ell_2))(v). \end{aligned}$$

So

$$T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2).$$

□

Proposition. Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps. Consider $V^* \leftarrow W^* : T^*$, $W^* \leftarrow U^* : S^*$. Then

$$(S \circ T)^* = T^* \circ S^*.$$

Proof. The proof is left as an exercise for the reader. $\square$

Exercise — Let V, W be finite-dimensional vector spaces.

1. $(O : V \rightarrow W)^* = (O : W^* \rightarrow V^*)$
2. $(\text{id}_V : V \rightarrow V)^* = (\text{id}_{V^*} : V^* \rightarrow V^*)$

Lemma. Let $S : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. Let $\mathcal{B}$ be a basis of V and $\mathcal{C}$ a basis of W . Consider $S^* : W^* \rightarrow V^*$, and the dual bases $\mathcal{B}^*, \mathcal{C}^*$. Then:

$$[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^{\top}.$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n)$, $\mathcal{C} = (w_1, \dots, w_m)$. $A = [S]_{\mathcal{C}}^{\mathcal{B}}$, $A = (a_{ij})$. $Sv_j = \sum_{i=1}^m a_{ij}w_i$. Now, for $w_j^* \in \mathcal{C}^*$, we have

$$\begin{aligned} S^*(w_j^*) &= w_j^* \circ S \underset{\text{by lemma 18.a.7}}{=} \sum_{i=1}^m (w_j^* \circ S)(v_i)v_i^* \\ &= \sum_{i=1}^m w_j^* \left(\sum_{k=1}^m a_{ki}w_k \right) v_i^* = \sum_{i=1}^m a_{ji}v_i^*. \end{aligned}$$

$\Rightarrow$ The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^*$ are given $\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix} = \text{row}\#j \text{ of } A$, made into a column. $\square$

Linear Algebra II

Semester 2

Lecture 28: Semester 2 Kickoff

18 Feb 2026

18.a New vector spaces out of old ones: Semester 2 Setup

Recap

This recap reviews key facts from Semester 1 that we use throughout this lecture.

Dual space and functionals. Let V be a vector space over K . We define its dual space by

$$V^* := \text{Hom}(V, K).$$

An element $\ell \in V^*$ is a linear map, also called a functional, of the form

$$\ell : V \rightarrow K.$$

Dual basis and coordinate expansion. To relate coordinates in V and V^* , let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V , and let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis for V^* . Each map $v_i^* : V \rightarrow K$, for $i = 1, \dots, n$, is characterized by

$$v_i^*(v_j) = \delta_{ij} \quad \boxed{\mathcal{B}^* \text{ is useful!}}$$

Therefore, for every $\ell \in V^*$, we can write

$$\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*.$$

Dual map. We now recall the dual-map construction. Let V, W be vector spaces over K , and let $T : V \rightarrow W$ be a linear map. Then there exists a linear map $T^* : W^* \rightarrow V^*$, defined by

$$T^*(\ell) := \ell \circ T \quad \text{for every } \ell \in W^*.$$

This construction is summarized by the following diagram:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow \ell \circ T & \downarrow \ell \\ & & K \end{array}$$

18.a.1 Reflexivity

Let V be a vector space. Define $V^{**} := (V^*)^*$. V^{**} is sometimes called the *bidual* of V .

Theorem. Let V be a finite-dimensional vector space. Then $\exists$ a canonical isomorphism $\tau : V \rightarrow V^{**}$ given by:

$$\forall v \in V, \quad \tau(v) : V^* \rightarrow K, \quad \tau(v)(\ell) := \ell(v).$$

In other words:

$$V \ni v \mapsto (V^* \ni \ell \mapsto \ell(v)).$$

Proof. Linearity. We claim that τ is linear. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned} \tau(\alpha v) &= (V^* \ni \ell \mapsto \ell(\alpha v)) \\ &= (V^* \ni \ell \mapsto \alpha \ell(v) \in K) \\ &= \alpha (V^* \ni \ell \mapsto \ell(v) \in K) \\ &= \alpha \tau(v). \end{aligned}$$

Exercise — Show that $\tau(v' + v'') = \tau(v') + \tau(v'') \forall v', v'' \in V$.

Injectivity. We now claim τ is injective. We will show that $\ker \tau = \{0\}$. Indeed, assume $\tau(v) = 0$. $\Rightarrow \ell(v) = 0 \forall \ell \in V^*$. (*) choose a basis $\mathcal{B} = (v_1, \dots, v_n)$ for V . Write $v = \sum_{i=1}^n c_i v_i$ with $c_i \in K$. Apply (*) to $\ell = v_i^*$. We get

$$\begin{aligned} v_k^*(c_1 v_1 + \dots + c_n v_n) &= c_k = 0 \quad (\text{kernel: } v_k^*(v) = \delta_{ki}) \\ &= 0 \quad \text{this holds } \forall k \\ \Rightarrow c_k &= 0 \quad \forall k \Rightarrow v = 0. \end{aligned}$$

This proves $\ker \tau = \{0\} \Rightarrow \tau$ is injective. Recall now $\dim V = \dim V^* = \dim V^{**}$ since $\tau : V \rightarrow V^{**}$ is injective, it must also be an isomorphism. $\square$

28

18.a.2 naturality

The map $\tau : V \rightarrow V^{**}$ can be defined for every vect. sp. V . To emphas. dependence on V , we'll write here τ_V instead of τ .

Let V, W be vect. sp. over K . Then $\forall$ lin. map $T : V \rightarrow W$ we have the following commut. diagram:

$$\boxed{\text{Naturality}} \longrightarrow \begin{array}{ccc} V & \xrightarrow{\tau_V} & V^{**} \\ T \downarrow & & \downarrow (T^*)^* \\ W & \xrightarrow{\tau_W} & W^{**} \end{array}$$

Exc. : prove this.

Exercise:

Let V be a *f.dim.* Vect. sp. over K .
 Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V .
 Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis,
 and $\mathcal{B}^{**} = (v_1^{**}, \dots, v_n^{**})$ be the dual basis of $\mathcal{B}^*$.

18.b New vector spaces out of old ones: Direct Sums

Proposition (18.b.1). Let V, W be vector spaces over K . Then the cartesian product $V \times W$ becomes a vector space over K if we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2) \quad \forall v_1, v_2 \in V, w_1, w_2 \in W.$
- $\alpha(v, w) := (\alpha v, \alpha w) \quad \forall \alpha \in K, v, w \in V \times W.$
- $0_{V \times W} := (0_V, 0_W).$

Proof. Just go over the axioms of vector spaces one by one. Exercise! □

$V \times W$ is denoted by $V \oplus W$ and is called the *direct sum* of V and W . Its elements are denoted by v, w (or sometimes (ugly!) $(v \oplus w)$).

Remark. $V \oplus W$ connects with the following canonical maps:

$$\left. \begin{array}{ll} 1) i_V : V \longrightarrow V \oplus W, & i_V(v) := (v, 0), \\ 2) i_W : W \longrightarrow V \oplus W, & i_W(w) := (0, w) \end{array} \right\} \text{ embeddings, or inclusions.}$$

$$\left. \begin{array}{ll} 3) p_V : V \oplus W \longrightarrow V, & p_V(v, w) := v, \\ 4) p_W : V \oplus W \longrightarrow W, & p_W(v, w) := w \end{array} \right\} \text{ projections.}$$

Using i_V & i_W we can view V as a subspace of $V \oplus W$. $V \subseteq V \oplus W$ and also $W \subseteq V \oplus W$.

28

Proposition (18.b.2). Let V be a vector space over K and $U \subseteq V$ be a subspace. Let $U' \subseteq V$ be another subspace which is a *complement* of U (recall: this means $U + U' = V$ and $U \cap U' = \{0\}$). Then the map:

$$\varphi : U \oplus U' \rightarrow V, \quad \text{defined by } \varphi(u, u') := u + u'$$

is linear and is an isomorphism.

Proof. Linearity of φ : exercise.

Injectivity of φ : Suppose $\varphi(u, u') = 0$. Then

$$u + u' = 0 \implies u = -u'.$$

Now $u \in U_b$ and $-u' \in U'_b$, hence $u \in U_b \cap U'_b$. Since U'_b is a complement of U_b , we have $U_b \cap U'_b = \{0\}$. Therefore $u = 0$, and then $u' = 0$. So $(u, u') = (0, 0)$, i.e. $\ker(\varphi) = \{0\}$.

Surjectivity of φ : By assumption, $U_b + U'_b = V$. Let $v \in V$. Then there exist $u \in U_b$ and $u' \in U'_b$ such that $v = u + u'$. Hence

$$\varphi(u, u') = u + u' = v.$$

Thus φ is surjective. □

Generalization: to arbitrarily many spaces.

Let $U_1, \dots, U_n$ be finitely many vector spaces over K . We can define

$$U_1 \oplus \dots \oplus U_n := U_1 \times \dots \times U_n$$

similarly to what we have done before.

Let $\{U_i\}_{i \in I}$ be a family of vector spaces over K , parameterized by $i \in I$.

$$\prod_{i \in I} U_i := \left\{ t : I \rightarrow \bigcup_{i \in I} U_i \mid t(i) \in U_i, \forall i \in I \right\}$$

We can endow this with the structure of a vector space over K . Define also $\bigoplus_{i \in I} U_i$:

$$\bigoplus_{i \in I} U_i := \left\{ t \in \prod_{i \in I} U_i \mid t(i) \neq 0 \text{ only for finitely many } i \in I \right\}$$

If I is finite, then $\prod_{i \in I} U_i = \bigoplus_{i \in I} U_i$. But consider the example $I = \mathbb{Z}_{\geq 0}$:

$$U_i = K \quad \forall i \in I$$

$$\prod_{i \in I} U_i = \{(a_1, a_2, \dots, a_n, \dots) \mid a_i \in K, \forall i\}$$

$$\bigoplus_{i \in I} U_i = \{(a_1, a_2, \dots, a_n, \dots) \mid a_i \in K, \forall i, \text{ and } a_j = 0 \text{ for all large enough } j\}$$

Lecture 29: Quotient spaces

23 Feb 2026

Recap

Dual space. For a vector space V , its dual space is V^* .

Bidual. The bidual is $V^{**} = (V^*)^*$.

Canonical map. There exists a canonical map $V \rightarrow V^{**}$. If V is finite-dimensional, then this map is an isomorphism.

Direct sum. The direct sum of vector spaces is denoted $U \oplus V$.

18.c New vector spaces out of old ones: Quotient spaces

Let V be a vector space over a field K and let $U \subseteq V$ be a subspace. We will define a new vector space V/U together with a linear map $\pi : V \rightarrow V/U$ such that π is surjective and $\ker(\pi) = U$.

We begin by defining an equivalence relation on the set V . Let $v_1, v_2 \in V$. We declare $v_1 \sim v_2$ if $v_1 - v_2 \in U$.

Claim. $\sim$ is an equivalence relation.

Proof. *Reflexivity.* For every $v \in V$, we have $v - v = 0 \in U$, hence $v \sim v$.

Symmetry. If $v_1 \sim v_2$, then $v_1 - v_2 \in U$. Therefore

$$v_2 - v_1 = -(v_1 - v_2) \in U,$$

so $v_2 \sim v_1$.

Transitivity. If $v_1 \sim v_2$ and $v_2 \sim v_3$, then

$$v_1 - v_3 = (v_1 - v_2) + (v_2 - v_3) \in U,$$

so $v_1 \sim v_3$ (because $U \subseteq V$ is a subspace). $\square$

Denote by $[v]$ the equivalence class of $v \in V$. One can also think of $[v]$ as

$$[v] = v + U = \{v + u \mid u \in U\}.$$

Definition (18.c.1). Let V & U be as above. We define

$$V/U := V/\sim := \text{the set of equivalence classes of } \sim.$$

Elements of V/U are $[v]$, where $v \in V$. V/U is called the V quotient U or V modulo U . We will turn the set V/U into vector space over K . Addition: Let $x_1, x_2 \in V/U$. Pick representatives $v_1, v_2 \in V$ of x_1, x_2 respectively, i.e. $x_1 = [v_1]$ and $x_2 = [v_2]$. Define

$$x_1 + x_2 := [v_1] + [v_2] \in V/U.$$

Multiplication by scalars: Let $\alpha \in K$ and $x \in V/U$. Pick $v \in V$, with $[v] = x$. Define

$$\alpha x := [\alpha v] \in V/U.$$

Zero vector: Define $0_{V/U} := [0_V] \in V/U$.

Proposition (18.c.2). The operations above are well-defined. Moreover, they give V/U the structure of a vector space over K .

Proof. *Well definedness.*

Multiplication by a scalar. Suppose $v, v' \in V$ represent the same class $x \in V/U$, i.e. $v \sim v'$. Let $\alpha \in K$. Since $v - v' \in U$, we get

$$\alpha v - \alpha v' = \alpha(v - v') \in U.$$

Hence $\alpha v \sim \alpha v'$, so $[\alpha v] = [\alpha v']$. Therefore αx is well-defined.

Addition. Let $v_1, v'_1, v_2, v'_2 \in V$ such that $[v_1] = [v'_1] = x$ and $[v_2] = [v'_2] = x'$. We

need to show $[v_1 + v_2] = [v'_1 + v'_2]$. Since $v_1 \sim v'_1$ and $v_2 \sim v'_2$,

$$(v_1 + v_2) - (v'_1 + v'_2) = (v_1 - v'_1) + (v_2 - v'_2) \in U.$$

So $v_1 + v_2 \sim v'_1 + v'_2$, hence $[v_1 + v_2] = [v'_1 + v'_2]$. Therefore $x + x'$ is well-defined. $\square$

Define now a map $\pi : V \rightarrow V/U$ by $\pi(v) = [v] \forall v \in V$.

Proposition (18.c.3). π is a linear map. Moreover, $\text{Im}(\pi) = V/U$ and $\ker(\pi) = U$.

Proof. *Linearity.* Let $v_1, v_2 \in V$. Then

$$\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2).$$

Let $\alpha \in K$ and $v \in V$. Then

$$\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v).$$

Surjectivity. Let $x \in V/U$. By definition, there exists $v \in V$ such that $x = [v]$. Hence $x = \pi(v)$, so $\text{Im}(\pi) = V/U$.

Kernel. If $v \in \ker(\pi)$, then $\pi(v) = 0_{V/U}$, so $[v] = [0_V]$. Thus $v \sim 0_V$, i.e. $v - 0_V = v \in U$. Therefore $\ker(\pi) \subseteq U$.

Conversely, let $u \in U$. Then $u \sim 0_V$, so $[u] = [0_V]$, hence $\pi(u) = 0_{V/U}$. Therefore $u \in \ker(\pi)$, and $U \subseteq \ker(\pi)$.

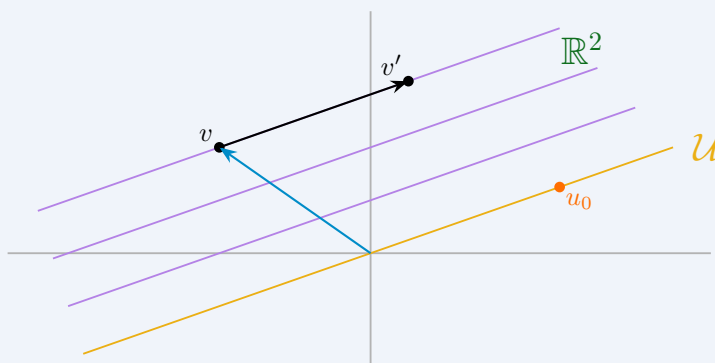
So $\ker(\pi) = U$. $\square$

$$V/\{0\} = V/\{0\} = V, \quad V/V = \{0\}.$$

Example. A geometric way to think about V/U is the following. Let $K = \mathbb{R}$ and $V = \mathbb{R}^2$. Choose $0 \neq u_0 \in \mathbb{R}^2$, and define

$$U := \text{span}(u_0),$$

which is the line through 0 and u_0 . The elements of $\mathbb{R}^2/U$ are in one-to-one



correspondence with lines $\ell \subseteq \mathbb{R}^2$ that are parallel to U :

$$[v] \in \mathbb{R}^2/U \longleftrightarrow v + U.$$

Proposition (18.c.4). Suppose V is finite-dimensional. Then V/U is also finite-dimensional, and

$$\dim(V/U) = \dim V - \dim U.$$

Proof. Since $\pi : V \rightarrow V/U$ is surjective, V/U is finite-dimensional. Indeed, if $T : W' \rightarrow W''$ is a surjective linear map and W' is finite-dimensional, then W'' is finite-dimensional: if $w'_1, \dots, w'_m$ is a basis of W' , then $T(w'_1), \dots, T(w'_m)$ spans $\text{Im}(T) = W''$.

We now apply the rank theorem to π :

$$\dim V = \dim \ker(\pi) + \dim \text{Im}(\pi) = \dim U + \dim(V/U).$$

□

Theorem (The isomorphism theorem 18.c.5). Let V, W be vector spaces over K , and let $T : V \rightarrow W$ be a linear map. Define a new map

$$\tilde{T} : V/\ker(T) \rightarrow \text{Im}(T)$$

as follows. Let $x \in V/\ker(T)$. Choose $v \in V$ with $[v] = x$, and define

$$\tilde{T}(x) := T(v).$$

Then:

1. $\tilde{T}$ is well-defined and linear.
2. $\tilde{T} : V/\ker(T) \rightarrow \text{Im}(T)$ is an isomorphism.

3. The following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \pi \downarrow & \textcircled{\subset} & \uparrow \iota \text{ (inclusion)} \\ V/\ker(T) & \xrightarrow{\tilde{T}} & \text{Im}(T) \end{array}$$

Proof. Well-definedness. Suppose $x = [v] = [v']$ in $V/\ker(T)$. We must show that $T(v) = T(v')$. Since $[v] = [v']$, we have $v - v' \in \ker(T)$. Therefore

$$T(v - v') = 0,$$

which implies $T(v) = T(v')$. Hence $\tilde{T}$ is well-defined.

Linearity. Let $\alpha \in K$ and $v \in V$. Then

$$\tilde{T}(\alpha[v]) = \tilde{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \tilde{T}([v]).$$

Let $v_1, v_2 \in V$. Then

$$\tilde{T}([v_1] + [v_2]) = \tilde{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \tilde{T}([v_1]) + \tilde{T}([v_2]).$$

So $\tilde{T}$ is linear.

Injectivity. Let $x \in \ker(\tilde{T})$, and write $x = [v]$ for some $v \in V$. Then $\tilde{T}(x) = T(v) = 0$, so $v \in \ker(T)$. Hence $x = [v] = [0]$, and therefore $\tilde{T}$ is injective.

Surjectivity onto $\text{Im}(T)$. Let $w \in \text{Im}(T)$. Then there exists $v \in V$ with $T(v) = w$. Thus

$$\tilde{T}([v]) = T(v) = w,$$

so $w \in \text{Im}(\tilde{T})$. Therefore $\text{Im}(T) \subseteq \text{Im}(\tilde{T})$. On the other hand, by definition of $\tilde{T}$, we have $\text{Im}(\tilde{T}) \subseteq \text{Im}(T)$. Hence $\text{Im}(\tilde{T}) = \text{Im}(T)$, and $\tilde{T}$ is surjective onto $\text{Im}(T)$.

Commutativity of the diagram. For every $v \in V$,

$$(\iota \circ \tilde{T} \circ \pi)(v) = \iota(\tilde{T}([v])) = \iota(T(v)) = T(v).$$

So the diagram commutes. $\square$

29

Theorem (18.c.6). Let V be a vector space over $\mathbb{K}$ and let $U \subseteq V$ be a subspace. The quotient V/U has the following universal property: $\exists$ a space W and every linear map $T : V \rightarrow W$ with $T(U) = 0$ (i.e. $\ker(T) \supseteq U$) $\exists!$ a linear map $\tilde{T} : V/U \rightarrow W$ such that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \pi \downarrow & \textcircled{\subset} & \nearrow T' (\exists!) \\ V/U & & \end{array}$$

commutes. Moreover

$$\ker(T') = \ker(T)/U.$$

Jargon: Every lin. map $T : V \rightarrow W$ that sends U to 0 factors through V/U , i.e., $\overline{T} = T' \circ \pi$

Lecture 30: Quotient spaces & complements

25 Feb 2026

Recap

Last time: quotient spaces. If $U \subseteq V$ is a subspace, then V/U is a quotient space.

Last time: induced isomorphism. If $T : V \rightarrow W$ is linear, then it induces an isomorphism

$$\overline{T} : V/\ker(T) \xrightarrow{\cong} \text{Im}(T), \quad [v] \mapsto T(v).$$

30

Theorem (18.c.6). Let V be a vector space over K and let $U \subseteq V$ be a subspace. The quotient V/U has the following universal property: $\exists$ a space W and every linear map $T : V \rightarrow W$ with $T(U) = 0$ (i.e. $\ker(T) \supseteq U$) $\exists!$ a linear map $T' : V/U \rightarrow W$ such that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \pi \downarrow & \textcircled{\circlearrowleft} & \nearrow T' (\exists!) \\ V/U & & \end{array}$$

commutes. Moreover

$$\ker(T') = \ker(T)/U.$$

Jargon: Every lin. map $T : V \rightarrow W$ that sends U to 0 factors through V/U , i.e., $T = T' \circ \pi$ T descends to $T' : V/U \rightarrow W$.

Proof. Define $T' : V/U \rightarrow W$ in the following way. Let $x \in V/U$. Choose $v \in V$ with $[v] = x$, and define

$$T'(x) := T(v).$$

Exercise 1.: Show that T' is well-defined, i.e. if $[v] = [v']$, then $T(v) = T(v')$.

Exercise 2.: Show that T' is linear.

We claim that the diagram commutes. Indeed, $\forall v \in V$,

$$T'(\pi(v)) = T'([v]) = T(v).$$

Uniqueness of T' : Suppose that $T = T' \circ \pi = T'' \circ \pi$. We will show that $T' = T''$. Indeed, let $x \in V/U$. Choose $v \in V$ with $[v] = x$. Then

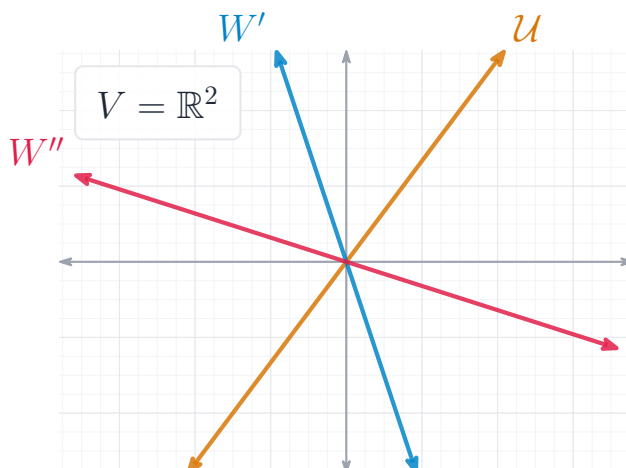
$$\begin{aligned} \Rightarrow T'(v) &= T'(\pi(v)) \text{ and } T''(v) = T''(\pi(v)) \\ &\Rightarrow T'(x) = T''(x). \end{aligned}$$

□

18.c.1 Quotient spaces & complements

Let V be a vector space and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U , i.e., $W \subseteq V$ is a subspace and $W \cap U = \{0\}$ and $W + U = V$.

Proposition (18.c.7). $\exists$ a *canonical* isomorphism. $Q : W \xrightarrow{\cong} V/U$, defined by $Q(w) := [w] \in V/U \forall w \in W$.



Proof. Let's consider the inclusion $i : W \rightarrow V$, given by $i(w) = w \ \forall w \in W$, and the projection map $\pi : V \rightarrow V/U$. Recall that $\pi(v) = [v]$, so $Q = \pi \circ i$. Hence Q is linear.

We claim that $\ker Q = 0$. Indeed, if $w \in \ker Q$, then $[w] = 0 \Rightarrow w \in U$. But $w \in W$ and $U \cap W = \{0\} \Rightarrow w = 0$.

We claim now that Q is surjective. Let $x \in V/U$. Choose $v \in V$ with $[v] = x$. Since $V = W + U$, $\exists w \in W$ and $u \in U$ such that $v = w + u$.

$$\begin{aligned} & \Rightarrow [v] = [w + u] \in V/U \\ & = \underbrace{[w]}_{=Q(w)} = x. \end{aligned}$$

Thus Q is surjective. □

18.c. Relations to the dual space

Proposition (18.c.8). Let V be a vector space and let $U \subseteq V$ be a subspace. Define $U^\perp := \{\ell \in V^* \mid \ell|_U = 0\}$. $U^\perp$ = all the functionals $\ell : V \rightarrow K$ that vanish on U .

1. $U^\perp \subseteq V^*$ is a linear subspace.
2. $\exists$ a canonical isomorphism.

$$(V/U)^* \cong U^\perp.$$

If we assume that V is finite-dimensional, then $\exists$ also a canonical isomorphism $V^*/U^\perp \cong U^*$.

Proof. 1. Clearly $0 \in U^\perp$. If $\ell_1, \ell_2 \in U^\perp$ and $\alpha, \beta \in K$, then $(\alpha\ell_1 + \beta\ell_2) \in U^\perp$, because $\forall u \in U$, we have:

$$(\alpha\ell_1 + \beta\ell_2)(u) = \underbrace{\alpha\ell_1(u)}_{=0} + \underbrace{\beta\ell_2(u)}_{=0} = 0.$$

2. Define a map $L : (V/U)^* \rightarrow U^\perp$ as follows: let $s \in (V/U)^*$, i.e. $s : V/U \rightarrow K$ is linear. Define $L(s) \in U^\perp$ to be $L(s) := s \circ \pi : V \rightarrow K$. So, L is: $s \mapsto s \circ \pi$.

$$\begin{array}{ccc} V & \xrightarrow{s \circ \pi} & K \\ \pi \downarrow & \nearrow s & \\ V/U & & \end{array}$$

Exercise. 1) Show that $s \circ \pi$ is indeed in $U^\perp$ (i.e. $s \circ \pi|_U \equiv 0$). 2) Show that L is linear. We claim that L is an isomorphism. To see this, we define another map

$$P : U^\perp \rightarrow (V/U)^*.$$

Let $t \in U^\perp$, i.e. $t : V \rightarrow K$ such that $t|_U \equiv 0$. By the universal property of quotient spaces, $\exists! t' : V/U \rightarrow K$ such that $t = t' \circ \pi$.

$$\begin{array}{ccc} V & \xrightarrow{t} & K \\ \pi \downarrow & \nearrow t' & \\ V/U & & \end{array}$$

Define $P(t) := t'$. We claim that

$$L \circ P = \text{id}_{U^\perp} \quad \text{and} \quad P \circ L = \text{id}_{(V/U)^*}.$$

Indeed, if $s : V/U \rightarrow K$, then

$$(P \circ L)(s) = P(s \circ \pi) = s,$$

so $P \circ L = \text{id}_{(V/U)^*}$. Also, for every $t \in U^\perp$, we have

$$(L \circ P)(t) = L(t') = t' \circ \pi = t,$$

so $L \circ P = \text{id}_{U^\perp}$. Since L is also a linear map and both of L & P are isomorphisms, assume now that V is finite-dimensional. Consider the map $R : V^* \rightarrow U^*$, defined by

$$R(\varphi) := \varphi|_U \in U^*, \quad \forall \varphi \in V^*.$$

(Restriction map). (BTW check that if $i : U \rightarrow V$ is the inclusion, then $R = i^* : V^* \rightarrow U^*$). Exercise: Check that R is linear. We claim that R is surjective. Indeed, let $\psi \in U^*$. We need to show that $\exists \varphi \in V^*$ such that $\varphi|_U = \psi$. Pick a basis $\{u_1, \dots, u_k\}$ for U , and extend it to a basis $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$ of V . Define $\varphi : V \rightarrow K$ by

$$\varphi(u_i) = \psi(u_i) \quad \forall 1 \leq i \leq k, \quad \varphi(v_j) = 0 \quad \forall k+1 \leq j \leq n.$$

30

Since $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$ is a basis for V , φ is well-defined and clearly $\varphi|_U = \psi$. This shows $\text{Im}(R) = U^*$. Also, $\ker(R) = U^\perp$ (by def.). By theorem 18.c.5 (the isomorphism theorem), we get that

$$V^*/\ker(R) \cong \text{Im}(R).$$

So $V^*/U^\perp \cong U^*$. □

Exc. Let $T : V \rightarrow W$ be a linear map. Then

$$(\text{Im}(T))^\perp = \ker(T^* : W^* \rightarrow V^*).$$

Filip Jaskovic

Chapter 6

Determinants and eigen theory

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(K) \rightsquigarrow \det(A) = ad - bc$$

Lecture 31: Introduction to determinants

02 Mar 2026

20.a Determinants: Introduction

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(K)$.

Proposition (20.2.1). A is invertible iff $ad - bc \neq 0$. Moreover if A is invertible

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar $ad - bc$ is called the determinant of A and is denoted by $\det(A)$ or sometimes $|A|$.

Proof. This is in the homework, but here is the outline A is not invertible iff the rank of A is strictly less than 2, $\Leftrightarrow$ the columns of A are linearly dependent $\Leftrightarrow$ either $\begin{pmatrix} a \\ c \end{pmatrix} = \lambda \begin{pmatrix} b \\ d \end{pmatrix}$ for some $\lambda \in K$ or $\begin{pmatrix} b \\ d \end{pmatrix} = \tau \begin{pmatrix} a \\ c \end{pmatrix}$ for some $\tau \in K$. In the first case

$$A = \begin{pmatrix} \lambda b & b \\ \lambda d & d \end{pmatrix}, \&a = \lambda b, c = \lambda d$$

and in the second case

$$A = \begin{pmatrix} a & \tau a \\ c & \tau c \end{pmatrix}, \&b = \tau a, d = \tau c.$$

Exc. Complete the proof. □

Definition (20.a.2). Let $D : M_n(K) \rightarrow K$ be a function. We say that D is n -linear if $\forall 1 \leq i \leq n$, D is a linear function of the i -th row, when the other $n - 1$ rows are fixed. We can think of $M_n(K)$ as $M_{n \times n}(K)$ as $K_{\text{row}}^n \times \cdots \times K_{\text{row}}^n$ and if

$$A = \begin{pmatrix} \text{---} \alpha_1 \text{---} \\ \vdots \\ \text{---} \alpha_n \text{---} \end{pmatrix}$$

we write $D(A) = D(\alpha_1, \dots, \alpha_n)$ saying that D is n -linear, means: $\forall 1 \leq i \leq n$,

1. $D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) = D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n)$
2. $D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) = c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) \forall c \in K$

Example. For a matrix A , write $A(i, j)$ for its (i, j) -entry. One also writes a_{ij} or A_{ij} .

Fix integers $k_1, \dots, k_n$ with $1 \leq k_i \leq n$ for all i . Fix also $a \in K$. Define $D : M_{n \times n}(K) \rightarrow K$ by

$$D(A) := a \cdot A(1, k_1) \cdots A(n, k_n).$$

Claim. The map D is n -linear.

Proof. Proof. Consider

$$\begin{pmatrix} \text{---} \alpha_1 \text{---} \\ \vdots \\ \text{---} \alpha_i \text{---} \\ \vdots \\ \text{---} \alpha_n \text{---} \end{pmatrix} = \begin{pmatrix} A(1, 1) & \cdots & A(1, n) \\ \vdots & \ddots & \vdots \\ A(i, 1) & \cdots & A(i, n) \\ \vdots & \ddots & \vdots \\ A(n, 1) & \cdots & A(n, n) \end{pmatrix}.$$

We have $D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) = b \cdot A(i, k_i)$, where $b \in K$ does NOT depend on row i . (b depends only on rows $1, \dots, i-1, i+1, \dots, n$).

Let $\alpha'_i = (\alpha'_{i,1}, \dots, \alpha'_{i,n})$. Then

$$\begin{aligned} D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) &= b \cdot (A(i, k_i) + \alpha'_{i, k_i}) \\ &= D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n). \end{aligned}$$

Let $c \in K$. Then

$$\begin{aligned} D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) &= b \cdot c A(i, k_i) \\ &= c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n). \end{aligned}$$

□

special case

$D(A) := A(1, 1) \cdot A(2, 2) \cdots A(n, n)$ (the product of the elements along the diagonal) is n -linear.

Let's find all 2-linear functions $D : M_2(K) \rightarrow K$.

Write

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ - & \varepsilon_2 & - \end{pmatrix}$$

If D is 2-linear, then

$$\begin{aligned} D \left(\begin{pmatrix} A(1, 1) & A(1, 2) \\ A(2, 1) & A(2, 2) \end{pmatrix} \right) &= \\ &= D((A(1, 1), A(1, 2)), (A(2, 1), A(2, 2))) \\ &= D(A(1, 1)\varepsilon_1 + A(1, 2)\varepsilon_2, A(2, 1)\varepsilon_1 + A(2, 2)\varepsilon_2) \\ &= A(1, 1)D(\varepsilon_1, A(2, 1)\varepsilon_1 + A(2, 2)\varepsilon_2) + A(1, 2)D(\varepsilon_2, A(2, 1)\varepsilon_1 + A(2, 2)\varepsilon_2) \\ &= A(1, 1)A(2, 1)D(\varepsilon_1, \varepsilon_1) + A(1, 1)A(2, 2)D(\varepsilon_1, \varepsilon_2) \\ &\quad + A(1, 2)A(2, 1)D(\varepsilon_2, \varepsilon_1) + A(1, 2)A(2, 2)D(\varepsilon_2, \varepsilon_2). \end{aligned}$$

D is completely determined by the values of $D(\varepsilon_1, \varepsilon_1)$, $D(\varepsilon_1, \varepsilon_2)$, $D(\varepsilon_2, \varepsilon_1)$, and $D(\varepsilon_2, \varepsilon_2)$.

31

Lemma (20.a.3). Any linear combination of n -linear function is again an n -linear function.

Proof. Enough to prove for a linear combination of two n -linear functions. (WHY?)

Let $D, E : M_n(K) \rightarrow K$ be two n -linear functions, and let $a, b \in K$. Consider $aD + bE : M_n(K) \rightarrow K$, where $(aD + bE)(A) := aD(A) + bE(A)$.

Let $1 \leq i \leq n$. For additivity in the i -th row, we have:

$$\begin{aligned} (aD + bE)(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) &= \\ &= aD(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) + bE(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) \\ &= a \cdot \left(D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n) \right) \\ &\quad + b \left(E(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + E(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n) \right) \\ &= (a \cdot D + bE)(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + (aD + bE)(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n). \end{aligned}$$

For scalar multiplication by $\lambda \in K$ in the i -th row, we have:

$$\begin{aligned}(aD + bE)(\alpha_1, \dots, \lambda \cdot \alpha_i, \dots, \alpha_n) &= \\ &= aD(\alpha_1, \dots, \lambda \alpha_i, \dots, \alpha_n) + bE(\alpha_1, \dots, \lambda \alpha_i, \dots, \alpha_n) \\ &= \lambda aD(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + \lambda bE(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) \\ &= \lambda \cdot (aD + bE)(\alpha_1, \dots, \alpha_i, \dots, \alpha_n).\end{aligned}$$

□

Example. Consider $D : M_{2 \times 2}(K) \rightarrow K$ $D(A) = A_{11}A_{22} - A_{12}A_{21}$. $(\det A)$ is 2-linear.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Proof. $D = D_1 - D_2$ where $D_1(A) = A_{11}A_{22}$ and $D_2(A) = -A_{12}A_{21}$. By a previous example, both D_1 and D_2 are 2-linear. $\Rightarrow D$ is also 2-linear. (b.c. its a linear combination of D_1 & D_2). □

Remark. $D = \det A$ as a function $M_{2 \times 2}(K) \rightarrow K$ has two additional properties:

1. $D(I) = D \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$.
2. If A' is obtained from A by interchanging two rows, then $\det A' = -\det A$.

Indeed, if $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, $A' = \begin{pmatrix} A_{21} & A_{22} \\ A_{11} & A_{12} \end{pmatrix}$, then $\det A' = A_{21}A_{12} - A_{22}A_{11} = -(A_{11}A_{22} - A_{12}A_{21}) = -\det A$.

Definition (20.a.4). Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear function. We say that D is alternating if for every matrix A , in which some two rows are equal we have $D(A) = 0$.

Proposition (20.a.5). Let D be an n -linear function. Assume that D has the property that whenever $A \in M_{n \times n}(K)$ has two equal adjacent rows then $D(A) = 0$. Then:

1. D is alternating.
2. $\forall$ matrix B , if B' is obtained from B by interchanging *two* of its rows then $D(B') = -D(B)$.

Proof. We begin with proving (emph2). Assume 1st, that B' is obtained from B

by interchanging two adjacent rows.

$$B = \begin{pmatrix} - & \beta_1 & - \\ - & \beta_k & - \\ - & \beta_{k+1} & - \\ - & \beta_n & - \end{pmatrix}, \quad B' = \begin{pmatrix} - & \beta_1 & - \\ - & \beta_{k+1} & - \\ - & \beta_k & - \\ - & \beta_n & - \end{pmatrix}.$$

Consider $D \begin{pmatrix} \beta_1, \dots, \beta_{k-1}, \underbrace{\beta_k + \beta_{k+1}}_{\text{entry } \#k}, \underbrace{\beta_k + \beta_{k+1}}_{\text{entry } \#(k+1)}, \beta_{k+2}, \dots, \beta_n \end{pmatrix}$.

By assumption $(*) = 0$, By n-linearity,

$$\begin{aligned} (*) &= D(\beta_1, \dots, \beta_k, \beta_k + \beta_{k+1}, \dots, \beta_n) + \\ &\quad + D(\beta_1, \dots, \beta_{k+1}, \beta_k + \beta_{k+1}, \dots, \beta_n) \\ &= D(\beta_1, -, \underline{\underline{\beta_k}}, \underline{\underline{\beta_k}}, -, \beta_n) + \\ &\quad + D(\beta_1, -, \underline{\underline{\beta_k}}, \underline{\underline{\beta_{k+1}}}, -, \beta_n) + \\ &\quad + D(\beta_1, -, \underline{\underline{\beta_{k+1}}}, \underline{\underline{\beta_k}}, -, \beta_n) + \\ &\quad + D(\beta_1, -, \underline{\underline{\beta_{k+1}}}, \underline{\underline{\beta_{k+1}}}, -, \beta_n) \end{aligned}$$

☐

Lecture 32: Determinants (continued)

04 Mar 2026

Recap

Last time: Prop. with 20.a.5. Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear map. Assume that $D(A) = 0$ whenever two adjacent rows of A are equal. Then:

1. Γ is alternating.
2. If B' is obtained from B by interchanging two rows, then $D(B') = -D(B)$.

Proof idea for (2), adjacent-row case. We first prove the statement when B' is obtained from B by swapping rows k and $k + 1$. Write

$$B = \begin{pmatrix} \text{---}\beta_1\text{---} \\ \vdots \\ \text{---}\beta_k\text{---} \\ \text{---}\beta_{k+1}\text{---} \\ \vdots \\ \text{---}\beta_n\text{---} \end{pmatrix}, \quad B' = \begin{pmatrix} \text{---}\beta_1\text{---} \\ \vdots \\ \text{---}\beta_{k+1}\text{---} \\ \text{---}\beta_k\text{---} \\ \vdots \\ \text{---}\beta_n\text{---} \end{pmatrix}.$$

Define

$$X := D(\beta_1, \dots, \beta_k + \beta_{k+1}, \beta_k + \beta_{k+1}, \dots, \beta_n).$$

The two middle rows in this input are equal, so $X = 0$ by assumption. Expand X by

n -linearity in the k -th and $(k+1)$ -st rows:

$$X = \underbrace{D(\beta_1, \dots, \beta_k, \beta_{k+1}, \dots, \beta_n)}_{(1)} + \underbrace{D(\beta_1, \dots, \beta_{k+1}, \beta_k, \dots, \beta_n)}_{(2)}.$$

Therefore $(1) + (2) = 0$, and hence $D(B') = -D(B)$ for one adjacent swap.

Suppose now that B' is obtained from B by interchanging rows $\#k$ and $\#\ell$, where $k < \ell$ (not necessarily $\ell = k+1$).

We can obtain B' from B by doing a sequence of interchanges of adjacent rows. Indeed, begin by exchanging rows k and $k+1$, and continue in this way until we get

$$\beta_1, \dots, \beta_{k-1}, \beta_{k+1}, \dots, \beta_\ell, \beta_k, \beta_{\ell+1}, \dots, \beta_n.$$

This requires $r := \ell - k$ interchanges of adjacent rows.

Next, move β_ℓ back by a sequence of interchanges of adjacent rows until it arrives at position $\#k$. This requires $(\ell - 1) - k = r - 1$ interchanges of adjacent rows.

So in total, we can obtain B' from B by performing $r + (r - 1) = 2r - 1$ interchanges of adjacent rows.

So $D(B') = (-1)^{2r-1} D(B) = -D(B)$. This concludes the proof of (2).

32

Proof of (1). Let A be an $n \times n$ matrix in which row $\#i =$ row $\#j$, where $i < j$. If $j = i+1$, then by assumption $D(A) = 0$. So assume $j > i+1$.

We interchange now rows $\#i+1$ and $\#j$. We get a new matrix A' , in which rows $\#i$ and $\#i+1$ are equal.

$$A = \begin{pmatrix} \text{---} & \alpha_1 & \text{---} \\ & \vdots & \\ \text{---} & \alpha_i & \text{---} \\ \text{---} & \alpha_{i+1} & \text{---} \\ & \vdots & \\ \text{---} & \alpha_j & \text{---} \\ & \vdots & \\ \text{---} & \alpha_n & \text{---} \end{pmatrix} \begin{matrix} \leftarrow \\ \\ \leftarrow \\ \text{equal} \\ \\ \leftarrow \end{matrix}$$

By (2): $D(A) = -D(A')$. By assumption, $D(A') = 0$. So $D(A) = 0$. This concludes the proof.

Definition (20.a.6). A function $D : M_{n \times n}(K) \rightarrow K$ is called a *determinant function* if it is n -linear, alternating, and $D(I_n) = 1$. Warmup: Let's find *all* determinant functions $D : M_{2 \times 2}(K) \rightarrow K$. Write $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ - & \varepsilon_2 & - \end{pmatrix}$. $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$.

We show that every 2-linear function D satisfies:

$$\begin{aligned} D(A) &= A_{11}A_{21}D(\varepsilon_1, \varepsilon_1) + A_{11}A_{22}D(\varepsilon_1, \varepsilon_2) \\ &\quad + A_{12}A_{21}D(\varepsilon_2, \varepsilon_1) + A_{12}A_{22}D(\varepsilon_2, \varepsilon_2). \quad (*) \end{aligned}$$

Since D is alternating, $D(\varepsilon_1, \varepsilon_1) = D(\varepsilon_2, \varepsilon_2) = 0$, and $D(\varepsilon_1, \varepsilon_2) = -D(\varepsilon_2, \varepsilon_1)$. So $D(A) = A_{11}A_{22}D(\varepsilon_1, \varepsilon_2) - A_{12}A_{21}D(\varepsilon_1, \varepsilon_2)$. Since $D(I) = 1$, we get

$$D(A) = A_{11}A_{22} - A_{12}A_{21} = \det \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Definition (20.a.7). Let $A \in M_{n \times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. Denote by $A(i|j) \in M_{(n-1) \times (n-1)}(K)$ the $(n-1) \times (n-1)$ -matrix obtained from A after deleting row $\#i$ and column $\#j$. If D is an $(n-1)$ -linear function, denote

$$D_{ij} := D(A(i|j)).$$

Theorem (20.a.8). Let $n \geq 2$, and D an $(n-1)$ -linear function, which is alternating. Let $1 \leq j \leq n$. Define a function $E_j : M_{n \times n}(K) \rightarrow K$ by

$$E_j(A) := \sum_{i=1}^n (-1)^{i+j} A_{ij} \cdot D_{ij}(A).$$

Then E_j is an n -linear function and is alternating. If D is a determinant function (i.e. $D(I_{n-1}) = 1$), then so is E_j .

Before the proof a quick

Corollary (20.a.9). $\forall n \geq 1, \exists$ at least one determinant function $M_{n \times n}(K) \rightarrow K$.

Proof. Induction on n . For $n = 1$, $M_{1 \times 1}(K) \rightarrow K, a \mapsto a$ is a determinant function. For $n = 2$, we have already seen that $\det : M_{2 \times 2}(K) \rightarrow K$ is a determinant function. Continue by induction using Theorem 20.a.8.: if $\exists$ a determinant function for $(n-1) \times (n-1)$ matrices, then $\exists$ a determinant function also for $n \times n$ matrices. $\square$

Proof of Theorem 20.a.8.. Let $A \in M_{n \times n}(K)$. $D_{ij}(A)$ is independent of row $\#i$ of A . Since D is $(n-1)$ -linear, $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each row of A , except row $\#i$. This implies that the function $M_{n \times n}(K) \ni A \mapsto A_{ij} \cdot D_{ij}(A)$ is n -linear. Since linear combinations of n -linear functions are n -linear, it follows that

$$E_j(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \cdot D_{ij}(A)$$

is n -linear. $\square$

Claim. E_j is alternating.

Proof. Indeed, by Proposition 20.a.5, it is enough to show that $E_j(A) = 0$ whenever A has two equal adjacent rows. So let

$$A = \begin{pmatrix} \text{---} \alpha_1 \text{---} \\ \vdots \\ \text{---} \alpha_n \text{---} \end{pmatrix}$$

be such a matrix, where $\alpha_k = \alpha_{k+1}$ for some $1 \leq k \leq n-1$. Let $1 \leq i \leq n$, $i \neq k, k+1$. The matrix $A(i|j)$ has two equal adjacent rows (because $i \neq k, k+1$), hence $D_{ij}(A) = 0$ (b.c. $D_{ij}(A) = D(A(i|j))$ and D is assumed to be alternating).

$$\Rightarrow E_j(A) = (-1)^{k+j} A_{kj} \cdot D_{kj}(A) + (-1)^{(k+1)+j} A_{(k+1)j} \cdot D_{(k+1)j}(A).$$

hence $D_{ij}(A) = 0$ (b.c. $D_{ij}(A) = D(A(i|j))$ and D is assumed to be alternating).

$$\textcircled{1} = \textcircled{2}$$

$$\Rightarrow E_j(A) = (-1)^{k+j} \overbrace{(A_{kj} D_{kj}(A))}^{\textcircled{1}} + (-1)^{k+1+j} \overbrace{(A_{k+1,j} D_{k+1,j}(A))}^{\textcircled{2}}$$

But $\alpha_k = \alpha_{k+1} \Rightarrow A_{kj} = A_{k+1,j}$ and $A(k|j) = A(k+1|j)$

But $\alpha_k = \alpha_{k+1} \Rightarrow A_{kj} = A_{k+1,j}$ and $A(i|j) = A(i+1|j)$

So $E_j(A) = (-1)^{k+j} \textcircled{1} + (-1)^{k+1+j} \textcircled{1}$

$$(-1)^{k+j} \cdot (-1)^{k+1+j} \cdot (\textcircled{1} - \textcircled{1}) = 0.$$

This proves E_j is alternating

Finally, assume $D(I_{n-1}) = 1$.

$$\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = I_n$$

$$I_n(j|j) = I_{n-1}, (I_n)_{ij} = \delta_{ij}$$

$$\text{So, } E_j(I_n) = \sum_{i=1}^n (-1)^{i+j} \delta_{ij} D(I_n(i|j))$$

$$= (-1)^{2j} \cdot 1 \cdot D(I_{n-1}) = 1.$$

□

Example. For a 2×2 mat. $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$

we have that $B \mapsto B_{11}B_{22} - B_{12}B_{21}$

is a determin funct. $\det(B)$ or $|B|$

32

Consider $A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \in M_{3 \times 3}(K)$.

$$E_1(A) = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21} \begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31} \begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}$$

$$E_2(A) = -A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{22} \begin{vmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{vmatrix} - A_{32} \begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix}$$

Lecture 33: Uniqueness of the determinant function

09 Mar 2026

Recap

Last time. Determinant functions $M_{n \times n}(K) \rightarrow K$ do exist.

In fact. If D is an $(n-1)$ -linear alternating function and if $i \leq j \leq n$ is fixed, then the map $E_j : M_{n \times n}(K) \rightarrow K$ is defined by

$$E_j(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} D \left(\begin{array}{c} \text{row } i \quad \text{column } j \\ \vdots \quad \vdots \\ \text{---} A_{ij} \text{ ---} \\ \vdots \quad \vdots \end{array} \right),$$

where we delete row i and column j . Then E_j is n -linear and alternating. If D is a determinant function, then $E_j = \det$.

33

20.b Determinants: Uniqueness of the determinant function

Let $D : M_n(K) \rightarrow K$ be an n -linear and alternating function.

$$A = \begin{pmatrix} - & \alpha_1 & - \\ - & \alpha_2 & - \\ - & \vdots & - \\ - & \alpha_n & - \end{pmatrix} \in M_n(K).$$

$$I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ - & \vdots & - \\ - & \varepsilon_n & - \end{pmatrix} \text{ this is the standard basis of } K_{row}^n.$$

We have $\alpha_i = \sum_{j=1}^n A_{ij} \varepsilon_j$

$$D(A) = D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^n A_{1j} \varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^n A_{1j} D(\varepsilon_j, \alpha_2, \dots, \alpha_n).$$

$$\begin{aligned} \sum_{j=1}^n A_{1j} D\left(\varepsilon_j, \sum_{k=1}^n A_{2k} \varepsilon_k, \dots, \alpha_n\right) &= \sum_{1 \leq j, k \leq n} A_{1,j} \cdot A_{2,k} \cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n). \\ \sum_{1 \leq k_1, \dots, k_n \leq n} A_{1,k_1} \cdot A_{2,k_2} \cdots A_{n,k_n} \cdot D(\varepsilon_{k_1}, \varepsilon_{k_2}, \dots, \varepsilon_{k_n}) &\quad (*) \end{aligned}$$

Since D is alternating, $D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = 0$ whenever two of the indices $k_1, \dots, k_n$ are equal. So in the previous sum, it's enough to sum up over $(k_1, \dots, k_n)$ with the property that $k_i \neq k_j, \forall i \neq j$ such a sequence is called a permutation of degree n . (or n -permutation) We can think of permutations as functions

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

which are bijective. $(k_1, \dots, k_n) = (\sigma(1), \dots, \sigma(n))$. So $(*) = \sum_{\sigma} A_{1,\sigma(1)} \cdot A_{2,\sigma(2)} \cdots A_{n,\sigma(n)} \cdot D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)})$. Where the sum $(*)$ runs over all permutations σ on $\{1, \dots, n\}$.

Number of permutations on n objects = $n!$ ($n \cdot (n-1) \cdots \cdots 2 \cdot 1$).

33

Every permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ can be written as a composition $\sigma = \theta_1 \circ \dots \circ \theta_m$, of transpositions. A transposition θ is a permutation such that $\exists i \neq j \in \{1, \dots, n\} i \neq j$ such that $\theta(i) = j, \theta(j) = i$ and $\theta k = k \forall k \neq i, j$.

Why? Start with $1, \dots, n$. If $\sigma(1) \neq 1$ then interchange 1 & $\sigma(1)$. Now we have $\sigma(1), 1, \dots, n$. If $\sigma(2) \neq 2$ then interchange them. Now we have to 3, etc.

Example (3,2,1). $\sigma(1) = 3, \sigma(2) = 2, \sigma(3) = 1$.

$(1, 2, 3) \rightarrow (3, 2, 1)$ We are done.

$$(1, 2, 3) \longrightarrow (2, 1, 3) \longrightarrow (2, 3, 1) \longrightarrow (3, 2, 1).$$

$$\sigma = \theta_1 = \theta'_1 \circ \theta'_2 \circ \theta'_3.$$

Every σ can be written as $\sigma = \theta_1 \circ \dots \circ \theta_m$ but not in a unique way, i.e. $\sigma = \theta'_1 \circ \dots \circ \theta'_{m'}$ with θ'_j transpositions.

Lemma (20.b.1). If $\sigma = \theta_1 \circ \dots \circ \theta_m = \theta'_1 \circ \dots \circ \theta'_{m'}$ with θ_i and θ'_j transpositions, then m and m' have the same parity, namely m is odd iff m' is odd $\Leftrightarrow (-1)^m = (-1)^{m'}$. Define $\text{sgn}(\sigma) := (-1)^m$.

Proof. We know that a determinant function $M_{n \times n}(K) \rightarrow K$ exists for all $n \geq 1$. Fix one such function E . Consider $E(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)})$. If $\sigma(1), \dots, \sigma(n)$ can be

obtained from $1, \dots, n$ by a sequence of m transpositions, then the matrix

$$\begin{pmatrix} - & \varepsilon_{\sigma(1)} & - \\ & \vdots & \\ - & \varepsilon_{\sigma(n)} & - \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}$$

by a sequence of m interchanges of pairs of rows. Since E is alternating, each interchange changes the sign, hence

$$E(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)}) = (-1)^m E(I) = (-1)^m,$$

because $E(I) = 1$.

By the same argument, using the decomposition $\sigma = \theta'_1 \dots \circ \theta'_{m'}$, we get

$$E(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)}) = (-1)^{m'}.$$

Therefore

$$(-1)^m = (-1)^{m'},$$

so m and m' have the same parity. □

Back to $D(A)$,

$$D(A) = \sum_{\sigma} A_{1,\sigma(1)} \cdots A_{n,\sigma(n)} D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)})$$

and by Lemma 2.1b.,

$$D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)}) = \underbrace{\text{sgn}(\sigma)}_{\text{depends only on } \sigma \text{ and } D(I)} \cdot D(I).$$

Theorem (20.b.2). For every n -linear and alternating function $D: M_n(K) \rightarrow K$, we have

$$D(A) = \sum_{\sigma} \text{sgn}(\sigma) A_{1,\sigma(1)} \cdots A_{n,\sigma(n)} D(I).$$

In particular, if D is a determinant function, then

$$D(A) = \sum_{\sigma} \text{sgn}(\sigma) A_{1,\sigma(1)} \cdots A_{n,\sigma(n)}.$$

So D is unique.

Proof. Substitute the previous formula for $D(\varepsilon_{\sigma(1)}, \dots, \varepsilon_{\sigma(n)})$ into the expression for $D(A)$. $\square$

Corollary (20.b.3). $\forall$ n -linear and alternating function $D: M_{n \times n}(K) \rightarrow K$, we have $D(A) = D(I) \cdot \det(A)$.

Denote by S_n the set of all permutations on n elements. S_n is called a group. We can compose permutations, $\sigma, \tau: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. We have $\sigma \circ \tau$. We can also invert every $\sigma \in S_n$.

$$\sigma^{-1}: \{1, \dots, n\} \rightarrow \{1, \dots, n\}.$$

1 identity/unit : the trivial permutation.

Lemma. (20.b.4) $\forall \sigma, \tau \in S_n$, we have $\text{sgn}(\sigma \circ \tau) = \text{sgn}(\sigma) \cdot \text{sgn}(\tau)$.

Proof. Suppose $\sigma = \theta_1 \circ \dots \circ \theta_m$ and $\tau = \lambda_1 \circ \dots \circ \lambda_\ell$, where θ_i and λ_j are transpositions. $\Rightarrow \sigma \circ \tau = \theta_1 \circ \dots \circ \theta_m \circ \lambda_1 \circ \dots \circ \lambda_\ell \Rightarrow \text{sgn}(\sigma \circ \tau) = (-1)^{m+\ell} = (-1)^m \cdot (-1)^\ell$. $\square$

Theorem (20.b.5). $\forall A, B \in M_{n \times n}(K)$, we have $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Proof. Fix $B \in M_{n \times n}(K)$, and consider the function $D: M_{n \times n}(K) \rightarrow K$ defined by $D(A) = \det(A \cdot B)$.

Claim. D is n -linear and alternating.

33

Proof. Write $A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$. Then

$$A \cdot B = \begin{pmatrix} - & \alpha_1 \cdot B & - \\ & \vdots & \\ - & \alpha_i \cdot B & - \\ & \vdots & \\ - & \alpha_n \cdot B & - \end{pmatrix},$$

so $D(A) = \det(\alpha_1 \cdot B, \dots, \alpha_n \cdot B)$.

Fix $1 \leq i \leq n$. If α_i and α'_i are two row vectors, then $(\alpha_i + \alpha'_i) \cdot B = \alpha_i \cdot B + \alpha'_i \cdot B$, so

$$D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) = D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n),$$

because $\det$ is n -linear.

Similarly, for all $c \in K$, $(c \cdot \alpha_i) \cdot B = c \cdot (\alpha_i \cdot B)$, hence

$$D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) = c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n).$$

If $\alpha_i = \alpha_j$, then $\alpha_i \cdot B = \alpha_j \cdot B$, hence $D(A) = \det(\alpha_1 \cdot B, \dots, \alpha_n \cdot B) = 0$. So D is alternating. $\square$

By a previous corollary (20.b.3) we have $D(A) = \det(A) \cdot D(I)$. But $D(I) = \det(B)$. Therefore,

$$\det(A) = \det(A \cdot B) = \det(A) \cdot \det(B).$$

 $\square$

Corollary (20.b.6). If A is invertible, then $\det(A) \neq 0$. Moreover $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof. If A is invertible, then $A \cdot A^{-1} = I$. $\Rightarrow \det(A \cdot A^{-1}) = \det(I) = 1$. But $\det(A \cdot A^{-1}) = \det(A) \cdot \det(A^{-1})$. $\square$

Lecture 34: Notes pending

11 Mar 2026

34

20.c Determinants: Lecture 34 Placeholder

Recall: $\forall M \in M_{m \times n}(K)$ we have the transposed matrix $M^\top \in M_{n \times m}(K)$ with $(M^\top)_{ij} = M_{ji}$.

Theorem (20.C.1). $\forall A \in M_{n \times n}(K)$ we have $\det(A) = \det(A^\top)$.

Proof. If σ is a permutation of $\{1, \dots, n\}$ then $(A^\top)_{i\sigma(i)} = A_{\sigma(i)i}$. Hence, $\det(A^\top) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$. If $\sigma_i = j$ then $A_{\sigma(i),i} = A_{j,\sigma^{-1}j}$.

Claim. $A_{\sigma(1),1} \cdots A_{\sigma(n),n} = A_{1,\sigma^{-1}1} \cdots A_{n,\sigma^{-1}n}$.

Proof. Every factor on LHS of the equation appears will appear and exactly once on the RHS of the equation. So the product on the LHS is the same as the product on the RHS. This proves the claim. $\square$

$(\text{sgn}(\sigma)) \cdot \text{sgn}(\sigma \circ \sigma^{-1}) = 1$ so $\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1})$ (recall: $\text{sgn} \in \{-1, 1\}$). $\Rightarrow$
 $\det(A^\top) = \sum_{\sigma \in S_n} (\text{sgn} \sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n} = \sum_{\sigma \in S_n} (\text{sgn} \sigma^{-1}) A_{1,\sigma^{-1}1} \cdots A_{n,\sigma^{-1}n} =$

$$\sum_{\tau \in S_n} (\text{sgn} \tau) A_{1,\tau 1} \cdots A_{n,\tau n} = \det(A).$$
 When σ runs over S_n , $\tau = \sigma^{-1}$ also runs over S_n and vice versa. $\square$

Corollary (20.c.2). If $D : M_{n \times n}(K) \rightarrow K$ is an alternating n -linear function (of the rows of the matrix) then D is also alternating n -linear when viewed as a function of the columns of the matrix. In fact, $D(A^\top) = D(A) \forall A \in M_{n \times n}(K)$.

Proof. $\forall A, D(A) = \det(A) \cdot D(I) = \det(A^\top) = \det(A^\top) \cdot D(I) = \det(A) \cdot D(I) = D(A)$. $\square$

Theorem (20.C.3). 1. If B is obtained from A by adding a multiple of one row of A to another row

$$A \xrightarrow{R_i + c \cdot R_j \rightarrow R_i} (i \neq j(!)) B,$$

then $\det(B) = \det(A)$.

2. If $A \xrightarrow{R_i \leftrightarrow R_j (i \neq j(!))} B$ then $\det(B) = -\det(A)$.

3. If $A \xrightarrow{c \cdot R_i \rightarrow R_i (c \in K)} B$, then $\det(B) = c \cdot \det(A)$.

Proof. 2 Follows from $\det$ being alternating.

3 Follows from n -linearity of $\det$.

1 Assume $j > i$ first. Write $A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$.

$$\det B = \det(\alpha_1, \dots, \alpha_i + c \cdot \alpha_j, \dots, \alpha_j, \dots, \alpha_n) = \det(\alpha_1, \dots, \alpha_i, \dots, \alpha_j, \dots, \alpha_n) + c \cdot \underbrace{\det(\alpha_1, \dots, \alpha_j, \dots, \alpha_j, \dots, \alpha_n)}_{=0} = \det(\alpha_1, \dots, \alpha_n) = \det(A).$$

$\square$

Determinants of block matrices.

Example of a block matrix:

$$M = \begin{pmatrix} \xrightarrow{r} & & \\ \uparrow r & A & B \\ & \xrightarrow{\quad} & \\ & O & C \\ & & \xleftarrow{s} \\ & & \downarrow s \end{pmatrix}$$

Consider M like here, where

$$A \in M_{r \times r}(K), \quad C \in M_{s \times s}(K), \quad B \in M_{r \times s}(K).$$

Proposition (20.c.4).

$$\det(M) = \det(A) \cdot \det(C).$$

Proof. Define

$$D : M_{r \times r}(K) \times M_{r \times s}(K) \times M_{s \times s}(K) \rightarrow K,$$

by

$$D\left(\begin{array}{c|c} A & B \\ \hline O & C \end{array}\right) := \det\left(\begin{array}{c|c} A & B \\ \hline O & C \end{array}\right).$$

If we fix A, B , then the function

$$M_{s \times s}(K) \rightarrow K, \quad C \mapsto D(A, B, C)$$

is linear and alternating. By a previous theorem (Corollary 20.b.3), we have

$$D(A, B, C) = \det(C) \cdot D(A, B, I_s). (*)$$

$$D(A, B, I_s) = \det\left(\begin{array}{c|c} A & B \\ \hline O & I_s \end{array}\right)$$

By applying row operations of the type $R_i + c \cdot R_j \rightarrow R_i$ where $1 \leq i \leq r, r+1 \leq j \leq r+s$ we can get to the matrix

$$\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}, \text{ and } \det\left(\begin{array}{c|c} A & B \\ \hline O & I_s \end{array}\right) = \det\left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}\right). (**)$$

The function $M_{r \times r}(K) \ni A \mapsto \det \left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array} \right) \in K$ is r -linear and alternating.

$$\Rightarrow \det \left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array} \right) = \det(A) \cdot \det \left(\underbrace{\begin{array}{c|c} I_r & O \\ \hline O & I_s \end{array}}_{I_{r+s}} \right) = \det(A).$$

Back to (*), and using (**): $\det \left(\begin{array}{c|c} A & B \\ \hline O & C \end{array} \right) = \det(C) \cdot \det(A)$ □

Example.

Exp: $A = \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & 1 & -1 \\ 1 & 2 & 3 & 0 \end{pmatrix} \in M_{4 \times 4}(\mathbb{R})$

$$A \xrightarrow{\substack{R_2 - 2R_1 \rightarrow R_2 \\ R_3 - 4R_1 \rightarrow R_3 \\ R_4 - R_1 \rightarrow R_4}} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix} \xrightarrow{\substack{R_3 - \frac{5}{4}R_2 \rightarrow R_3 \\ R_4 - \frac{3}{4}R_2 \rightarrow R_4}} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{pmatrix}$$

$$\det(A) = 4 \cdot (-4) \cdot 4 = 4 \cdot 32 = 128.$$

Let D be the determinant function on $M_{(n-1) \times (n-1)}(K)$. Recall that $\forall 1 \leq j \leq n$, $E_j : M_{n \times n}(K) \rightarrow K$, is an

$$E_j(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} D(A(i | j)).$$

We know now that $\forall 1 \leq j \leq n$, $E_j(A) = \det(A)$.

The scalars $C_{ij} = (-1)^{i+j} \cdot \det A(i | j)$ are called the i, j cofactors of A . We have

$$\det(A) = \sum_{i=1}^n A_{ij} C_{ij},$$

(*)

Claim. If we replace A_{ij} in (*) with A_{ik} (k is fixed), where $k \neq j$, then (*) becomes 0.

Namely,

$$\sum_{i=1}^n A_{ik} C_{ij} = \begin{cases} \det(A) & k = j, \\ 0 & k \neq j. \end{cases}$$

Proof. Let $B \in M_{n \times n}(K)$ be the matrix obtained from A col $\#j$ of A by col $\#k$ of A . (Assume $k \neq j$) Clearly $\det(B) = 0$. (b.c. B has two equal columns).

$$\Rightarrow 0 = \det(B) = \sum_{i=1}^n (-1)^{i+j} B_{ij} \det(B(i | j))$$

$$= \sum_{i=1}^n (-1)^{i+j} A_{ik} \det(A(i | j)) = \sum_{i=1}^n A_{ik} C_{ij}$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ A_{ik} = B_{ij} & & A(i | j) = B(i | j) \end{array}$$

□

Summary: $\forall A \in M_{n \times n}(K), \forall j, k$ we have $\sum_{i=1}^n A_{ik} C_{ij} = \delta_{jk} \cdot \det(A)$. The matrix $(C_{ij})^\top$ is called the classical adjoint of A and it is denoted by $\text{adj}(A) \in M_{n \times n}(K)$.

$$(\text{adj}(A))_{ij} = C_{ji} = (-1)^{j+i} \cdot \det A(j | i).$$

The formula (the one with the delta) says $\boxed{(\text{adj}(A)) \cdot A = (\det(A)) \cdot I.}$

Claim. $A \cdot (\text{adj } A) = (\det A) \cdot I.$

Proof. $A^T(i | j) = (A(j | i))^T \forall i, j.$

$$\begin{aligned} \Rightarrow (\text{adj } A^T)_{ij} &= (-1)^{i+j} \det A^T(j | i) = (-1)^{i+j} \det(A(j | i))^T = (-1)^{i+j} \det A(j | i) = \\ &= (\text{adj } A)_{j,i} = (\text{adj } A)_{i,j}^T. \end{aligned}$$

$$\Rightarrow \text{adj}(A^T) = (\text{adj } A)^T.$$

□

Lecture 35: Notes pending

16 Mar 2026

Recap

Last time.

$$A \in M_{n \times n}(K), \quad \text{adj } A \in M_{n \times n}(K).$$

Adjugate matrix. For $A \in M_{n \times n}(K)$, the adjugate matrix $\text{adj } A$ is defined by

$$(\text{adj } A)_{i,j} = (-1)^{i+j} \det A(j | i).$$

We have seen.

We have

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I.$$

Also.

Transpose compatibility. Also,

$$\operatorname{adj}(A^\top) = (\operatorname{adj} A)^\top.$$

Claim.

Goal for today. We will prove

$$A \cdot (\operatorname{adj} A) = (\det A) \cdot I.$$

Theorem (20.c.6). Let $A \in M_{n \times n}(K)$. A is invertible iff $\det A \neq 0$. When A is invertible, we have,

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A.$$

20.c.1 Cramer's rule

Let $A \in M_{n \times n}(K)$ and $b \in K^n$. Consider the system of equations, $A \cdot x = b$, where

$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$. $Ax = b \Rightarrow (\operatorname{adj} A) \cdot A \cdot x = (\operatorname{adj} A) \cdot b \Rightarrow (\det A) \cdot x = (\operatorname{adj} A) \cdot b$ (*). If $\det A \neq 0$, then:

$$\Rightarrow x = \frac{1}{\det A} (\operatorname{adj} A) \cdot b.$$

$\Rightarrow \forall 1 \leq j \leq n$ we have:

$$(\det A) \cdot x_j = \sum_{i=1}^n (\operatorname{adj} A)_{j,i} \cdot b_i = \sum_{i=1}^n (-1)^{i+j} b_i \cdot \det A(i | j) (**).$$

Consider the matrix

$$A(j, b) := \left(A \mid b \mid A \right) \quad \leftarrow \text{col } \# j$$

the matrix obtained from A by replacing col $\# j$ with the vector b . Entry (i, j) of this matrix is exactly b_i .

So, by previous formulas (formula for E_j , but applied to $A(j, b)$), we have:

$$(**) = \det A(j, b).$$

Cramer's formula:

$$x_j = \frac{\det A(j, b)}{\det A}.$$

20.c.2 Determinants and Endomorphisms

Warm up. Let $A \in M_{n \times n}(K)$, $P \in GL(n, K)$ invertible.

$$\Rightarrow \det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \det(A)$$

So, similar matrices

$$\left(\frac{1}{\det P} \right)$$

have the same det. Let V be a f. dim. vect. sp. over K .

Put $n := \dim V$. Let $T : V \rightarrow V$ be a lin. map. Pick a basis B for V .

$\rightsquigarrow$ We get a matrix $[T]_B^B$.

Consider $\det[T]_B^B \in K$. Q. Does $\det[T]_B^B$ depend on B ?

A. Let B' be another basis for V .

Recap

Recall.

Change of basis. If B' is another basis of V , then

$$[T]_{B'}^{B'} = [\text{id}_V]_B^{B'} [T]_B^B [\text{id}_V]_{B'}^B = I^{-1} [T]_B^B P, \quad P = [\text{id}_V]_{B'}^B.$$

Recall that

$$[\text{id}_V]_{B'}^B = ([\text{id}_V]_{B'}^{B'})^{-1},$$

so

$$\det[T]_{B'}^{B'} = \det[T]_B^B,$$

so $\det(T)$ is independent of the choice of basis for V .

Composition and inverses. Recall also that for linear maps $S, T : V \rightarrow V$, we have

$$[S \circ T]_B^B = [S]_B^B \cdot [T]_B^B, \quad [T^{-1}]_B^B = ([T]_B^B)^{-1}$$

whenever T is an isomorphism. Applying $\det$, we get

$$\det(S \circ T) = \det(S) \cdot \det(T), \quad \det(T^{-1}) = \frac{1}{\det(T)}.$$

Special cases. So

$$\det(\text{id}_V) = 1, \quad \det(O_{V \rightarrow V}) = 0.$$

Important distinction. For a general linear map $T : V \rightarrow W$, the quantity $\det[T]_C^B$ depends on the choices of bases B and C . Consequently, we cannot define $\det(T)$ for arbitrary maps $T : V \rightarrow W$. It is only well-defined for endomorphisms $T : V \rightarrow V$.

T is an iso if and only if, in some basis or in every basis—doesn't matter—which we choose here and that we choose here, the determinant of T , sorry, the matrix of T , is an invertible matrix.

So, once we know that T can be represented in some basis here and some basis here by

an invertible matrix, you know this is equivalent to requiring that T is an iso.

Now, forget about T . We know how to decide whether a matrix is invertible or not. We just take the determinant of this matrix and see if the number is zero or not. The value of this determinant, assuming it's not zero, is going to depend on the choices. However, the property of this number not being zero is not going to matter.

As someone else in here phrased it, and as my dad phrased it, if in some basis here and here the matrix has determinant zero, then this will continue to be the case in all other bases. But if the determinant is not zero in some basis here and some basis here, it's going to continue not to be zero in every other choice of basis; however, the value which is not zero may depend on your choice.

Okay? So, let's get to the next question. I don't know; I have, now it's just confusing.

Right, so we begin a new topic—an extremely important, central topic when we are providing those sorts of genetics. This is quite a strange word game.

So, I am now using English, but the name in English is Eigen Vectors.

21.a Eigenvectors: Eigen Vectors

Definition (21.a.1). Let V be a vector space over K and $T: V \rightarrow V$ a linear map. We say that $\lambda \in K$ is an eigenvalue of T if $\lambda \neq 0$ and $v \in V$ s.t. $Tv = \lambda v$. Any vector $v \in V$ with this property (i.e. $Tv = \lambda v$) is called an eigenvector of T for the eigenvalue λ .

Definition (21.a.2). Let V be a finite dimensional vector space over K . Let $T: V \rightarrow V$ be a linear map. We say that T is diagonalizable if $\exists$ a basis B of V such that

$$[T]_B^B = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} (*)$$

Lemma (21.a.3). T is diagonalizable iff $\exists$ a basis $B = \{v_1, \dots, v_n\}$ consisting of eigenvectors of T . In this case, if $Tv_i = \lambda_i v_i$ for some $\lambda_i \in K$. Moreover, if for some basis $B = \{v_1, \dots, v_n\}$ the matrix $[T]_B^B$ has the shape $(*)$, then $\forall k, \lambda_i$ is an eigenvalue of T and v_i is an eigenvector of T .

Definition. A matrix $A \in M_{n \times n}(K)$ is called diagonalizable if the $T_A: K^n \rightarrow K^n$ is diagonalizable.

Example. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $K = \mathbb{R}$.

$$T_A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = A \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\lambda = 3$ is an eigenvalue of T_A ,

and

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ is an eigenvect. of } \lambda_1 = 3.$$

$$T_A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

so

$$v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ is an eigenvect. of the eigenvalue } \lambda_2 = 1.$$

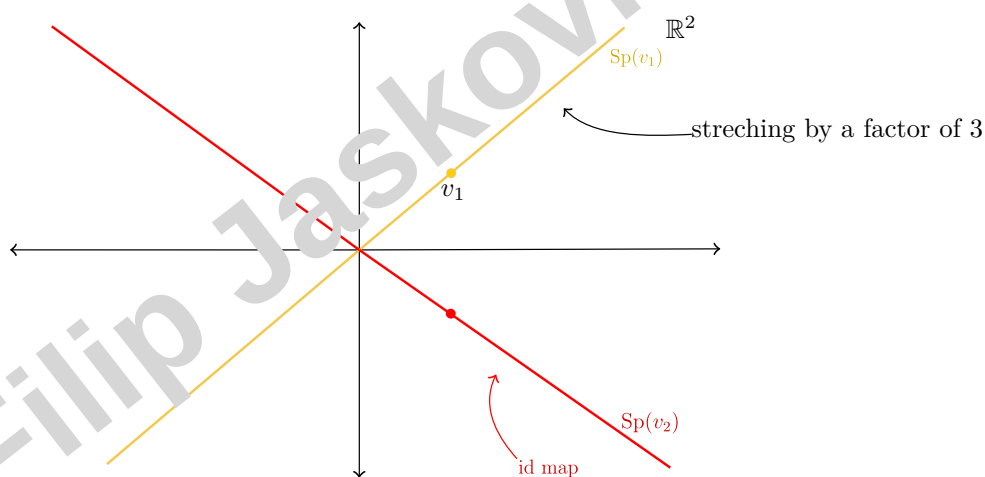


Figure 6.1: FigEigen1

Quick Sidequest

Goal: prove the following statement.

Theorem (Completeness). Let $\mathcal{L}$ be an arbitrary (uncountable) signature, and let T be a consistent set of $\mathcal{L}$ -sentences. Then T has a model.

Subgoal: construct an $\mathcal{L}$ -structure M^* with domain A^* .

Lemma (Compactness). Let T be an $\mathcal{L}$ -theory such that every finite subset $\Phi \subseteq T$ has an $\mathcal{L}$ -structure M_Φ satisfying $M_\Phi \models \Phi$. Then T has a model.

Definition. Let S be a set. An ultrafilter $\mathcal{U} \subseteq \mathcal{P}(S)$ over S is a filter on S such that

1. $\mathcal{U}$ is a filter on S .
2. For every $X \in \mathcal{P}(S)$, we have $X \in \mathcal{U}$ or $S \setminus X \in \mathcal{U}$.

Theorem. Every filter can be extended to an ultrafilter.

Definition. Let $\mathcal{L}$ be an arbitrary fixed signature, let $I \neq \emptyset$ be a set, and for each $i \in I$ let

$$M_i$$

be an $\mathcal{L}$ -structure with domain A_i . Put

$$A := \prod_{i \in I} A_i,$$

and identify elements of A with functions

$$g : I \rightarrow \prod_{i \in I} A_i$$

such that $g(i) \in A_i$ for each $i \in I$.

Definition. Let $\mathcal{U} \subseteq \mathcal{P}(I)$ be an ultrafilter over I . Define a relation on A by

$$f \sim g \Leftrightarrow \{i \in I \mid f(i) = g(i)\} \in \mathcal{U}.$$

Let $A^* = A / \sim$.

Definition. For each constant symbol $c \in \mathcal{L}$, define $f_c \in A$ by

$$f_c(i) := c^{M_i} \quad (i \in I),$$

and define

$$c^{M^*} := [f_c].$$

Definition. For every n -ary function symbol $F \in \mathcal{L}$, define

$$F^{M^*} : (A^*)^n \rightarrow A^*$$

by

$$F^{M^*}([f_0], \dots, [f_{n-1}]) = [f] \Leftrightarrow \{u \in I \mid F^{M_u}(f_0(u), \dots, f_{n-1}(u)) = f(u)\} \in \mathcal{U}.$$

Definition. For every n -ary relation symbol $R \in \mathcal{L}$, define $R^{M^*} \subseteq (A^*)^n$ by

$$\langle [f_0], \dots, [f_{n-1}] \rangle \in R^{M^*} \Leftrightarrow \{ u \in I \mid \langle f_0(u), \dots, f_{n-1}(u) \rangle \in R^{M_u} \} \in \mathcal{U}.$$

Definition. The $\mathcal{L}$ -structure M^* with domain A^* is then an ultraproduct of the $\mathcal{L}$ -structures M_i ($i \in I$) with respect to $\mathcal{U}$.

Definition. If $M_i = M$ for every $i \in I$, then M^* is called an ultrapower of M (with respect to $\mathcal{U}$).

Theorem (Łoś). Let σ be an $\mathcal{L}$ -sentence. Then

$$M^* \models \sigma \Leftrightarrow \{ i \in I \mid M_i \models \sigma \} \in \mathcal{U}.$$

Proof of compactness. Let

$$I := \{ \Phi \subseteq T \mid \Phi \text{ finite} \}.$$

For every $\Phi \in I$, choose an $\mathcal{L}$ -structure M_Φ with domain A_Φ such that

$$M_\Phi \models \Phi.$$

For every $\Phi \in I$, define

$$\Delta(\Phi) := \{ \Phi' \in I \mid \Phi \subseteq \Phi' \}.$$

Then $\Phi \in \Delta(\Phi)$, so $\Delta(\Phi) \neq \emptyset$. Moreover, for all $\Phi_1, \Phi_2 \in I$,

$$\Delta(\Phi_1) \cap \Delta(\Phi_2) = \Delta(\Phi_1 \cup \Phi_2),$$

and $\Phi_1 \cup \Phi_2 \in I$, hence the family

$$\mathcal{F} := \{ X \subseteq I \mid \exists \Phi \in I : \Delta(\Phi) \subseteq X \}$$

is a filter on I . By the previous theorem, $\mathcal{F}$ extends to an ultrafilter $\mathcal{U}$ on I .

Let M^* be the ultraproduct of the structures M_Φ ($\Phi \in I$) with respect to $\mathcal{U}$.

Now fix an arbitrary sentence $\sigma_0 \in T$. Then $\{\sigma_0\} \in I$ and

$$M_{\{\sigma_0\}} \models \sigma_0.$$

For every $\Phi \in \Delta(\{\sigma_0\})$, we have $\sigma_0 \in \Phi$, and since $M_\Phi \models \Phi$, it follows that

$$M_\Phi \models \sigma_0.$$

Thus

$$\Delta(\{\sigma_0\}) = \{ \Phi \in I \mid \sigma_0 \in \Phi \} \subseteq \{ \Phi \in I \mid M_\Phi \models \sigma_0 \}.$$

Observe that $\Delta(\{\sigma_0\}) \in \mathcal{F} \subseteq \mathcal{U}$, so

$$\{\Phi \in I \mid M_\Phi \models \sigma_0\} \in \mathcal{U}.$$

By Łoś's theorem, it follows that

$$M^* \models \sigma_0.$$

Because $\sigma_0 \in T$ was arbitrary, $M^* \models T$. Thus T has a model. $\square$

Recap of the strategy. To prove completeness for arbitrary signatures, it is enough to show that every finite subset $\Phi \subseteq T$ has a model. For such a finite Φ , only finitely many non-logical symbols occur, so we pass to the finite subsignature $\mathcal{L}'_\Phi$, construct a model there, extend it back to an $\mathcal{L}$ -structure, and then apply compactness.

Proof of completeness. Assume that T is consistent. Let $\Phi \subseteq T$ be finite. Then Φ is consistent as well.

Let $\mathcal{L}'_\Phi \subseteq \mathcal{L}$ be the finite subsignature consisting of all non-logical symbols appearing in sentences of Φ . Then Φ is a consistent $\mathcal{L}'_\Phi$ -theory. Since $\mathcal{L}'_\Phi$ is finite, the finite-signature model-existence theorem applies, so there exists an $\mathcal{L}'_\Phi$ -structure

$$M'_\Phi$$

with domain A_Φ such that

$$M'_\Phi \models \Phi.$$

Now extend M'_Φ to an $\mathcal{L}$ -structure M_Φ with the same domain A_Φ by choosing arbitrary interpretations for symbols in $\mathcal{L} \setminus \mathcal{L}'_\Phi$. Since the sentences in Φ only use symbols from $\mathcal{L}'_\Phi$, we still have

$$M_\Phi \models \Phi.$$

Thus for every finite subset $\Phi \subseteq T$, there exists an $\mathcal{L}$ -structure M_Φ such that

$$M_\Phi \models \Phi.$$

By Lemma 1, T has a model. $\square$

Lecture 36: Notes pending

18 Mar 2026

21.a.1 Lecture 36 Placeholder

Proposition (21.a.4). Let V be a vector space over a field K and let $T : V \rightarrow V$ be a linear map. Suppose $\lambda_1, \dots, \lambda_k$ are eigenvalues of T , s.t. they are pairwise distinct (i.e. $\lambda_i \neq \lambda_j \forall i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent. (put differently, eigenvectors corresponding to distinct eigenvalues are linearly independent).

Proof. Induction on n . For $n = 1$, v_1 is and eigenvector of λ_1 . By definition of eigenvector, $v_1 \neq 0$. So v_1 is linearly independent. Let $n > 1$ and suppose the

proposition holds for every list of n eigenvalues and eigenvectors. We will show the proposition holds also for any list of $n + 1$ eigenvalues and eigenvectors. Indeed, let $\lambda_1, \dots, \lambda_{n+1}$ be pairwise distinct eigenvalues of T and let $v_1, \dots, v_n, v_{n+1}$ be eigenvectors corresponding to $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$. Assume

$$a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1} = 0(*)$$

We will show that $a_1 = \dots = a_n = a_{n+1} = 0$. Applying T to $(*)$ we get

$$a_1 T v_1 + \dots + a_n T v_n + a_{n+1} T v_{n+1} = 0$$

But $T v_i = \lambda_i v_i$, so we get

$$a_1 \lambda_1 v_1 + \dots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0$$

Consider $(**) - \lambda_{n+1} \cdot (*)$:

$$a_1 (\lambda_1 - \lambda_{n+1}) v_1 + \dots + a_n (\lambda_n - \lambda_{n+1}) v_n = 0.$$

By the induction hypothesis, $v_1, \dots, v_n$ are linearly independent, hence,

$$a_1 (\lambda_1 - \lambda_{n+1}) = \dots = a_n (\lambda_n - \lambda_{n+1}) = 0.$$

By assumption $\lambda_{n+1} \neq \lambda_i \forall 1 \leq i \leq n$ so $\lambda_i - \lambda_{n+1} \neq 0 \forall 1 \leq i \leq n$. Hence, $a_1 = \dots = a_n = 0$. Substituting this back in $(*)$:

$$a_{n+1} v_{n+1} = 0$$

. Since $v_{n+1} \neq 0$ (it is an eigenvector!), we deduce that $a_{n+1} \cdot v_{n+1} = 0$. This completes the induction step and the proof. $\square$

An eigenvector for a given eigenvalue is not unique. If v is an eigenvector for λ and $c \in K^\times$, then $c v$ is again an eigenvector for λ . More generally, all eigenvectors for λ , together with 0, form the eigenspace $\text{Eig}_T(\lambda)$.

21.a.2 The characteristic polynomial

Proposition (21.a.5). Let V be a vector space over a field K and let $T : V \rightarrow V$ be a linear map. Then $\lambda \in K$ is an eigenvalue of T iff the map $T - \lambda \cdot \text{id}_V : V \rightarrow V$ is NOT injective.

Proof. λ is an eigenvalue of $T \Leftrightarrow \exists 0 \neq v \in V$ s.t. $T v = \lambda v \Leftrightarrow \exists 0 \neq v \in V$ s.t. $(T - \lambda \cdot \text{id}_V)(v) = 0 \Leftrightarrow \ker(T - \lambda \cdot \text{id}_V) \neq \{0\} \Leftrightarrow T - \lambda \cdot \text{id}_V$ is NOT injective. $\square$

Definition (21.a.6). Let V be a vector space over a field K and let $T : V \rightarrow V$ be a linear map. Define $P_T(x) := \det(T - x \cdot \text{id}_V)$. We call $P_T(x)$ the *characteristic polynomial* of T . If $A \in M_{n \times n}(K)$, we define the characteristic polynomial of A , as $P_A(x) := \det(A - x \cdot I_n)$.

Definition (21.a.7). Let V be a vector space over a field K and let $T : V \rightarrow V$ be a linear map. Let $\lambda \in K$ be an eigenvalue of T . The eigenspace of T corresponding to λ is $\text{Eig}_T(\lambda) := \{v \in V : Tv = \lambda v\} = \ker(T - \lambda \cdot \text{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } T \text{ corresponding to } \lambda\} \cup \{0\}$. Clearly $\text{Eig}_T(\lambda) \subseteq V$ is a linear subspace (b.c. it is a kernel of a linear map).

Lemma (21.a.8). Let $A \in M_{n \times n}(K)$. Then:

1. $P_A(x)$ is a polynomial of degree n , with leading coefficient $(-1)^n$, namely,

$$P_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$$

with $c_i \in K$ for all i , and $c_n = (-1)^n$.

2. $P_{A^T}(x) = P_A(x)$.

3. If

$$A = \begin{pmatrix} \lambda_1 & \cdots & * \\ 0 & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

has upper triangular shape, then

$$P_A(x) = (\lambda_1 - x) \cdots (\lambda_n - x).$$

4. If

$$A = \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array} \right) \begin{matrix} \uparrow r \\ \downarrow s \end{matrix}$$

is a block matrix where

$$B_1 \in M_{r \times r}(K), \quad B_2 \in M_{s \times s}(K),$$

then

$$P_A(x) = P_{B_1}(x) \cdot P_{B_2}(x).$$

5. If A and B are similar matrices, $B = P^{-1}AP$ for some invertible P , then

$$P_B(x) = P_A(x).$$

Proof. 1)

$$P_A(x) = \det(A - xI) = \det \begin{pmatrix} A_{11} - x & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} - x & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & \cdots & \cdots & A_{nn} - x \end{pmatrix}$$

Denote this matrix by B . Then

$$P_A(x) = \sum_{\sigma \in S_n} (\text{sgn } \sigma) B_{1,\sigma(1)} \cdots B_{n,\sigma(n)}.$$

Hence

$$P_A(x) = (A_{11} - x)(A_{22} - x) \cdots (A_{nn} - x) + \sum_{\sigma \neq \text{id}} (\text{sgn } \sigma) B_{1,\sigma(1)} \cdots B_{n,\sigma(n)}.$$

Here the first term corresponds to $\sigma = \text{id}$, and

$$\text{sgn}(\text{id}) = 1.$$

If $\sigma \neq \text{id}$, then at least one of the factors

$$B_{1,\sigma(1)}, \dots, B_{n,\sigma(n)}$$

will not have an x in it. Hence

$$P_A(x) = (-1)^n x^n + \text{polynomial of deg} \leq n-1.$$

This proves 1.

2)

$$P_{A^T}(x) = \det(A^T - xI) = \det((A - xI)^T) = \det(A - xI) = P_A(x).$$

(3+4) Exc.

5)

$$\begin{aligned} P_B(x) &= \det(B - xI) = \det(P^{-1}AP - xI) \\ &= \det(P^{-1}(A - xI)P) = \det(A - xI) = P_A(x). \end{aligned}$$

□

As already suggested, the roots of the characteristic polynomial are the candidates for eigenvalues. However, this depends on the field. For example, the polynomial $x^2 + 1$ has no roots over $\mathbb{R}$, but it has roots $\pm i$ over $\mathbb{C}$. Thus a real matrix may have no real eigenvalues, even though it has complex eigenvalues after extending scalars to $\mathbb{C}$.

Corollary. If V is an n -dimensional vector space over a field K and $T : V \rightarrow V$ is a linear map, then $P_T(x)$ is a polynomial in x of degree n with leading coefficient $(-1)^n$.

Proof. Let $\mathcal{B}$ be a basis of V . Then $[T - x\text{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}} - xI_n$. Hence

$$P_T(x) = \det(T - x\text{id}_V) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = P_A(x),$$

where $A = [T]_{\mathcal{B}}^{\mathcal{B}}$. By Corollary follows from (1) of Lemma 21.a.8. □

Definition (21.a.9). Let $A \in M_{n \times n}(K)$. Define the *trace* of A as $\text{Tr}(A) := \sum_{i=1}^n A_{ii}$. If V is finite-dimensional and $T : V \rightarrow V$ is a linear map, define $\text{Tr}(T) := (\text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}))$, where $\mathcal{B}$ is any basis of V .

Lemma (21.a.10). 1. $\forall A, B \in M_{n \times n}(K)$, $\text{Tr}(A \cdot B) = \text{Tr}(B \cdot A)$.

2. If A and B are similar, then $\text{Tr}(A) = \text{Tr}(B)$. In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V , then $\text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \text{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'})$. Hence, the trace of a linear map is well-defined.

Proof. 1. $\text{Tr}(A \cdot B) = \sum_{i=1}^n (A \cdot B)_{ii} = \sum_{i=1}^n \left(\sum_{j=1}^n A_{ij} \cdot B_{ji} \right) = \sum_{ij} A_{ij} \cdot B_{ji} =$
 $\sum_{ij} A_{ji} \cdot B_{ij} = \sum_{i=1}^n \sum_{j=1}^n B_{ij} \cdot A_{ji} = \sum_{i=1}^n (B \cdot A)_{ii} = \text{Tr}(B \cdot A)$.

2. If $B = P^{-1}AP$, then

$$\text{Tr}(B) = \text{Tr}(P^{-1}AP) = \text{Tr}(P^{-1}(AP)) = \text{Tr}((AP)P^{-1}) = \text{Tr}(A),$$

by (1).

□

Lecture 37: Notes pending

20 Mar 2026

21.b Eigenvectors: Direct sums of eigenspaces

Recap

Characteristic polynomial. For $A \in M_{n \times n}(K)$, we define

$$p_A(x) := \det(A - xI) = (-1)^n x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0.$$

Trace. For $A = (A_{ij}) \in M_{n \times n}(K)$, we set

$$\text{Tr}(A) := A_{11} + A_{22} + \cdots + A_{nn}.$$

Moreover,

$$\text{Tr}(AB) = \text{Tr}(BA) \quad \text{and} \quad \text{Tr}(P^{-1}AP) = \text{Tr}(A)$$

for every invertible P .

Endomorphisms. If $T: V \rightarrow V$ is a linear map on a finite-dimensional vector space and $A = [T]_{\mathcal{B}}^{\mathcal{B}}$ for a basis $\mathcal{B}$ of V , then

$$p_T(x) := p_A(x), \quad \text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \text{Tr}(A).$$

Both quantities are independent of the choice of basis $\mathcal{B}$.

37

Somehow nonlinear algebra enters when you start to study linear algebra.

Claim. $\text{Tr}(T)$ is independent of $\mathcal{B}$. Indeed, if $\mathcal{B}'$ is another basis of V , then

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \underbrace{[\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'}}_{=P^{-1}} [T]_{\mathcal{B}}^{\mathcal{B}} \underbrace{[\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}}}_{=P}$$

So $\text{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}) = \text{Tr}[T]_{\mathcal{B}}^{\mathcal{B}}$.

Lemma (21.a.11). $C_{n-1} = (-1)^{n-1} \text{Tr}(A)$ and $c_0 = \det A$.

Proof.

$$p_A(x) = \det \begin{pmatrix} A_{11} - x & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} - x & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} - x \end{pmatrix}$$

Denote this matrix by B . Then

$$p_A(x) = (A_{11} - x)(A_{22} - x) \cdots (A_{nn} - x) + \sum_{\substack{\sigma \neq \text{id} \\ \sigma \in S_n}} (\text{sgn } \sigma) \cdot \underbrace{B_{1,\sigma 1} \cdots B_{n,\sigma n}}_{(*)}.$$

If $\sigma \neq \text{id}$, then $\exists$ at least two indices $i \neq j$ s.t. $\sigma i \neq i$, $\sigma j \neq j$. Therefore, in the

product $(*)$, $\exists$ at least two factors

$$(B_{i,\sigma i} \& B_{j,\sigma j})$$

that are not from the diag. of B . So these two factors do NOT contain x . Hence

$$\deg(\circledast) \leq n - 2.$$

So

$$\begin{aligned} p_A(x) &= (A_{11} - x)(A_{22} - x) \cdots (A_{nn} - x) + (\text{polyn. of deg } \leq n - 2) = \\ &= (-1)^n x^n + A_{11} \cdot (-1)^{n-1} \cdot x^{n-1} + A_{22}(-1)^{n-1} x^{n-1} \\ &\quad + \cdots + A_{nn}(-1)^{n-1} x^{n-1} + (\text{another polyn. of deg } \leq n - 2) = \\ &= (-1)^n x^n + \underbrace{(-1)^{n-1} \text{Tr}(A)}_{= c_{n-1}} x^{n-1} + c_{n-2} x^{n-2} + \cdots + c_1 x + c_0. \end{aligned}$$

$$c_0 = p_A(0) = \det(A - 0 \cdot I) = \det(A).$$

Substitute $x = 0$. □

Definition (21,2,12). Let V be a vect. sp. over K and $U_1, \dots, U_r \leq V$ lin. subspaces. Denote $W := U_1 + \cdots + U_r$. We say that W is a direct sum of $U_1, \dots, U_r$ if every $w \in W$ has a unique representation as a sum

$$w = u_1 + \cdots + u_r \quad \text{with } u_i \in U_i \ \forall i.$$

Remark. Exercise. Show that $U_1 + \cdots + U_r$ is a direct sum iff any of the following conditions hold:

- 1) The only way to write 0 as a sum

$$0 = u_1 + \cdots + u_r, \text{ with } u_i \in U_i$$

is as

$$0 = 0 + \cdots + 0.$$

- 2) $\forall 2 \leq j \leq r$ we have

$$U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\}.$$

Under the additional assumption that $U_1, \dots, U_r$ are f. dimensional, we have:

$$U_1 + \cdots + U_r \text{ is a direct sum iff } \dim(U_1 + \cdots + U_r) = \dim U_1 + \cdots + \dim U_r.$$

$$\Leftrightarrow$$

wherever $B_1, \dots, B_r$ are bases for $U_1, \dots, U_r$ respectively, then $B_1, \dots, B_r$ forms a basis for $U_1 + \cdots + U_r$.

Remark. Exercise. If $U_1, \dots, U_r$ is a direct sum, then $\exists$ a canonical isomorphism.

$$\begin{array}{ccc} U_1 \oplus \dots \oplus U_r & \xrightarrow{\cong} & U_1 + \dots + U_r \\ \Psi & & \Psi \\ (u_1, \dots, u_r) & \longmapsto & u_1 + \dots + u_r \end{array}$$

Proposition (21.a.13). Let V be a vector space over K , and let $T \in \text{End}(V)$. Let $\lambda_1, \dots, \lambda_r \in K$ be pairwise distinct eigenvalues of T . Then $\text{Eig}_T(\lambda_1), \dots, \text{Eig}_T(\lambda_r)$ is a direct sum.

Proof. Assume $u_1 + \dots + u_r = u'_1 + \dots + u'_r (*)$ where $u_i, u'_i \in \text{Eig}_T(\lambda_i) \forall i$. Rewrite $(*)$ as:

$$(u_1 - u'_1) + \dots + (u_r - u'_r) = 0 (**).$$

Remove/ignore in the sum $(**)$ all the summands $u_j - u'_j$ that are zero. $(**)$ becomes:

$$(u_{i_1} - u'_{i_1}) + \dots + (u_{i_s} - u'_{i_s}) = 0 (**').$$

where $0 \leq \ell, 1 \leq i_k \leq r \forall k$ and $u_{i_k} - u'_{i_k} \neq 0 \forall k$. Note that $u_{i_k} - u'_{i_k} \in \text{Eig}_T(\lambda_{i_k}) \Rightarrow u_{i_k} - u'_{i_k}$ is an eigenvector λ_{i_k} . But $\lambda_{i_1}, \dots, \lambda_{i_s}$ are pairwise distinct. By Proposition 21.a.4, $u_{i_1} - u'_{i_\ell}$ must be linearly independent. This contradicts the $(**')$, unless $\ell = 0$, id est $u_j = u'_j \forall j$. $\square$

Corollary (21.a.14). Assume V is finite-dimensional, $T : V \rightarrow V$ linear. Let $\lambda_1, \dots, \lambda_r$ be all the eigenvalues of T . (we assume that $\lambda_i \neq \lambda_j \forall i \neq j$) Then T is diagonalizable iff

$$\sum_{i=1}^r \dim(\text{Eig}_T(\lambda_i)) = \dim V (*).$$

Proof. Assume $(*) \Rightarrow T$ is diagonalizable. Choose a basis $\mathcal{B}_i$ for $\text{Eig}_T(\lambda_i)$. We have then that

$$W := \text{Eig}_T(\lambda_1) + \dots + \text{Eig}_T(\lambda_r)$$

is a direct sum. $\Rightarrow (\mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_r)$ is a basis for W . $\Rightarrow \dim W = \sum_{i=1}^r |\mathcal{B}_i| =$

$$\sum_{i=1}^r \dim(\text{Eig}_T(\lambda_i)) \underbrace{=}_{\text{by } (*)} \dim V. \Rightarrow W = V. V \text{ has a basis consisting of eigenvectors}$$

of T . $\Rightarrow T$ is diagonalizable. T is diagonalizable $\Rightarrow (*)$. This is left as an exercise to the reader. $\square$

Example.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

$$p_A(x) = (1 - x)^2 = (x - 1)^2.$$

$$p_B(x) = \dots = (x - 1)^2.$$

Example. Pick $\lambda \in K$, consider

$$J := J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ 0 & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}.$$

Jordan block.

$$p_{J_{\lambda,n}}(x) = (\lambda - x)^n.$$

The only eigenvalue of $J_{\lambda,n}$ is λ ,

Note (Eigenvectors).

$$J - \lambda I = \begin{pmatrix} 0 & 1 & & 0 \\ & 0 & 1 & \\ & & \ddots & \ddots \\ 0 & & & 0 & 1 \\ & & & & 0 \end{pmatrix}$$

The cols. $\#1, \dots, \#n$ of $\uparrow$ are lin. indep. And we get that $\text{rank}(J - \lambda I_n) = n - 1$.

$$\dim \ker(J - \lambda I) = 1,$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \text{ is an eigenvector of } J_{\lambda,n}.$$

$$\text{Eig}_J(\lambda) = \text{Sp}(e_1).$$

Proof. By assumption $\exists$ an invertible matrix P , s.t.

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} =: \Lambda$$

$$\Rightarrow p_A(x) = p_\Lambda(x) = \det \begin{pmatrix} \lambda_1 - x & & \\ & \ddots & \\ & & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \cdots (\lambda_n - x).$$

□

Remark.

$$\frac{\det(xI - A)}{\det(A - xI)}$$

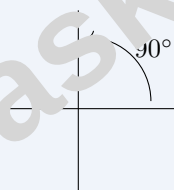
37

Example. $K = \mathbb{R}$, $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

$$T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$(x, y) \mapsto (-y, x), \quad \nexists \text{ eigenvalues, vec.}$$

$$p_A(x) = \det \begin{pmatrix} -x & -1 \\ 1 & -x \end{pmatrix} = x^2 + 1, \quad \nexists \text{ roots to } p_A(x) \text{ in } K = \mathbb{R}.$$

Hence $\nexists$ eigenvalues.**Lecture 38: Notes pending**

25 Mar 2026

21.c Eigenvectors: Geometric and algebraic multiplicities

We first need a quick digression on polynomials. Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$, we get can write

$$p(x) = (x - \lambda)q(x) + R,$$

38

where $q(x) \in K[x]$ with $\deg q = \deg p - 1$ and $R \in K$. The process by which one is able to compute this manually is similar to integer long division. Notice that in that case, $p(\lambda) = R$.

Using this, we can conclude that

$$p(\lambda) = 0 \Leftrightarrow p(x) = (x - \lambda)q(x).$$

We say that λ is a zero (or root) of $p(x)$.

Sometimes λ is a zero also of $q(x)$, which means that we can write $q(x) = (x - \lambda)h(x)$ with $h(x)$ such that $\deg h = \deg p - 2$. This motivates the following definition.

Definition (Multiplicity). Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. We say that λ is a zero of multiplicity $m \in \mathbb{Z}_{\geq 1}$ if $p(x) = (x - \lambda)^m g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$.

If $m = 1$, we say λ is a simple zero of $p(x)$. If $m \geq 2$ we say that λ is a multiple zero.

Definition (Geometric and algebraic multiplicity). Let V be a vector space over K and $T : V \rightarrow V$ linear and $\lambda \in K$ an eigenvalue of T . We define

$$m_g(T, \lambda) := \dim \operatorname{Eig}_T(\lambda).$$

Assume, in addition, that $\dim V < \infty$ and define the algebraic multiplicity of λ to be multiplicity of the zero λ in $P_T(x)$. We denote this number $m_a(T, \lambda)$.

Proposition (Inequality for multiplicities). If V is finite dimensional, it always holds $m_g(T, \lambda) \leq m_a(T, \lambda)$.

Proof. Let $v_1, \dots, v_k$ be a basis for $\operatorname{Eig}_T(\lambda)$, so $k = m_g(T, \lambda)$. Extend this to a basis $\mathcal{B} = (v_1, \dots, v_k, v_{k+1}, \dots, v_n)$ of V .

It's not hard to realise that $[T]_{\mathcal{B}}^{\mathcal{B}}$ will be a block matrix

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c} \Lambda & * \\ \hline 0 & C \end{array} \right)$$

where $\Lambda = \lambda \cdot I_k$ and C is $(n - k) \times (n - k)$ and could be anything. Then, we have

$$P_{[T]_{\mathcal{B}}^{\mathcal{B}}}(x) = P_{[T]}(x) = P_{\Lambda}(x) \cdot P_C(x) = (\lambda - x)^k \cdot P_C(x).$$

Hence $m_a(T, \lambda) \geq k = m_g(T, \lambda)$. □

Example. Consider the Jordan block $J = J_{\lambda, n}$. We saw that $P_J(x) = (\lambda - x)^n$. We also saw that $\operatorname{Eig}_J(\lambda)$ is 1-dimensional. This gives $m_a(J, \lambda) = n$ and $m_g(J, \lambda) = 1$. In a way, this is the most extreme example, m_g being minimal and m_a maximal.

21.c.1 Diagonalisability

Now the goal is to find clear criteria for diagonalisability.

Theorem (Broader criteria for diagonalisation). Let V be an n -dimensional vector space over K and $T \in \operatorname{End}(V)$. The following statements are equivalent.

1. T is diagonalisable.

2. $\exists$ basis for V whose elements are eigenvectors.
3. $P_T(x)$ splits into a product of linear factors and $m_g(T, \lambda) = m_a(T, \lambda)$.
4. $\sum_{i=1}^k \dim \text{Eig}_T(\lambda_i) = \dim V$.
5. $V \cong \bigoplus_{i=1}^k \text{Eig}_T(\lambda_i)$.

Proof. (1) $\Leftrightarrow$ (2) has been proven already.

(1) $\Rightarrow$ (3). Let $\mathcal{B}$ be a basis in which $[T]_{\mathcal{B}}^{\mathcal{B}} = \Lambda$ is diagonal. By changing the order of the elements of $\mathcal{B}$, we can arrange that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \text{diag}(\underbrace{\lambda_1, \dots, \lambda_1}_{\times \ell_1}, \underbrace{\lambda_2, \dots, \lambda_2}_{\times \ell_2}, \dots, \underbrace{\lambda_k, \dots, \lambda_k}_{\times \ell_k})$$

where $\lambda_1, \dots, \lambda_k$ are the eigenvalues of T , and

$$\lambda_i \neq \lambda_j \quad \forall i \neq j.$$

Write $\mathcal{B} = (v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, v_1^{(2)}, \dots, v_{\ell_2}^{(2)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$,

$$\Rightarrow p_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k}. \quad (*)$$

So $p_T(x)$ splits as a product of linear terms. It remains to show that $m_g(T, \lambda_i) = m_a(T, \lambda_i)$ for all i . Clearly, $v_1^{(i)}, \dots, v_{\ell_i}^{(i)} \in \text{Eig}_T(\lambda_i)$ are linearly independent, so

$$\ell_i \leq \dim \text{Eig}_T(\lambda_i) = m_g(T, \lambda_i).$$

But $\ell_i = m_a(T, \lambda_i)$, so

$$m_a(T, \lambda_i) \leq m_g(T, \lambda_i).$$

But we also always have $m_g(T, \lambda_i) \leq m_a(T, \lambda_i)$. Therefore

$$m_g(T, \lambda_i) = m_a(T, \lambda_i).$$

(3) $\Rightarrow$ (4). Assume

$$p_T(x) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k},$$

and

$$m_g(T, \lambda_i) = m_a(T, \lambda_i) \quad \forall i.$$

Clearly $\lambda_1, \dots, \lambda_k$ are all the eigenvalues of T because these are all the zeros of $p_T(x)$. Also by definition $\ell_i = m_a(T, \lambda_i)$, so

$$n = \dim V = \deg p_T(x) = \ell_1 + \cdots + \ell_k = \sum_{i=1}^k m_a(T, \lambda_i) \stackrel{\text{by assumption}}{=} \sum_{i=1}^k m_g(T, \lambda_i) = \sum_{i=1}^k \dim \text{Eig}_T(\lambda_i).$$

(4) $\Rightarrow$ (5). Recall that

$$\text{Eig}_T(\lambda_1) + \cdots + \text{Eig}_T(\lambda_k) \subseteq V$$

is a direct sum.

$$\sum_{i=1}^k \dim \operatorname{Eig}_T(\lambda_i) = \dim(\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k))$$

By assumption, the left-hand side equals $\dim V$. Hence

$$\dim(\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k)) = \dim V.$$

Since

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k)$$

is a subspace of V , we get that

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) = V.$$

But since it is also isomorphic to

$$\operatorname{Eig}_T(\lambda_1) \oplus \cdots \oplus \operatorname{Eig}_T(\lambda_k),$$

it is a direct sum. Hence

$$V \cong \bigoplus_{i=1}^k \operatorname{Eig}_T(\lambda_i)$$

(5) $\Rightarrow$ (1). If

$$V \cong \operatorname{Eig}_T(\lambda_1) \oplus \cdots \oplus \operatorname{Eig}_T(\lambda_k),$$

then, since

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) \subseteq V$$

is a direct sum, we must have that

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) = V.$$

So, if

$$v_1^{(i)}, \dots, v_{\ell_i}^{(i)} \text{ is a basis for } \operatorname{Eig}_T(\lambda_i),$$

then

$$(v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$$

is a basis for V whose elements are eigenvectors of T . $\square$

Corollary (21.C.2). If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ has $m_a(T, \lambda) = 1$, then T is diagonalizable.

Proof. We have $1 \leq m_g(T, \lambda) \leq m_a(T, \lambda) = 1$.

$$\Rightarrow \forall \text{ eigenvalue } \lambda, \quad m_g(T, \lambda) = m_a(T, \lambda),$$

The corollary follows from the previous theorem. $\square$

21.c.2 Trigonalization

This means bringing a matrix/endomorphism to upper triangular form. For example,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad p_A(x) = (x - 1)^2.$$

$$m_a(A, 1) = 2, \quad m_g(A, 1) = 1.$$

Definition (21.C.3). We say that $T \in \text{End}(V)$ is trigonalizable if $\exists$ a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} \quad (n = \dim V).$$

Theorem (21.C.4). Let V be a vector space over K and $T \in \text{End}(V)$. Then T is trigonalizable iff

$$p_T(x) \text{ splits as } p_T(x) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k}.$$

Note. Digression.

Theorem (Fundamental theorem of algebra (21.C.5)). Every $p(z) \in \mathbb{C}[x]$ (polynomial with $\mathbb{C}$ -coefficients) splits as a product of linear factors and a constant factor, i.e.

$$p(z) = c(z - a_1)^{n_1} \cdots (z - a_r)^{n_r},$$

with $c \in \mathbb{C}, r \geq 0, a_j \in \mathbb{C}, n_j \in \mathbb{Z}_{\geq 1}$.

Lecture 39: Notes pending

37 Mar 2026

21.d Eigenvectors: Minimal polynomial & Cayley–Hamilton theorem

Corollary. Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \text{End}(V)$ is trigonalizable.

Proof of Theorem (21.C.4). $\Rightarrow$. Suppose T is trigonalizable. Let $\mathcal{B}$ be a basis for

V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ 0 & & \ddots & * \\ & & & \lambda_n \end{pmatrix}.$$

Then

$$p_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - xI) = \det \begin{pmatrix} \lambda_1 - x & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ 0 & & \ddots & * \\ & & & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \cdots (\lambda_n - x).$$

$\Leftarrow$. Induction on $n := \dim V$.

$n = 1$. In this case $[T]_{\mathcal{B}}^{\mathcal{B}} = (\lambda)$ (for any basis), and this is diagonal. (Nothing to prove).

Let $n \geq 1$. Assume that the theorem holds for all vector spaces of dimension $\leq n$ and all their endomorphisms. Let V be an $(n+1)$ -dimensional vector space, and $T \in \text{End}(V)$, s.t. $p_T(x) = \text{product of linear factors}$.

$\Rightarrow \exists$ one eigenvalue $\lambda_1 \in K$ of T .

Choose an eigenvector, $0 \neq u_1 \in \ker(T - \lambda_1 \text{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$ of V . We have

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} \lambda_1 & & & \\ & \boxed{*} & & \\ & & \ddots & \\ & & & 0 \end{pmatrix} = \left(\begin{array}{c|ccc} \lambda_1 & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array} \right) =: M,$$

with A the $n \times n$ block.

We have

$$p_T(x) = (\lambda_1 - x) \cdot p_A(x).$$

Since $p_T(x)$ splits as a product of linear factors, so does $p_A(x)$.

Consider $S_A : K^n \rightarrow K^n$, $S_A(u) = Au$. By the induction hypothesis $\exists$ a basis $\mathcal{E}$ of K^n s.t.

$$[S_A]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

Denote by $\mathcal{E}$ the standard basis of K^n .

We have

$$p_T(x) = (\lambda_1 - x) \cdot p_A(x).$$

Since $p_T(x)$ splits as a product of linear factors, so does $p_A(x)$.

Consider $S_A : K^n \rightarrow K^n$, $S_A(u) = Au$. By the induction hypothesis $\exists$ a basis $\mathcal{E}$ of K^n s.t.

$$[S_A]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

Denote by $\mathcal{C}$ the standard basis of K^n .

$$[S_A]_{\mathcal{E}}^{\mathcal{E}} = [\text{id}]_{\mathcal{C}}^{\mathcal{E}} [S_A]_{\mathcal{C}}^{\mathcal{C}} [\text{id}]_{\mathcal{E}}^{\mathcal{C}} = \underbrace{[\text{id}]_{\mathcal{C}}^{\mathcal{E}}}_{Q^{-1}} \underbrace{[S_A]_{\mathcal{C}}^{\mathcal{C}}}_A \underbrace{[\text{id}]_{\mathcal{E}}^{\mathcal{C}}}_{\text{denote by } Q}.$$

So

$$Q^{-1}AQ = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}$$

Define

$$P := \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline \vdots & & Q & \\ 0 & & & \end{array} \right) \in M_{(n+1) \times (n+1)}(K).$$

Claim.

1) P is invertible and

$$P^{-1} = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array} \right)$$

2)

$$P^{-1}MP = \left(\begin{array}{c|cccc} \lambda_1 & *' & \cdots & \cdots & *' \\ \hline 0 & \lambda_2 & * & \cdots & * \\ \vdots & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & * \\ 0 & 0 & \cdots & 0 & \lambda_{n+1} \end{array} \right)$$

Proof of claim. For (1),

$$P \cdot \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array} \right) = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q \cdot Q^{-1} & \\ 0 & & & \end{array} \right) = I.$$

Therefore

$$\left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array} \right) = P^{-1}.$$

For (2),

$$\begin{aligned} P^{-1}MP &= \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array} \right) \left(\begin{array}{c|ccccc} \lambda_1 & *' & \cdots & \cdots & *' \\ \hline 0 & & & A & \\ \vdots & & & & \\ 0 & & & & \end{array} \right) \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q & \\ 0 & & & \end{array} \right) \\ &= \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array} \right) \left(\begin{array}{c|ccccc} \lambda_1 & \cdots & \cdots & \cdots & *' \\ \hline 0 & & & AQ & \\ \vdots & & & & \\ 0 & & & & \end{array} \right) \\ &= \left(\begin{array}{c|ccccc} \lambda_1 & \cdots & \cdots & \cdots & *' \\ \hline 0 & & & & \\ \vdots & & Q^{-1}AQ & & \\ \lambda_1 & *' & \cdots & \cdots & *' \\ \hline 0 & \lambda_2 & * & \cdots & * \\ \vdots & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & * \\ 0 & 0 & \cdots & 0 & \lambda_{n+1} \end{array} \right). \end{aligned}$$

Claim. $\exists$ a basis for V , s.t.

$$[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = P.$$

Proof of claim. Write

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1k} & \cdots & p_{1m} \\ \vdots & \vdots & & \vdots & & \vdots \\ \vdots & & \ddots & & & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mk} & \cdots & p_{mm} \end{pmatrix}, \quad (m := n+1 = \dim V).$$

Write $\mathcal{B}'$ as $\mathcal{B}' = (u_1, \dots, u_m)$. Define

$$\begin{aligned} v_1 &= p_{11}u_1 + p_{21}u_2 + \cdots + p_{m1}u_m, \\ v_2 &= p_{12}u_1 + p_{22}u_2 + \cdots + p_{m2}u_m, \\ &\vdots \\ v_k &= p_{1k}u_1 + p_{2k}u_2 + \cdots + p_{mk}u_m, \\ &\vdots \\ v_m &= p_{1m}u_1 + p_{2m}u_2 + \cdots + p_{mm}u_m. \end{aligned} \tag{*}$$

Exercise. $\mathcal{B} := (v_1, \dots, v_m)$ is a basis for V .

Hint. Since $\dim V = m$, it is enough to show $v_1, \dots, v_m$ are linearly independent. Suppose

$$c_1v_1 + \cdots + c_mv_m = 0. \tag{**}$$

Substitute (*) into (**), and obtain

$$P \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

But P is invertible,

$$\Rightarrow \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This will prove claim B.

We now have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}} [T]_{\mathcal{B}'}^{\mathcal{B}'} [\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = \underbrace{[\text{id}_V]_{\mathcal{B}'}^{\mathcal{B}}}_{P^{-1}} \underbrace{[T]_{\mathcal{B}'}^{\mathcal{B}'} }_M \underbrace{[\text{id}_V]_{\mathcal{B}}^{\mathcal{B}'} }_P = P^{-1}MP.$$

By claim A,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

□

21.d.1 Minimal polynomial & Cayley–Hamilton theorem

Definition (Polynomial in a matrix). Let $A \in M_{n \times n}(K)$. For $\forall k \geq 1$, define

$$A^k := \underbrace{A \cdots A}_{\times k}.$$

Let

$$g(x) = a_d x^d + \cdots + a_1 x + a_0 \in K[x].$$

Define

$$g(A) := a_d A^d + a_{d-1} A^{d-1} + \cdots + a_1 A + a_0 I,$$

where $I = I_n$ is the identity matrix.

Claim.

$$\forall A \in M_{n \times n}(K), \exists g \in K[x], g \neq 0, \text{ s.t. } g(A) = 0_{n \times n}.$$

Proof. Consider

$$I_n, A, A^2, \dots, A^r \in M_{n \times n}(K).$$

But

$$\dim M_{n \times n}(K) = n^2.$$

So, if $r > n^2$, the list

$$I_n, A, A^2, \dots, A^r$$

must be linearly dependent. Therefore

$$\exists a_0, \dots, a_r \in K, \text{ not all zero, s.t. } a_0 I_n + a_1 A + \cdots + a_r A^r = 0.$$

So take

$$g(x) = a_r x^r + \cdots + a_1 x + a_0.$$

Then $g \neq 0$ and $g(A) = 0_{n \times n}$. □

Note. Everything here applies also for $T \in \text{End}(V)$. Define

$$T^0 := \text{id}_V, \quad T^k := \underbrace{T \circ \cdots \circ T}_{\times k}.$$

Then we can define

$$g(T) := a_d T^d + a_{d-1} T^{d-1} + \cdots + a_1 T + a_0 \text{id}_V.$$

Also, if V is finite-dimensional, then

$$\dim \text{End}(V) = (\dim V)^2.$$

Definition (21.d.1). Let $T \in \text{End}(V)$. A minimal polynomial for T is a polynomial

$$0 \neq g(x) \in K[x]$$

of minimal possible degree, s.t. $g(T) = 0$.

Theorem (Cayley–Hamilton, 21.d.2). 1) Let $A \in M_{n \times n}(K)$. Then

$$p_A(A) = 0.$$

2) Let V be a finite-dimensional vector space over K , and $T \in \text{End}(V)$. Then

$$p_T(T) = 0.$$

Example. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$p_A(x) = \det \begin{pmatrix} a-x & b \\ c & d-x \end{pmatrix} = (a-x)(d-x) - bc = x^2 - (a+d)x + (ad-bc),$$

where $a+d = \text{Tr}(A)$ and $ad-bc = \det(A)$. Hence

$$p_A(A) = A^2 - (a+d)A + (ad-bc) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

“Easy proof” of CH^[1] (21.d.2):

$$p_A(x) = \det(A - xI),$$

so

$$p_A(A) = \det(A - A \cdot I) = \det(0) = 0.$$

[1]:set theorists hate this trick

Analytic proof. Assume first that $K = \mathbb{C}$. Prove 1) first for $A \in M_{n \times n}(\mathbb{C})$ which is diagonalizable. Indeed, if $\exists$ an invertible matrix Q , s.t.

$$Q^{-1}AQ = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} =: \Lambda,$$

and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ are the eigenvalues of A . Write

$$p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0.$$

Then

$$\begin{aligned}
 Q^{-1}p_A(A)Q &= c_n \underbrace{Q^{-1}A^nQ}_{(Q^{-1}AQ)^n} + c_{n-1} \underbrace{Q^{-1}A^{n-1}Q}_{(Q^{-1}AQ)^{n-1}} + \cdots + c_1 Q^{-1}AQ + c_0 I \\
 &= c_n (Q^{-1}AQ)^n + c_{n-1} (Q^{-1}AQ)^{n-1} + \cdots + c_1 (Q^{-1}AQ) + c_0 I \\
 &= p_A(Q^{-1}AQ) = p_A(\Lambda) \quad (\text{check}) \\
 &= \begin{pmatrix} p_A(\lambda_1) & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & p_A(\lambda_n) \end{pmatrix} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 \end{pmatrix},
 \end{aligned}$$

because $p_A(\lambda_i) = 0 \forall i$. □

TODO: Add biran quote rambling rant here!!!

Lecture 40: Euclidean and Hermitian (unitary spaces)

01 Apr 2026

Preparations for the proof

Definition (21.d.7). Let V be a vector space and $T \in \text{End}(V)$. Let $v \in V$. Define

$$\langle v \rangle := \text{Sp}\{T^k v \mid k \geq 0\} = \text{Sp}\{v, Tv, T^2v, \dots\}.$$

$\langle v \rangle$ is called the T -cyclic subspace generated by v . Exc. 1) $\langle v \rangle_T$ is invariant under T (i.e. $T(\langle v \rangle_T) \subseteq \langle v \rangle_T$). 2) $\langle v \rangle_T$ is the smallest linear subspace of V which contains v and is T -invariant. i.e. if $U \subseteq V$ is a linear subspace and $v \in U$ and $T(U) \subseteq U$, then $\langle v \rangle_T \subseteq U$.

Lemma (21.d.8). Suppose that $v, Tv, T^2v, \dots, T^{d-1}v$, $d \geq 1$, are linearly independent, and $T^d v = c_0 v + c_1 Tv + \cdots + c_{d-1} T^{d-1}v$ for some $c_i \in K$.

1. $B = v, Tv, \dots, T^{d-1}v$ forms a basis for $\langle v \rangle$.
2. Denote $S := T|_{\langle v \rangle_T}: \langle v \rangle_T \rightarrow \langle v \rangle_T$ (exc.1). Then

$$[S]_B^B = \begin{pmatrix} 0 & \cdots & 0 & c_0 \\ \boxed{\begin{matrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{matrix}} & \begin{matrix} c_1 \\ \vdots \\ c_{d-1} \end{matrix} \end{pmatrix} \in M_{d \times d}(K), \quad \boxed{\forall d \geq 2.}$$

with lower-left block I_{d-1} .

If $d = 1$, then $B = (v)$ and $[S]_B^B = (c_0)$. (where $Tv = c_0 v$)

Proof. 1. Denote $U = \text{Sp}(\mathcal{B})$. Note that $w \in \mathcal{B}$ then $Tw \in U$. Indeed,

$$Tv, T(Tv), \dots, T(T^{d-2}v) \in U_0$$

and

$$T(T^{d-1}v) = T^d v = c_0 v + c_1 Tv + \dots + c_{d-1} T^{d-1}v \in U_0$$

(all v are in U_0).

$$\Rightarrow T(U_0) \subseteq U_0,$$

$$\Rightarrow U_0 \text{ is } T\text{-invariant. But also } v \in U_0,$$

$$\Rightarrow \text{by exc. 2, } \langle v \rangle_T \subseteq U_0. \text{ But clearly,}$$

$$U_0 \subseteq \langle v \rangle_T,$$

It follows that $U_0 = \langle v \rangle_T$.

Since, by assumption, the elements of $\mathcal{B}$ are lin. indep.

& $U_0 := \text{Sp}(\mathcal{B})$, then $\mathcal{B}$

is a basis for $U_0 = \langle v \rangle_T$,

2. Direct calculations. (check!),

□

Lemma (21.d.9). Make the same assumptions

as in Lemma 21.d.8. Then

$$p_S(x) = (-1)^d (x^d - c_{d-1}x^{d-1} - \dots - c_1x - c_0) \quad \text{if } d \geq 2$$

$$p_S(x) = -(x - c_0) \quad \text{if } d = 1.$$

Proof.

$$\text{Proof.} \quad [S]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_d = \begin{pmatrix} \boxed{-x} & 0 & \cdots & 0 & c_0 \\ 1 & \boxed{-x} & \cdots & 0 & c_1 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & \boxed{-x} & \vdots \\ 0 & \cdots & 0 & 1 & c_{d-1} - x \end{pmatrix}$$

Induction on d . $d = 1$ and $d = 2$ can be done by direct calculations.

$$\text{Let } d \geq 3. \quad \det([S]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_d)$$

$$\begin{aligned}
&= -x \cdot \det \underbrace{\begin{pmatrix} -x & 0 & \cdots & 0 & c_1 \\ 1 & -x & \cdots & 0 & c_2 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & -x & \vdots \\ 0 & \cdots & 0 & 1 & c_{d-1}-x \end{pmatrix}}_{\text{shape as } [S]_B^B \text{ but } (d-1) \times (d-1)} + (-1)^{d+1} c_0 \det \underbrace{\begin{pmatrix} 1 & -x & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -x & 0 \\ \vdots & & \ddots & 1 & -x \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix}}_{1=}, \\
&= -x \cdot (-1)^{d-1} (x^{d-1} - c_{d-1}x^{d-2} - \cdots - c_2x - c_1) \quad \text{by induct. hypoth.} \\
&\quad + (-1)^{d+1} c_0 = (-1)^d (x^d - c_{d-1}x^{d-1} - \cdots - c_1x - c_0), \\
&\text{as wished. The proof follows by induct.} \quad \square
\end{aligned}$$

Proof of Cayley-Hamilton. Let V be f.dim. and $T \in \text{End}(V)$.

$$p_T(x) := \det(T - x \cdot \text{id}_V).$$

We want to show $p_T(T) = 0$.

This is equiv. to showing that

$$p_T(T)(v) = 0 \quad \forall v \in V$$

So, let $v \in V$, w.l.o.g. assume $v \neq 0$, consider $\langle v \rangle_T$.

Take the maximal $d \geq 1$ s.t. $v, Tv, \dots, T^{d-1}v$ are lin. independent. Since V is f.dim., d is well defined.

Since d is maximal,

$$T^d v = \text{lin. combin. of } v, Tv, \dots, T^{d-1}v, \text{ i.e.,}$$

$$T^d v = c_0 v + c_1 Tv + \cdots + c_{d-1} T^{d-1}v,$$

where

$$c_0, \dots, c_{d-1} \in K, \quad (\dim \langle v \rangle_T = d).$$

Put $n = \dim V$. If $d < n$, extend

$$B = (v, Tv, \dots, T^{d-1}v)$$

to a basis

$$E = (v, Tv, \dots, T^{d-1}v, w_1, \dots, w_{n-d})$$

for V .

We have:

$$\begin{aligned}
[T]_E^E &= \left(\begin{array}{c|c} [S]_B^B & * \\ \hline 0 & C \end{array} \right) \begin{array}{l} \text{size } n-d \\ \text{size } n-d \end{array} \\
&\Rightarrow p_T(x) = p_S(x) \cdot p_C(x) = p_C(x) \cdot p_S(x),
\end{aligned}$$

But $R := p_T(T) \in \text{End}(V)$.

$$R = p_S(T) \cdot p_C(T) = p_C(T) \cdot p_S(T),$$

$$\Rightarrow R(v) = p_C(T) \cdot p_S(T)(v). \quad (*)$$

But

$$p_S(T)(v) = (-1)^d \left(T^d v - c_{d-1} T^{d-1} v - \dots - c_1 T v - c_0 v \right),$$

$$\underbrace{T^d v - c_{d-1} T^{d-1} v - \dots - c_1 T v - c_0 v}_{= 0 \text{ by def. of } c_0, \dots, c_{d-1}}$$

So from (*),

$$R(v) = p_C(T)(0) = 0.$$

This shows $R(v) = 0 \forall v \in V$,

$$\Rightarrow R = 0,$$

□

Outline of the analytic proof of Cayley-Hamilton:

1. If $A \in M_{n \times n}(K)$, $Q = \text{invertible } n \times n \text{ matrix}$

$$(Q^{-1} A Q)^\ell = Q^{-1} A^\ell Q, \quad \forall \ell \geq 1.$$

(just b.c.

$$(Q^{-1} A Q)(Q^{-1} A Q) \dots (Q^{-1} A Q) = \dots = Q^{-1} A^\ell Q.)$$

2. $\forall f(x) \in K[x]$,

$$f(Q^{-1} A Q) = Q^{-1} f(A) Q.$$

3. If

$$A = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

is diagonal, then $\forall f(x) \in K[x]$,

$$f(A) = \begin{pmatrix} f(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & f(\lambda_n) \end{pmatrix},$$

(b.c.

$$A^\ell = \begin{pmatrix} \lambda_1^\ell & & 0 \\ & \ddots & \\ 0 & & \lambda_n^\ell \end{pmatrix} \quad \forall \ell \geq 1,$$

)

4. C-H holds for diagonal matrices. Indeed if

$$A = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

then $p_A(\lambda_i) = 0 \forall i$

(b.c. λ_i are the eigenvals. of A), hence

$$p_A(A) = \begin{pmatrix} p_A(\lambda_1) & & 0 \\ 0 & \ddots & \\ 0 & & p_A(\lambda_n) \end{pmatrix} = 0.$$

5. If A is diagonalizable, i.e., $\exists$ invertible Q s.t.

$$Q^{-1}AQ = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

denote this matrix by Λ .

Then

$$p_A(A) = p_A(Q\Lambda Q^{-1}) = Q p_\Lambda(\Lambda) Q^{-1} = 0,$$

b.c. λ_i are the eigenval. of A .

40 6. Assume $K = \mathbb{C}$. Let $A \in M_{n \times n}(\mathbb{C})$.

We saw that $\exists Q$ invertible s.t.

$$Q^{-1}AQ = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix},$$

So, by the prev. steps, it's enough to prove C-H for upper triang. matrices.

7. Let $A \in M_{n \times n}(\mathbb{C})$ be upper triangular. Then $\exists$ a sequence $A_k \in M_{n \times n}(\mathbb{C})$ of diagonalizable matrices s.t.

$$\lim_{k \rightarrow \infty} A_k = A,$$

$$A = \begin{pmatrix} \lambda_1 & \cdots & a_{ij} \\ 0 & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix},$$

In fact we can take

$$A_k = \begin{pmatrix} \lambda_1^{(k)} & \cdots & a_{ij} \\ 0 & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{(k)} \end{pmatrix},$$

where

$$|\lambda_j^{(k)} - \lambda_j| \xrightarrow[k \rightarrow \infty]{} 0.$$

Chapter 7

Inner product and spectral theory

Lecture 41: Notes pending

13 Apr 2026

22.a Euclidean-Hermitian spaces. Euclidean and Hermitian (unitary spaces)

At last (!) We endow our space with a geometric structure. We work here with vector spaces V over $K=\mathbb{R}$ or $K=\mathbb{C}$. Over $\mathbb{R}$ we'll have Euclidean spaces, over $\mathbb{C}$ we'll have Hermitian (a.k.a. unitary) spaces.

Definition (22.a.1). Let V be a vect. sp. over $\mathbb{R}$. A scalar product (a.k.a. inner product) is a function

$$V \times V \longrightarrow \mathbb{R}, \quad (v, w) \mapsto \langle v, w \rangle \quad (\text{or } (v, w))$$

that has the following properties:

1. Linearity in the 1st variable:

$$\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle \quad \forall v_1, v_2, w \in V$$

$$\langle \alpha v, w \rangle = \alpha \langle v, w \rangle \quad \forall \alpha \in \mathbb{R}, v, w \in V.$$

2. Linearity in the 2nd variable:

$$\langle v, w_1 + w_2 \rangle = \langle v, w_1 \rangle + \langle v, w_2 \rangle \quad \forall v, w_1, w_2 \in V$$

$$\langle v, \alpha w \rangle = \alpha \langle v, w \rangle \quad \forall \alpha \in \mathbb{R}, v, w \in V,$$

3. Symmetry:

$$\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V.$$

41

4. Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v \rangle > 0.$$

We call $(V, \langle \cdot, \cdot \rangle)$ an Euclidean space, or a space endowed with a scalar product.

Definition (22.a.2). Let V be a vect. space over $\mathbb{C}$. A Hermitian product (or inner product) on V is a funct.

$$\begin{aligned} V \times V &\longrightarrow \mathbb{C} \\ v, w &\longmapsto \langle v, w \rangle \end{aligned}$$

with the following properties:

1. Linearity in the 1st variable:

$$\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle \quad \forall v_1, v_2, w \in V$$

$$\langle \alpha v, w \rangle = \alpha \langle v, w \rangle \quad \forall \alpha \in \mathbb{C}, v, w \in V.$$

2. Sesquilinearity (complex anti-linearity) in the 2nd variable:

$$\langle v, w_1 + w_2 \rangle = \langle v, w_1 \rangle + \langle v, w_2 \rangle \quad \forall v, w_1, w_2 \in V$$

$$\langle v, \alpha w \rangle = \overline{\alpha} \langle v, w \rangle \quad \forall \alpha \in \mathbb{C}, v, w \in V,$$

3. Hermitian property:

$$\langle v, w \rangle = \overline{\langle w, v \rangle}, \quad \forall v, w \in V.$$

4. Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v \rangle > 0.$$

We call $(V, \langle \cdot, \cdot \rangle)$ a Hermitian (or unitary) space.

Remark. The definition of Hermitian product “coincides” with scalar prod. if V is over $\mathbb{R}$ and we allow α only in $\mathbb{R}$.

Recall that if

$$\alpha = a + ib \in \mathbb{C},$$

where $a, b \in \mathbb{R}$, $i = \sqrt{-1}$, then

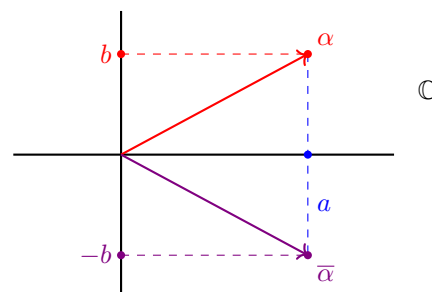
$$\overline{\alpha} := a - ib,$$

the complex conjugate of α .

Recall also that

$$\alpha \cdot \overline{\alpha} = a^2 + b^2 = |\alpha|^2 \geq 0,$$

with equality iff $\alpha = 0$.



Lemma (22.a.3). Let V be an Euclid./Hermit. vect. space. Then:

Exc.

- 1) $\forall v \in V, \langle 0, v \rangle = \langle v, 0 \rangle = 0.$
- 2) Let $w \in V$. If $\langle v, w \rangle = 0 \forall v \in V$, then $w = 0$.
- 3) Let $w_1, w_2 \in V$. If $\langle v, w_1 \rangle = \langle v, w_2 \rangle \forall v \in V$, then $w_1 = w_2$.

Definition (22.a.4).

- 1) Let V be an Euclid./Hermit. vect. space.

$$\forall v \in V, \quad \|v\| := \sqrt{\langle v, v \rangle} \in \mathbb{R}_{\geq 0}.$$

$\|-\|$ is called the norm on V induced by $\langle -, - \rangle$.

- 2) A vector $u \in V$ is called a unit vect. if $\|u\| = 1$.

Note that if $0 \neq v \in V$ is any vector, then $\frac{1}{\|v\|}v$ is a unit vector (which is positively proportional to v , so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the normalization of v .

- 3) For $v, w \in V$ we define the distance between v and w by

$$d(v, w) = \|v - w\|.$$

- 4) Two vectors $u, v \in V$ are called orthogonal (or perpendicular one to the other) if $\langle u, v \rangle = 0$.

If this is the case we sometimes write $u \perp v$.

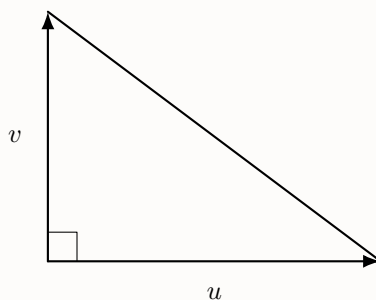
- 5) A subset $S \subseteq V$ is called an orthogonal system if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.

- 6) An orthogonal system $S \subseteq V$ is called orthonormal if

$$\forall v \in S \text{ we have } \|v\| = 1.$$

Theorem (Pythagoras theorem 22.a.5). Let V be an Euclidean/Hermitian space. Let $u, v \in V$ be two orthogonal vectors s.t. $u \perp v$. Then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$



Proof.

Assume that $u \perp v$. Then

$$\|u + v\|^2 = \langle u + v, u + v \rangle.$$

Expanding the inner product gives

$$\langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle.$$

Since $u \perp v$, we have

$$\langle u, v \rangle = 0 \quad \text{and} \quad \langle v, u \rangle = 0.$$

Also,

$$\langle u, u \rangle = \|u\|^2 \quad \text{and} \quad \langle v, v \rangle = \|v\|^2.$$

Therefore,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

□

Exercise

1. Let V be an Euclidean space. Show that for every $u, v \in V$ we have

$$\langle u, v \rangle = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

41

Example.

1. The standard scalar product on $\mathbb{R}^n$. $\mathbb{R}^n = \mathbb{R}^n \text{ col.}$ $\forall u, v \in \mathbb{R}^n, \langle u, v \rangle := \underbrace{u^\top}_{1 \times n} \cdot \underbrace{v}_{n \times 1} \in {}^1I_{1 \times n}(\mathbb{R}) = \mathbb{R}.$

$$\left\langle \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\rangle.$$

$$\left\| \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\| = \sqrt{\sum_{k=1}^n v_k^2}.$$

2. The standard Hermitian product on $\mathbb{C}^n$. $\forall u, v \in \mathbb{C}^n, (= \mathbb{C}^n \text{ col.}),$

$$\langle u, v \rangle := u^\top \cdot \bar{v} = \sum_{k=1}^n u_k \cdot \overline{v_k} \in \mathbb{C},$$

$$\|v\| = \sqrt{\sum_{k=1}^n v_k \cdot \overline{v_k}}.$$

3. If V is an Euclidean/Hermitian space and $W \subseteq V$ is a subspace, then by

restricting $\langle -, - \rangle$ from $V \times V$ to $W \times W$ we get an Euclidean/Hermitian structure on W .

Definition (22.a.6). A matrix $A \in M_{n \times n}(\mathbb{R})$ is called symmetric if $A^\top = A$. (i.e. $a_{ij} = a_{ji} \forall i, j$).

Definition (22.a.7). 1. Let $A \in M_{n \times n}(\mathbb{C})$. Write $\bar{A}$ for matrix where entry at a_{ij} is $\bar{a}_{ij}$. ($(\bar{A})_{ij} = \bar{a}_{ij}$).

2. Let $A \in M_{n \times n}(\mathbb{C})$. define

$$A^* := (\bar{A})^\top.$$

A^* is called the adjoint matrix of A . (has nothing to do with $\text{adj } A$ form determinants).

3. $A \in M_{n \times n}(\mathbb{C})$ is called Hermitian if $A^* = A$. ($\Leftrightarrow A^\top = \bar{A}$).

Example (Important example). Define

$$\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\langle u, v \rangle_A := \underbrace{v^\top}_{1 \times n} \underbrace{A}_{n \times n} \underbrace{\cdot}_{n \times 1} \in \mathbb{R}.$$

Exercise

$\langle \cdot, \cdot \rangle_A$ is bilinear (i.e. linear in both variables). Assume that A is symmetric. ($A^\top = A$). Then $\langle u, v \rangle_A = \langle w, u \rangle_A \forall u, w$. Indeed, $\langle w, u \rangle_A = w^\top A v = w^\top A^\top v = (v^\top A w)^\top = (\underbrace{\langle v, w \rangle_A}^{\text{this is } 1 \times 1})^\top = \langle v, w \rangle_A$.

Question: Is $\langle \cdot, \cdot \rangle_A$ a scalar product?

In general, no! e.g. $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_A = (0, 1) \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 < 0..$$

Example (Examples). 1) Let $a_1, \dots, a_n > 0$, where $a_k \in \mathbb{R}$ for all k .

Then

$$D = \begin{pmatrix} a_1 & & & 0 \\ & \ddots & & \\ & & \ddots & \\ 0 & & & a_n \end{pmatrix}$$

41

is positive definite.

Indeed, if

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n,$$

then

$$\langle v, v \rangle_D = v^T D v = a_1 v_1^2 + \cdots + a_n v_n^2.$$

If $v \neq 0$, then this is > 0 .

2) Let

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}).$$

Exercise

Show that A is positive definite iff $a > 0$ and $\det A > 0$.

Definition (22.a.8). Let $A \in M_{n \times n}(\mathbb{C})$ be a Hermitian matrix.

We say that A is positive definite if

$$\forall 0 \neq v \in \mathbb{C}^n, \quad v^T A \bar{v} > 0.$$

Remark. Note that $v^T A \bar{v} \in \mathbb{R}$ for all $v \in \mathbb{C}^n$. This is b.c.,

Lecture 42: Notes pending

15 Apr 2026

42

Lecture 42 Placeholder

Notes for this lecture will be added here.

Lecture 43: Notes pending

17 Apr 2026

43

Lecture 43 Placeholder

Notes for this lecture will be added here.

Lecture 44: Notes pending

22 Apr 2026

44

Lecture 44 Placeholder

Notes for this lecture will be added here.

Lecture 45: Notes pending

24 Apr 2026

45

Recap

Last time.Let $U \subseteq V$ be a finite-dimensional subspace.Then $U^\perp$ is a complement of U in V . That is,

$$U + U^\perp = V \quad \text{and} \quad U \cap U^\perp = \{0\}.$$

Suppose $e_1, \dots, e_r$ is an orthonormal basis for U . Then for every $v \in V$, define

$$\tilde{P}_U(v) := \langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r.$$

Then

$$\tilde{P}_U(v) \in U.$$

Moreover,

$$v - \tilde{P}_U(v) \in U^\perp.$$

Therefore,

$$v = \tilde{P}_U(v) + (v - \tilde{P}_U(v)),$$

where

$$\tilde{P}_U(v) \in U \quad \text{and} \quad v - \tilde{P}_U(v) \in U^\perp.$$

Digression: Let V be a vector space over any field K . Let $W_1 \subseteq V$ be a subspace and let $W_2 \subseteq V$ be a complement of W_1 in V . ($W_1 + W_2 = V$ and $W_1 \cap W_2 = \{0\}$.) So $\forall v \in V \exists!$ a decomposition $v = w_1 + w_2$ with $w_1 \in W_1$ and $w_2 \in W_2$. Define maps $P_{W_1} : V \rightarrow W_1$ and $P_{W_2} : V \rightarrow W_2$ by

$$P_{W_1}(v) := w_1 \quad \text{and} \quad P_{W_2}(v) := w_2.$$

Claim. P_{W_1} and P_{W_2} are linear, $\text{Im}(P_{W_1}) = W_1$, $\text{Ker}(P_{W_1}) = W_2$. $\forall v \in V, v - P_{W_1}(v) \in W_2$. So, $v = P_{W_1}(v) + (P_{W_2}(v))$.

Definition (22.b.4). Let V be an inner product space. Let $U \subseteq V$ be a finite-dimensional subspace. We have $U^\perp$ is a complement of U in V . The map $P_U : V \rightarrow$

P_U is called the orthogonal projection to U .

Proposition. P_U satisfies:

1. $P_U : V \rightarrow U$ is linear.
2. $\text{Im}(P_U) = U, \text{Ker}(P_U) = U^\perp$.
3. $\forall v \in V, v - P_U(v) \in U^\perp$ (i.e. $P_U + P_{U^\perp} = \text{Id}_V$).
4. $\forall u \in U, P_U(u) = u$.
5. If $e_1, \dots, e_r$ is an orthonormal basis for U , then $P_U(v) = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r = \tilde{P}_U(v)$.

Proof. 1-4 follow from the "digression" discussion before and is a priori trivial. 5) Let $v \in V$. Write

$$v = \left(\sum_{i=1}^r \langle v, e_i \rangle e_i \right) + \left(v - \sum_{i=1}^r \langle v, e_i \rangle e_i \right).$$

Clearly $(1) \in U$. We saw that $(2) \perp U$. So $(2) \in U^\perp$. So (1) must be equal to $P_U(v)$. $\square$

Corollary (22.b.6). Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space. Let $U \subseteq V$ be a finite-dimensional subspace. Then

$$(U^\perp)^\perp = U.$$

Proof. By previous statement (22.b.2) $U \subseteq (U^\perp)^\perp$. Let $v \in (U^\perp)^\perp$. Since $U^\perp$ is a complement of U , write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Since $u \in U \subseteq (U^\perp)^\perp$, it follows that $w = v - u \in (U^\perp)^\perp$. But $w \in U^\perp$, and we know that $U^\perp \cap (U^\perp)^\perp = \{0\}$. So $w = 0$, and $v = u \in U$. So $(U^\perp)^\perp \subseteq U$. $\square$

Corollary (22.b.7). If V is a finite-dimensional inner product space and $U \subseteq V$ is a subspace, then $\dim U + \dim U^\perp = \dim V$.

22.a.1 Orthogonal & unitary matrices

Definition (22.b.8). 1. A matrix $A \in M_{n \times n}(\mathbb{R})$ is called orthogonal if its columns form an orthonormal basis for $\mathbb{R}^n$, w.r.t. the standard scalar product on $\mathbb{R}^n$.

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \langle v_i, v_j \rangle = \delta_{ij}.$$

The set of all orthogonal matrices is denoted by $O(n)$.

2. A matrix $A \in M_{n \times n}(\mathbb{C})$ is called unitary if its columns form an orthonormal

basis for $\mathbb{C}^n$, w.r.t. the standard scalar product on $\mathbb{C}^n$. The set of unitary $n \times n$ matrices is denoted by $U(n)$.

Lemma (22.b.9). Let $A \in M_{n \times n}(\mathbb{R})$. Then

$$\begin{aligned} A \in O(n) &\iff A^T A = I_n \iff AA^T = I_n \\ &\iff A^{-1} = A^T. \end{aligned}$$

Lemma (22.b.10). Let $A \in M_{n \times n}(\mathbb{C})$. Then

$$A \in U(n) \iff A^* A = I_n \iff AA^* = I_n \iff A^{-1} = A^*.$$

Since

$$A^* = \overline{A}^T,$$

equivalently, after complex conjugating these equalities, we also have

$$A \in U(n) \iff A^T \overline{A} = I_n \iff \overline{\overline{A}^T} = I_n.$$

Proof of Lemma 22.b.10. Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \quad v_k \in \mathbb{C}_{\text{col}}^n, \quad k = 1, \dots, n.$$

Then

$$A^T \overline{A} = \begin{pmatrix} \text{---} & v_1^T & \text{---} \\ \vdots & & \vdots \\ \text{---} & v_n^T & \text{---} \end{pmatrix} \begin{pmatrix} | & & | \\ \overline{v_1} & \cdots & \overline{v_n} \\ | & & | \end{pmatrix} = (v_i^T \overline{v_j})_{ij} = (\langle v_i, v_j \rangle_{\text{std}})_{ij}.$$

Indeed, for $u \in \mathbb{C}_{\text{col}}^n$, we have

$$\langle v, u \rangle_{\text{std}} := v^T \overline{u} = (v_1, \dots, v_n) \begin{pmatrix} \overline{u_1} \\ \vdots \\ \overline{u_n} \end{pmatrix}.$$

Thus A is unitary if and only if $v_1, \dots, v_n$ are an orthonormal basis for $\mathbb{C}^n$. Equivalently,

$$\langle v_i, v_j \rangle = \delta_{ij} \iff A^T \overline{A} = I_n.$$

By linear algebra 1, we also have $\overline{A} A^T = I_n$, etc. □

Corollary (22.b.11). 1. Let $A \in M_{n \times n}(\mathbb{R})$. Then

$$A \in O(n) \iff A^T \in O(n).$$

2. Let $A \in M_{n \times n}(\mathbb{C})$. Then

$$A \in U(n) \iff A^* \in U(n).$$

Equivalently, in both cases, the rows of A form an orthonormal basis for $\mathbb{R}^n$ or $\mathbb{C}^n$, respectively.

QR-decomposition

Theorem (22.b.12). 1. Let $A \in GL_n(\mathbb{R})$. Then there exist $Q \in O(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{R})$ such that

$$A = QR.$$

2. Let $A \in GL_n(\mathbb{C})$. Then there exist $Q \in U(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{C})$ such that

$$A = QR.$$

Proof. We prove (1) and (2) together. Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \quad \text{with } v_j \in \mathbb{K}_{\text{col}}^n, \quad \mathbb{K} = \begin{cases} \mathbb{R}, \\ \mathbb{C}. \end{cases}$$

Since $A \in GL_n(\mathbb{K})$, the columns $v_1, \dots, v_n$ of A form a basis for $\mathbb{K}_{\text{col}}^n$.

Apply G-M to $v_1, \dots, v_n$ and obtain an orthonormal basis $e_1, \dots, e_n$ for $\mathbb{K}_{\text{col}}^n$. Recall that for every $1 \leq j \leq n$,

$$\text{Sp}(v_1, \dots, v_j) = \text{Sp}(e_1, \dots, e_j).$$

We also have

$$\begin{aligned} v_1 &= \langle v_1, e_1 \rangle e_1, \\ v_2 &= \langle v_2, e_1 \rangle e_1 + \langle v_2, e_2 \rangle e_2, \\ &\vdots \\ v_j &= \langle v_j, e_1 \rangle e_1 + \cdots + \langle v_j, e_j \rangle e_j, \\ &\vdots \\ v_n &= \langle v_n, e_1 \rangle e_1 + \cdots + \langle v_n, e_n \rangle e_n. \end{aligned}$$

Therefore

$$\begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix} \begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \cdots & \langle v_j, e_1 \rangle & \cdots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \cdots & \langle v_j, e_2 \rangle & \cdots & \langle v_n, e_2 \rangle \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \langle v_j, e_j \rangle & \cdots & \langle v_n, e_j \rangle \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & \langle v_n, e_n \rangle \end{pmatrix}.$$

In other words,

$$A = QR,$$

where

$$Q = \begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix}$$

and

$$R = \begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \cdots & \langle v_j, e_1 \rangle & \cdots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \cdots & \langle v_j, e_2 \rangle & \cdots & \langle v_n, e_2 \rangle \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \langle v_j, e_j \rangle & \cdots & \langle v_n, e_j \rangle \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & \langle v_n, e_n \rangle \end{pmatrix}.$$

The matrix R is upper triangular. Also, $Q \in O(n)$ in the real case and $Q \in U(n)$ in the complex case, because $e_1, \dots, e_n$ form an orthonormal basis for $\mathbb{K}_{\text{col}}^n$. This completes the proof. $\square$

Extension of QR decomposition

Let $A \in M_{n \times n}(\mathbb{K})$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Let

$$r = \text{rank}(A).$$

Then there exist matrices $Q \in O(n)$ or $U(n)$, and

$$R = \left(\begin{array}{c|c} C & * \\ \hline 0 & 0 \end{array} \right) \in M_{n \times n}(\mathbb{K}),$$

where $C \in M_{r \times r}(\mathbb{K})$ is upper triangular, such that

$$A = Q \cdot R.$$

Dual spaces and inner products

Let V be a vector space over $\mathbb{K}$. Recall that

$$V^* := \text{Hom}(V, \mathbb{K})$$

is the space of all linear maps, a.k.a. functionals,

$$\varphi : V \rightarrow \mathbb{K}.$$

Assume $V = \mathbb{K}_{\text{col}}^n$, and let

$$\varphi : \mathbb{K}_{\text{col}}^n \rightarrow \mathbb{K}$$

be a functional. Then there exists a unique

$$A = (a_1, \dots, a_n) \in M_{1 \times n}(\mathbb{K})$$

such that

$$\varphi(v) = A \cdot v \quad \forall v \in \mathbb{K}_{\text{col}}^n.$$

45 Indeed, if $e_1, \dots, e_n$ is the standard basis for $\mathbb{K}_{\text{col}}^n$, then

$$a_i = \varphi(e_i), \quad i = 1, \dots, n.$$

Lecture 46: Notes pending

29 Apr 2026

46 Lecture 46 Placeholder

Notes for this lecture will be added here.

Lecture 47: Notes pending

06 May 2026

47 Lecture 47 Placeholder

Notes for this lecture will be added here.

Lecture 48: Notes pending

08 May 2026

48 Lecture 48 Placeholder

Notes for this lecture will be added here.

Lecture 49: Notes pending

13 May 2026

49 Lecture 49 Placeholder

Notes for this lecture will be added here.

Lecture 50: Notes pending

15 May 2026

50 Lecture 50 Placeholder

Notes for this lecture will be added here.

Lecture 51: Notes pending

20 May 2026

51

Lecture 51 Placeholder

Notes for this lecture will be added here.

Lecture 52: Notes pending

22 May 2026

52

Lecture 52 Placeholder

Notes for this lecture will be added here.

Lecture 53: Notes pending

27 May 2026

53

Lecture 53 Placeholder

Notes for this lecture will be added here.

Lecture 54: Notes pending

29 May 2026

54

Lecture 54 Placeholder

Notes for this lecture will be added here.